



Choupo

The Open Source Chemical Process Simulator

Tutorials Guide

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Authors: Vítor Geraldês.

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How to use this guide

Every Choupo tutorial is a complete, runnable case:

```
source etc/bashrc
runCase tutorials/<category>/<group>/<name>
```

This guide gives each tutorial the CONTEXT the case file cannot: the theory block at the head of each group states the physics and the equations you need (condensed – the Theory Guide carries the full derivations), and each entry tells you **what the case does** and **what to expect** before you run it. The per-tutorial text is assembled from the cases' own headers – the case file is always the source of truth, and reading it IS part of the exercise.

Three reading rules:

1. **Run first, read second, re-run third.** The log at verbosity 3 shows every Newton iteration, every K-value, every announcement – the glass box is the point.
2. **Expect the announcements.** Choupo never silently helps: initial guesses, bounds that bind, model fallbacks, unit overrides – all are printed. If a run is quiet about something, it did not happen.
3. **Golden numbers are commitments.** Cases carrying an **expected** file pin their KPIs; the validation AADs quoted in the entries are locked by the regression suite and re-verified on every change.

Learning paths

The MCFT path (computational transfer phenomena). Start with `flash01_benzene_toluene` (the Rachford–Rice loop, visible), then `flash` non-ideal cases (activity models bend the K-values), `distillation` (stages couple; Wang–Henke vs simultaneous), one `flowsheets` recycle case (tears and convergence), and close with a `gibbs` flame (equilibrium without mechanism). Then the batch side: an `ignition` case for stiffness – run the `_rk4` twin and watch the explicit method die – and a `ctrl` loop for feedback.

The electrolytes path. `pitzer01_nacl` ($\gamma_{\pm}$, φ , a_w of one salt), then `pitzer_gamma_hamer_wu` (the catalogue vs NBS data, zero refit), `pitzer_nacl_sp77_hot` (temperature, to 200°C), speciation (`scaling` cases: SI, pH, Davies-vs-Pitzer forking an engineering answer), the eNRTL series (`enrtl_mixed_nacl_ethanol_esteso` mixed-solvent; `enrtl_multisalt_nacl_kcl` two salts, two formulations, an experimental verdict), and the process payoff: evaporators and crystallisers where all of it sets duties and yields.

The combustion path. `gibbs06_h2_flame_radicals` (the radical pool at equilibrium; the c_p extrapolation trap), `ignition01_h2o2` (GRI kinetics, the stiffness lesson via the `_rk4` twin), `ignition02_h2air_slack` (quantitative: τ_{ign} vs shock-tube data), `nox01_zeldovich_postflame` (thermal NO; kinetics meets Gibbs at 98.9%).

The props-bench path. `estimate_acetone` (CREATE: groups $\rightarrow$ constants, error shown), a `scan` case (SEE), `vle01_consistency` (TEST the data), a `fit` case (Levenberg–Marquardt with identifiability), then the promote-and-use loop into a steady case. The Props Guide is this path’s companion.

Category I: Steady-state flowsheets (choupoSolve)

Solved as one nonlinear system $F(\mathbf{x}) = \mathbf{0}$: units in sequence, recycles as tears, everything announced.

absorption

The theory you need. Gas absorption transfers a solute from gas to liquid across stages; Henry's law ($p_i = H_i x_i$) anchors the equilibrium at dilution. Expect the L/G ratio to control the operating line's slope, and a minimum solvent rate at the pinch with the equilibrium line. *Full derivation: Theory Guide, the absorption section.*

absorber01_NH3_water

tutorials/steady/absorption/absorber01_NH3_water



What it does: NH3 absorption from air into water (Kremser, 6 stages)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cleanGas	F	0.0259019084984
stream	gasFeed	F	0.02777777777778
stream	richLiquid	F	0.043542535946
stream	solvent	F	0.0416666666667
kpi	scrubber	A_N2	0.00225907961951
kpi	scrubber	A_NH3	1.53522727273
kpi	scrubber	F_cleanGas	0.0259019084984
kpi	scrubber	F_richLiquid	0.043542535946
kpi	scrubber	K_N2	663.987221631
kpi	scrubber	K_NH3	0.977054034049
kpi	scrubber	L_in	0.0416666666667
kpi	scrubber	L_over_V	1.5
kpi	scrubber	V_in	0.02777777777778
kpi	scrubber	recovery_N2	0
... and 2 more reference values in the case's <i>expected</i> .			

absorption01_CO2_water

tutorials/steady/absorption/absorption01_CO2_water



What it does: Single-stage CO2 absorption in water (Henry’s law)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cleanGas	F	0.0183200303715
stream	feed	F	1.41666666667
stream	richLiquid	F	1.3983466363
kpi	scrubber	F_alpha	1.3983466363
kpi	scrubber	F_beta	0.0183200303715
kpi	scrubber	F_in	1.41666666667
kpi	scrubber	P	500000
kpi	scrubber	T	298.15
kpi	scrubber	V_over_F	0.0129317861446
kpi	scrubber	phaseSet	0

extract01_ethanol_water_benzene

tutorials/steady/absorption/extract01_ethanol_water_benzene



What it does: Counter-current LL extraction: recover ethanol from water with benzene

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	extract	F	0.0359293973028
stream	feed	F	0.0277777777778
stream	raffinate	F	0.0251772962492
stream	solvent	F	0.0333333333333
kpi	extractor01	F_extract	0.0359293973028
kpi	extractor01	F_feed	0.0277777777778
kpi	extractor01	F_raffinate	0.0251772962492
kpi	extractor01	F_solvent	0.0333333333333
kpi	extractor01	P	101325
kpi	extractor01	T	298.15
kpi	extractor01	converged	1
kpi	extractor01	distribution_benzene	295.27509389
kpi	extractor01	distribution_ethanol	0.555106921422
kpi	extractor01	distribution_water	0.00706545462823
...and 9 more reference values in the case's expected.			

extraction01_waterButanol

tutorials/steady/absorption/extraction01_waterButanol

**What it does:** Water/n-butanol LL flash at 298 K (experimental)

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Water + 1-butanol at 25 °C and 1 atm. The pair has a well-known liquid-liquid miscibility gap below its upper critical solution temperature (125.5 °C), so a feed whose composition lies inside the gap spontaneously splits into two liquid phases. This case feeds a 70/30 water/nButanol mixture (inside the gap) to an isothermal decanter and lets the flash find the two-liquid split.

A single feed splits into two liquids, both at 298.15 K, $v_f = 0$:

The split is found by Michelsen TPD stability testing (to seed a robust initial guess) followed by direct Gibbs-energy minimisation — the LL path the engine uses precisely because a Newton/successive-substitution flash on a symmetric γ -model would collapse to the trivial $K = 1$ saddle.

- phases (...) syntax with TWO liquid phases; - phaseSet LL in the IsothermalFlash; - the Michelsen TPD stability test as the initial-guess generator.

- The LL flash reuses the VL solver's printout, so the two liquids are shown under Vapor flow V / Liquid flow L headings (and x/y/K labels). Read them as phase- α (organic) and phase- β (aqueous), not as a vapour and a liquid. - The decanter is isothermal and adiabatic in reality (an enthalpy hand-check gives $H_{in} \approx H_{out}$), so the physically correct duty is ≈ 0 . The KPI Q and any utility/cost the run reports for this LL split are a known artifact of the shared flash duty path — do not cost this unit from them.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	aqueous	F	0.0140665296362
stream	feed	F	0.02777777777778
stream	organic	F	0.0137112481416
kpi	decanter01	F_alpha	0.0140665296362
kpi	decanter01	F_beta	0.0137112481416
kpi	decanter01	F_in	0.02777777777778
kpi	decanter01	P	101325
kpi	decanter01	T	298.15
kpi	decanter01	betaFraction	0.493604933098
kpi	decanter01	phaseSet	1

stripper01_NH3_water

tutorials/steady/absorption/stripper01_NH3_water

**What it does:** NH3 stripping from water with air (Kremser, 6 stages, warm)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	liquidFeed	F	0.0277777777778
stream	richGas	F	0.029140357434
stream	strippedLiquid	F	0.0264151981216
stream	strippingGas	F	0.0277777777778
kpi	stripper	F_richGas	0.029140357434
kpi	stripper	F_strippedLiquid	0.0264151981216
kpi	stripper	K_NH3	3.62862883208
kpi	stripper	K_water	0.170562530402
kpi	stripper	L_in	0.0277777777778
kpi	stripper	S_NH3	3.62862883208
kpi	stripper	S_water	0.170562530402
kpi	stripper	T_bottom	313.524974667
kpi	stripper	T_top	315.973930965
kpi	stripper	V_in	0.0277777777778

... and 6 more reference values in the case's *expected*.**crystallisation**

The theory you need. Crystallisation removes solute when the solution crosses saturation: $m > m_{\text{sat}}(T)$, with $K_{\text{sp}}(T)$ from ΔG_{diss} and its T -dependence from ΔH_{diss} (van 't Hoff). The heat of crystallisation is ION-DERIVED – $\Delta H_{\text{cryst}} = -(\Delta H_{\text{soln}}^{\infty} + \bar{L}_2(m_{\text{sat}}))$ – never a stored constant (a solid's formation enthalpy is $\sum \nu_i h_{f,i}^{\text{aq}} - \Delta H_{\text{soln}}$, derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant, $\gamma_{\pm}$ rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

crystalliser01_sugar

tutorials/steady/crystallisation/crystalliser01_sugar

What it does: Cooling crystalliser: sugar solution 80 → 20 C, yield by solubility

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0261287129062
kpi	cryst	Q_removed	271677.114808
kpi	cryst	T_op	293.15
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.564469958333
kpi	cryst	crystal_mol	0.00164906487154
kpi	cryst	liquorFlow	0.0261287129062
kpi	cryst	solute_dissolved	0.861767541667
kpi	cryst	solute_in	1.4262375
kpi	cryst	yield	0.395775569169

crystalliser02_msmpr

tutorials/steady/crystallisation/crystalliser02_msmpr

What it does: MSMPR sugar crystalliser: population balance by method of moments

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0266203308749
kpi	cryst	CV	1
kpi	cryst	L_dominant	0.000223952804481
kpi	cryst	L_meanMass	0.000298603739308
kpi	cryst	L_meanNumber	7.46509348269e-05
kpi	cryst	Q_removed	269525.057151
kpi	cryst	T_op	293.15
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.396190602513
kpi	cryst	crystal_mol	0.00115744690287
kpi	cryst	growthRate	1.95272330048e-08
kpi	cryst	liquorFlow	0.0266203308749
kpi	cryst	magmaDensity	302.920529908
...and 10 more reference values in the case's <i>expected</i> .			

crystalliser03_pbe_size_independent

tutorials/steady/crystallisation/crystalliser03_pbe_size_independent

What it does: Crystalliser FVM-PBE: sanity vs MSMPR (size-independent growth)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0266086567804
kpi	cryst	CV	0.999540291345
kpi	cryst	L_dominant	0.000224609375
kpi	cryst	L_meanMass	0.000298633437171
kpi	cryst	L_meanNumber	7.48298114726e-05
kpi	cryst	Q_removed	269576.1605
kpi	cryst	T_op	293.15
kpi	cryst	bins	256
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.400186610054
kpi	cryst	crystal_mol	0.00116912099742
kpi	cryst	growthRate_L0	1.90635340048e-08
kpi	cryst	liquorFlow	0.0266086567804

...and 11 more reference values in the case's *expected*.

crystalliser04_size_dependent_growth

tutorials/steady/crystallisation/crystalliser04_size_dependent_growth

What it does: Crystalliser FVM-PBE with size-dependent growth (sizeFactor > 0)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	magma	F	0.0265894971524
kpi	cryst	CV	1.03562508094
kpi	cryst	L_dominant	0.000232421875
kpi	cryst	L_meanMass	0.000329374475484
kpi	cryst	L_meanNumber	7.45169890057e-05
kpi	cryst	Q_removed	269660.031771
kpi	cryst	T_op	293.15
kpi	cryst	bins	256
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.406744893212
kpi	cryst	crystal_mol	0.00118828062534
kpi	cryst	growthRate_L0	1.83025070562e-08
kpi	cryst	liquorFlow	0.0265894971524

...and 11 more reference values in the case's *expected*.

crystalliser05_nacl_pitzer

tutorials/steady/crystallisation/crystalliser05_nacl_pitzer

What it does: NaCl crystalliser: saturation from the PITZER ion activity ($K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

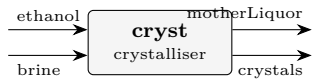
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | /// U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0271500572444
kpi	cryst	Ksp_activity	38.4846213082
kpi	cryst	Q_removed	1251.26049958
kpi	cryst	T_op	298.15
kpi	cryst	c_sat	0.35905536
kpi	cryst	crystal_flow	0.036683987968
kpi	cryst	crystal_mol	0.000627720533333
kpi	cryst	gamma_pm_sat	1.00970010562
kpi	cryst	liquorFlow	0.0271500572444
kpi	cryst	m_sat	6.144
kpi	cryst	solute_dissolved	0.158116012032
kpi	cryst	solute_in	0.1948
kpi	cryst	yield	0.18831616

crystalliser06_nacl_antisolvent

tutorials/steady/crystallisation/crystalliser06_nacl_antisolvent



What it does: Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises γ_{pm} , and DROPS the saturation molality m_{sat} – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent γ_{pm} .

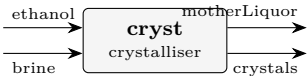
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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.0277777777778
stream	crystals	F	0.000329661969914
stream	ethanol	F	0.0166666666667
stream	motherLiquor	F	0.0424240933568
kpi	cryst	Ksp_activity	34.8439846653
kpi	cryst	Q_removed	6559.87377671
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	cakeLiquidFlow	0.000329661969914
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0988038720389
kpi	cryst	crystal_mol	0.00169068911771
...and 8 more reference values in the case's expected.			

crystalliser07_nacl_antisolvent_cooled

tutorials/steady/crystallisation/crystalliser07_nacl_antisolvent_cooled



What it does: Antisolvent + COOLING crystallisation of NaCl: the same drowning-out brine + ethanol as crystalliser06, but the crystalliser is COOLED to 5 C. The eNRTL is now temperature-dependent ($\tau(T)=\tau_{25}\cdot 298.15/T$, $A_{DH}(T)$ via $\epsilon_{ps_w}(T)/\rho_{w}(T)$, Born at T) and $K_{sp}(T)$ shifts by van't Hoff with NaCl's enthalpy of solution – so cooling adds to the antisolvent's effect (yield 61% → 70%). Exercises the T-dependent electrolyte path.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.0277777777778
stream	crystals	F	0.000380305320939
stream	ethanol	F	0.0166666666667
stream	motherLiquor	F	0.0421245706304
kpi	cryst	Ksp_activity	31.135192385
kpi	cryst	Q_removed	82721.44242
kpi	cryst	T_op	278.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0398065230351
kpi	cryst	cakeLiquidFlow	0.000380305320939
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.113348382739
kpi	cryst	crystal_mol	0.00193956849313
...and 8 more reference values in the case's expected.			

crystalliser08_nacl_antisolvent_msmpr

tutorials/steady/crystallisation/crystalliser08_nacl_antisolvent_msmpr



What it does: Continuous MSMPR antisolvent crystallisation of NaCl: a saturated brine + ethanol fed to a well-mixed vessel (residence time $\tau = V/Q$). The eNRTL antisolvent supersaturation $S = c/c_{\text{sat}}(\text{eNRTL})$ drives nucleation + growth; the method of moments closes the population balance and returns the CRYSTAL-SIZE DISTRIBUTION $n(L)$ (mean size, CV) – what the equilibrium model cannot give.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.0277777777778
stream	ethanol	F	0.0166666666667
stream	magma	F	0.0428576741462
kpi	cryst	CV	1
kpi	cryst	L_dominant	2.02534396647e-05
kpi	cryst	L_meanMass	2.70045862196e-05
kpi	cryst	L_meanNumber	6.75114655489e-06
kpi	cryst	Q_kW	-6.1566687571
kpi	cryst	Q_removed	6156.6687571
kpi	cryst	T_op	298.15
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	crystal_flow	0.092730856228
kpi	cryst	crystal_mol	0.00158677029822
kpi	cryst	growthRate	1.92664299147e-09
... and 12 more reference values in the case's expected.			

crystalliser09_KHT_KCl_series

tutorials/steady/crystallisation/crystalliser09_KHT_KCl_series



What it does: Two crystallisers in series separating KCl from potassium bitartrate (KHT, cream of tartar) by manipulating T and adding ethanol. CRYST1 COOLS the hot feed to 5 C → KHT (sparingly soluble, steep solubility curve) crystallises while KCl stays dissolved. CRYST2 doses ETHANOL into the KCl-rich mother liquor → KCl drops out by drowning-out (eNRTL mixed-solvent). Two salts, two saturation routes (curve vs ion-activity), one package – a stress test of the property system.

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization |

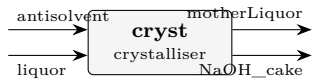
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	KCl_cake	F	0.000273622015015
stream	KHT_cake	F	2.54671494088e-05
stream	feed	F	0.0277777777792
stream	finalLiquor	F	0.0541131192255
stream	freshEthanol	F	0.00416666666667
stream	liquor1	F	0.0277239373151
stream	recoveredEthanol	F	0.0236111111111
stream	stillage	F	0.0305020081144
kpi	cryst1	Q_kW	-93.1162690166
kpi	cryst1	Q_removed	93116.2690166
kpi	cryst1	T_op	278.15
kpi	cryst1	c_sat	0.0034013435
kpi	cryst1	cakeLiquidFlow	2.54671494088e-05
kpi	cryst1	cakeMoisture	0.1

...and 39 more reference values in the case's *expected*.

crystalliser10_naoh_antisolvent

tutorials/steady/crystallisation/crystalliser10_naoh_antisolvent



What it does: NaOH antisolvent crystallisation (water + ethanol, eNRTL drowning-out) – the acceptance-case core

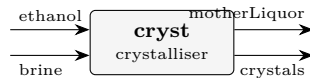
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | Heat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | Operations | Website: <https://choupo.org> | | \{\} \{\} | Units | Licence: GPL-3.0-or-later | | \{\} | Properties | Tutorial case file – part of Choupo | | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	NaOH_cake	F	0.000203756483719
stream	antisolvent	F	0.00833333333333
stream	liquor	F	0.0138888888889
stream	motherLiquor	F	0.020384504743
kpi	cryst	Ksp_activity	7365.8380048
kpi	cryst	Q_kW	72.7276039114
kpi	cryst	Q_removed	-72727.6039114
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.461538461538
kpi	cryst	c_sat	0.181200024556
kpi	cryst	cakeLiquidFlow	0.000203756483719
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0653535379386
...and 9 more reference values in the case's <i>expected</i> .			

crystalliser11_kcl_antisolvent_ethanol

tutorials/steady/crystallisation/crystalliser11_kcl_antisolvent_ethanol



What it does: Antisolvent (drowning-out) crystallisation of KCl: a SATURATED aqueous KCl brine (4.76 mol/kg) is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises $\gamma_{\pm}$, and DROPS the saturation molality – so KCl precipitates with NO cooling and NO evaporation. Saturation is the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent $\gamma_{\pm}$ at the feed's salt-free ethanol fraction. KCl-water-ethanol experimental reference: Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

The KCl counterpart of crystalliser06 (NaCl): a saturated aqueous KCl brine is mixed with ethanol, whose low dielectric collapses the KCl solubility, so KCl crystallises with no cooling and no evaporation. KCl is the salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)" – this case realises it.

Ethanol's relative permittivity (24.3) is far below water's (78.5). Dosing it lowers the mixed-solvent dielectric, which raises the mean ionic activity coefficient $\gamma_{\pm}$ (eNRTL mixed-solvent + Born term), so the saturation molality collapses. The saturation is the same ion-activity product $K_{sp} = (\gamma_{\pm} \cdot m_{sat})^2$ anchored in pure water, re-solved with the mixed-solvent $\gamma_{\pm}$ at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

100 kmol/h saturated brine (x_{water} 0.921, x_{KCl} 0.079 = 4.76 mol/kg) + 60 kmol/h ethanol → salt-free $x_{ethanol} \approx 0.39$:

The KCl–water–ethanol activity behaviour modelled here is measured in A. Lopes, F. Farelo & M. I. A. Ferra, "Activity Coefficients of Potassium Chloride in Water–Ethanol Mixtures", J. Solution Chem. 28 (1999) 117 – mean activity coefficients of KCl by emf in 5–20 % w/w ethanol, 25–45 °C. That $\gamma_{\pm}$ rise with ethanol content is exactly the driver of the drowning-out captured above.

- The eNRTL τ for K–Cl is the illustrative 1:1-alkali-chloride proxy (= Chen & Evans water–NaCl), so the numbers are qualitative. The Farelo data is the path to a quantitative refit of the KCl–water–ethanol interaction (the paper reports Pitzer $\beta^0/\beta^1/C^?$ per solvent composition). - The mass / yield is the reliable output. The crystalliser's heat term carries the shared electrolyte-enthalpy caveat (a dissolved salt's formation datum lives in its ions, so the stream enthalpy falls back to the sensible datum and is announced) – identical to crystalliser06.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.0277777777778
stream	crystals	F	0.000423217143227
stream	ethanol	F	0.0166666666667
stream	motherLiquor	F	0.0423393557491
kpi	cryst	Ksp_activity	7.69319557877
kpi	cryst	Q_kW	-28.9281906957
kpi	cryst	Q_removed	28928.1906957
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.394477317554
kpi	cryst	c_sat	0.0311001943033
kpi	cryst	cakeLiquidFlow	0.000423217143227
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.125385206079

... and 9 more reference values in the case's expected.

overlay02_nacl_calorimetric

tutorials/steady/crystallisation/overlay02_nacl_calorimetric

What it does: NaCl crystalliser: saturation from the PITZER ion activity ($K_{sp} = (\gamma_{pm} m_{sat})^2$). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

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A NaCl crystalliser (cloned from crystalliser05) whose case-local constant/components/NaCl.dat declares overlayOf NaCl; and recalibrates ONLY solidPhases.halite.calorimetric.dissolutionEnthalpy (3880 → 5880 J/mol).

The ThermoPackageBuilder deep-merges this delta over the standard record, so the crystalliser duty reflects it: $Q_{removed} = 3690.997$ W here vs 2435.556 in crystalliser05 (ratio 1.515 = 5880/3880). Every other KPI ($K_{sp_activity}$, m_{sat} , c_{sat} , $crystal_mol$, γ) is byte-identical to crystalliser05 – logK25, dH, the dissolution reaction and the crystal block all still come from the standard.

This is the second half of the roadmap Phase A acceptance test: the SINGLE unified reader applies the overlay at ALL three component-loading sites (SpeciationSolver, Database, ThermoPackageBuilder), not just the SI path.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0271500572444
kpi	cryst	Ksp_activity	38.4846213082
kpi	cryst	Q_kW	-2.50670156624
kpi	cryst	Q_removed	2506.70156624
kpi	cryst	T_op	298.15
kpi	cryst	c_sat	0.35905536
kpi	cryst	crystal_flow	0.036683987968
kpi	cryst	crystal_mol	0.000627720533333
kpi	cryst	gamma_pm_sat	1.00970010562
kpi	cryst	liquorFlow	0.0271500572444
kpi	cryst	m_sat	6.144
kpi	cryst	solute_dissolved	0.158116012032
kpi	cryst	solute_in	0.1948

... and 1 more reference values in the case's *expected*.

distillation

The theory you need. A column is N flash stages coupled by counter-current flows. Each stage obeys MESH equations: Material balance, Equilibrium ($y_i = K_i x_i$), Summation ($\sum y = \sum x = 1$), and Heat balance. Choupo ships two solution strategies, selectable per column: Wang–Henke (bubble-point, solving the tridiagonal material balances stage by stage; robust for near-ideal systems, slow near azeotropes) and **simultaneous** (one Newton on ALL the MESH residuals; ~5–6 iterations even through azeotropes). Expect: reflux and boilup set the split; a minimum reflux below which no stage count reaches the spec; and azeotropes as fixed points no column can cross (x_D pinned at the azeotropic composition). *Full derivation: Theory Guide, the distillation chapter (McCabe–Thiele + MESH).*

column01_benzene_toluene

tutorials/steady/distillation/column01_benzene_toluene



What it does: Benzene/toluene distillation column (Wang-Henke)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	column01	T_top	354.21089412
kpi	column01	T_bottom	382.893530951
kpi	column01	x_D_LK	0.981245045999
kpi	column01	x_B_HK	0.981245183716

column02_simultaneous

tutorials/steady/distillation/column02_simultaneous



What it does: Rigorous distillation by simultaneous MESH Newton (benzene/toluene; validates against column01)

The story: - Choupo - * | \{\}/ C hemicals | Choupo: open-source, glass-box simulation | | \{\}\{\}/ H eat-transfer | Version: Choupo-2607 | | \{\}\{\}\{\}/ O perations | Website: <https://choupo.org> | | \{\}\{\}/ U nits | Licence: GPL-3.0-or-later | | \{\}/ P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	R	2
kpi	column01	T_bottom	382.893527916
kpi	column01	T_top	354.210890137
kpi	column01	feedStage	8
kpi	column01	nStages	15
kpi	column01	x_B_HK	0.981245125322
kpi	column01	x_D_LK	0.981245125322

column03_azeotrope_mesh

tutorials/steady/distillation/column03_azeotrope_mesh



What it does: Azeotropic distillation (ethanol/water, NRTL) by the rigorous simultaneous MESH method

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.02083333333333
stream	distillate	F	0.006944444444444
stream	feed	F	0.02777777777778
kpi	column	B	0.02083333333333
kpi	column	D	0.006944444444444
kpi	column	R	2.5
kpi	column	T_bottom	357.713408957
kpi	column	T_top	351.388958309
kpi	column	feedStage	8
kpi	column	nStages	16
kpi	column	x_B_HK	0.868205515093
kpi	column	x_D_LK	0.804616545279

column04_multifeed_sidedraw

tutorials/steady/distillation/column04_multifeed_sidedraw



What it does: Simultaneous MESH column: two feeds + a liquid side draw (rigorous MESH)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

The rigorous simultaneous MESH column extended to multiple feeds and side draws — the most-used features of a commercial simulator column, on the glass-box Newton solver.

A benzene / toluene / o-xylene split (light / intermediate / heavy):

- primary feed — 100 kmol/h, benzene/toluene/o-xylene 0.40/0.40/0.20, enters high (stage 5); - second feed — 60 kmol/h, heavier (0.05/0.45/0.50), enters low (stage 11); - liquid side draw — 15 kmol/h pulled from stage 8, between the two feeds.

Mass closes ($100 + 60 = 42 + 15 + 103 = 160$ kmol/h). $Q_{\text{condenser}} = -1436$ kW, $Q_{\text{reboiler}} = +1527$ kW. The side draw does exactly what it should — bleed the intermediate component where it is most enriched on the trays.

The simultaneous MESH is a bubble-point-reduced Newton (CMO flows). The extension generalises the single fixed feed into a per-stage flow staircase: $L[j]$, $V[j]$ are rebuilt by a CMO march that adds each feed's liquid/vapour parts and subtracts each draw; every stage's material balance carries an F_j $z_{\{F,j\}}$ source and (L_j+U_j) , (V_j+W_j) outflows. The general balance reduces exactly to the old single-feed equations when there is one feed and no draw — so every existing distillation tutorial is unchanged (verified).

outputs (distillate bottoms <draw1> <draw2> ...) — side-draw streams follow distillate and bottoms, in stage order.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0286111111111
stream	distillate	F	0.0116666666667
stream	feed	F	0.0277777777778
stream	intermediateCut	F	0.0041666666667
kpi	radfracLite	B	0.0286111111111
kpi	radfracLite	D	0.0116666666667
kpi	radfracLite	Q_condenser_kW	-1435.80274013
kpi	radfracLite	Q_reboiler_kW	1516.14164522
kpi	radfracLite	R	3
kpi	radfracLite	T_bottom	395.926996928
kpi	radfracLite	T_top	354.978489481
kpi	radfracLite	feedStage	5
kpi	radfracLite	nFeeds	2
kpi	radfracLite	nSideDraws	1

... and 3 more reference values in the case's *expected*.

column05_reactive_methylacetate

tutorials/steady/distillation/column05_reactive_methylacetate



What it does: Reactive distillation: methyl acetate synthesis (validation vs P?pken et al. 2001, run S-1)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

The simultaneous MESH column with an equilibrium reaction on the catalytic stages (rigorous column rung 4), compared to a published experiment.

Mole-conserving, so the CMO internal-flow profile is untouched. Each catalytic stage carries one extra unknown — the molar extent ξ_j — closed by the activity-based equilibrium residual $\sum_i \nu_i \ln(\gamma_i x_i) = \ln K_a(T)$, with γ_i from the liquid activity model and $K_a(T)$ by van't Hoff: $K_a(T) = K_a(298) \cdot \exp(-\Delta h^\circ_r / R \cdot (1/T - 1/298.15))$. Converged by homotopy (solve non-reactive, then switch the reaction on).

> T. P?pken, S. Steinigeweg & J. Gmehling, "Synthesis and Hydrolysis of > Methyl Acetate by Reactive Distillation", Ind. Eng. Chem. Res. 40 > (2001) 1566. Run S-1: $N = 25$, reactive stages 11–19, HOAc fed above / > MeOH below the zone, reflux 2.1, $P \approx 1.017$ bar. Measured endpoints, cited, in > constant/experimental/methyl_acetate_RD_profile.csv.

The equilibrium constant and thermodynamics are from the companion paper P?pken, G?tze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601: $K_a(298) \approx 38.7$, $\Delta h^\circ_r \approx -5.67$ kJ/mol $\rightarrow K_a \approx 29$ in the reactive zone ($\sim 65^\circ\text{C}$).

*This is not a tight match, and the reason is physically important: with the correct equilibrium K_a (≈ 29) the model gives the equilibrium ceiling (96 %), and it correctly brackets the measurement from above. The measured 87.6 % is *below* the equilibrium because the real column is kinetically limited — the paper's recommended model is an adsorption-based finite *rate* (per gram of catalyst), not equilibrium. An earlier run with an (incorrect, too low) $K_a = 5.2$ gave 90.7 %, a closer *number* but for the wrong reason; using a wrong constant to fake agreement would be dishonest.*

So the path to a bit-tight match is the kinetic rate model, not the thermo.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	9.13888888889e-06
stream	distillate	F	8.36111111111e-06
stream	feed	F	8.94444444444e-06
kpi	rdColumn	B	9.13888888889e-06
kpi	rdColumn	D	8.36111111111e-06
kpi	rdColumn	Q_condenser_kW	-0.790521316742
kpi	rdColumn	Q_reboiler_kW	0.626813788646
kpi	rdColumn	R	2.1
kpi	rdColumn	T_bottom	363.229492539
kpi	rdColumn	T_top	330.062558898
kpi	rdColumn	conversion	0.960508806863
kpi	rdColumn	feedStage	20
kpi	rdColumn	nFeeds	2
kpi	rdColumn	nSideDraws	0

...and 4 more reference values in the case's expected.

column06_fullMESH

tutorials/steady/distillation/column06_fullMESH



What it does: Full-MESH (Naphtali-Sandholm) distillation – energy-consistent V_j/L_j , seeded from the CMO solution (benzene/toluene; compare column02)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

The same benzene/toluene column as column02_simultaneous, but solved by the full MESH (model fullMESH) instead of the bubble-point/CMO reduction.

The simultaneous method is a bubble-point-reduced MESH: it assumes constant molal overflow (CMO — equimolal latent heats), so the internal vapour/liquid flows V_j , L_j are fixed by a mass march and only the compositions x_j and stage temperatures T_j are solved.

The full MESH (Naphtali-Sandholm) promotes V_j and L_j to unknowns and adds, per stage, the total-mass balance and the energy balance — so the flows follow the *real* latent heats instead of assuming them equal. The extra per-stage energy equation is the enthalpy residual the CMO never carried:

Total condenser: $V_0 = 0$, $L_0 = R \cdot D$; reboiler: $L_{\{N-1\}} = B$; their energy balances give Q_{cond} / Q_{reb} as results.

The full-MESH is seeded from the converged CMO profile — the cheap model launches the rigorous one. Here it converges in ~4 Newton iterations from that seed (it would be far harder from a cold start).

Benzene ($\Delta h_{\text{vap}} \approx 30.8 \text{ kJ/mol}$) and toluene ($\approx 33.2 \text{ kJ/mol}$) have similar latent heats, so CMO is a good approximation and the two answers nearly coincide — the ~0.002 / ~0.1 K gap IS the energy-balance correction the full-MESH makes and the CMO drops. On a wide-boiling column (a depropaniser) the gap is much larger; that is where the full-MESH earns its cost.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column06	B	0.0138888888889
kpi	column06	D	0.0138888888889
kpi	column06	Q_condenser_kW	-1281.19539111
kpi	column06	Q_reboiler_kW	1299.72437334
kpi	column06	R	2
kpi	column06	T_bottom	382.774927808
kpi	column06	T_top	354.339103085
kpi	column06	feedStage	8
kpi	column06	nFeeds	1
kpi	column06	nSideDraws	0
kpi	column06	nStages	15

... and 2 more reference values in the case's *expected*.

column07_naphtaliSandholm

tutorials/steady/distillation/column07_naphtaliSandholm



What it does: Full-MESH (Naphtali-Sandholm) validation vs AIChE J. 17 (1971) 148, Problem 2: nC4/iC5/nC5

The story: - Choupo -*—————* VALIDATION – Naphtali & Sandholm, AIChE Journal 17(1), 148-153 (1971), DOI 10.1002/aic.690170130, Problem 2 (ideal hydrocarbon distillation). 20 plates; liquid feed ($T = 30^\circ\text{F} = 272.04\text{ K}$, 500 mol each of nC4/iC5/nC5) on plate 10; reflux ratio 1000; 80% recovered in bottoms ($B = 1200$). A liquid sidestream on plate 1 (condenser) takes 30% of the reflux.

EQUIVALENT SPEC (same internal flows, no condenser sidestream): the distillate and the sidestream leave the total condenser with the SAME composition, so they are combined into ONE top product of 300 with reflux $R = L_1/300 = 996.68/300 = 3.3223$. This gives identical $L_1 = 996.68$ and $V_2 = 1296.68$, hence the same column.

Paper result: $B = \text{nC4 } 200.1 / \text{iC5 } 499.9 / \text{nC5 } 500.0$; combined top $\sim 99.97\%$ nC4; plate $T \ 29.7^\circ\text{F}$ (top) $\rightarrow 77.0^\circ\text{F}$ (reboiler) $\sim 272.0 \rightarrow 298.2\text{ K}$.

Reproduces the ideal-hydrocarbon distillation of Naphtali & Sandholm, "Multicomponent Separation Calculations by Linearization", AIChE Journal 17(1), 148–153 (1971) — DOI [10.1002/aic.690170130](https://doi.org/10.1002/aic.690170130) — the paper that introduced the very full-MESH method Choupo's model fullMESH implements.

n-butane / isopentane / n-pentane, 20 plates, liquid feed ($T = 30^\circ\text{F} = 272.04\text{ K}$, 500 mol of each) on plate 10, reflux ratio 1000, 80 % of the feed recovered in the bottoms ($B = 1200$). A liquid sidestream on plate 1 (the condenser) takes 30 % of the reflux.

*Equivalent spec used here. The distillate (≈ 1) and the condenser sidestream (≈ 299) leave the *total* condenser with the same composition, so they are combined into ONE top product of 300 with reflux $R = L_1/300 = 996.68/300 = 3.3223$. This gives identical internal flows ($L_1 = 996.68$, $V_2 = 1296.68$), hence the identical column — without needing a condenser side draw.*

Converges in 3 full-MESH Newton iterations (seeded from the CMO profile).

Products Choupo — top product (300) 99.99 % n-C₄ bottoms n-C₄ / i-C₅ / n-C₅ 200 / 500 / 500 B 1200

*The profile shape — the flat n-butane-rich top, the sharp rise across the feed zone, the pentane-plateau, the reboiler climb — matches stage-by-stage. The $\sim 1\text{--}2\text{ K}$ residual is thermo, not method: Choupo uses fitted Antoine vapour pressures, the paper used the K-value tables in Brown's *Unit Operations* (1950).*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.416666666667
kpi	column07	B	0.333333333333
kpi	column07	D	0.0833333333333
kpi	column07	Q_condenser_kW	-8082.6120755
kpi	column07	Q_reboiler_kW	9374.37429576
kpi	column07	R	3.3223
kpi	column07	T_bottom	296.58946842
kpi	column07	T_top	272.663334648
kpi	column07	feedStage	10
kpi	column07	nFeeds	1
kpi	column07	nSideDraws	0
kpi	column07	nStages	20
kpi	column07	x_B_HK	0.416665691626
kpi	column07	x_D_LK	0.999874145869

column08_radfrac_multidraw

tutorials/steady/distillation/column08_radfrac_multidraw



What it does: rigorous column: multi-feed + multi-side-draw distillation (C3/C4/C5/C6), full-MESH

The story: - Choupo -*—————* Multi-feed, multi-side-draw rigorous column (the most-used rigorous column features at once). A C3/C4/C5/C6 mix split FOUR ways: distillate ~ propane, an upper liquid side draw ~ n-butane, a lower liquid side draw ~ n-pentane, bottoms ~ n-hexane. TWO feeds enter at different heights (a light cut high, a heavy cut low) – so the column sees the realistic case of feeds of different composition. Feeds are flowsheet STREAMS (‘inputs (...)’) so the plant-boundary mass+energy balance sees them (information follows the streams). Validation reference (propose): Hofeling & Seader, AIChE J. 24(6) (1978), DOI 10.1002/aic.690240631 – the Naphtali-Sandholm extension for general multi-feed / multi-draw / interlinked separators.

A C3/C4/C5/C6 depropanizer-style column at 16 bar, split four ways by the rigorous full-MESH, exercising the two most-used rigorous column features at once: TWO feeds (a light cut high, a heavy cut low) and TWO liquid side draws.

Two saturated-liquid feeds (100 kmol/h each, 290 K / 320 K): a light cut on stage 10, a heavy cut on stage 20. 30 stages, $R = 4$. Mass closes exactly ($200 = 45 + 55 + 55 + 45$ kmol/h) and the energy balance closes: the feed-to-product enthalpy gap (1137.7 kW) equals the net duty $Q_{\text{reb}} + Q_{\text{cond}} = 1942.3 - 804.6$ kW. Converges in a few full-MESH iterations from the CMO seed.

This case was first written at 1 atm, and the energy balance would not close — because at 1 atm and 290 K the light hydrocarbons (propane b.p. 231 K, n-butane 272 K) are above their boiling points: the feed streams flash to VAPOUR ($vf = 1.0$), yet the column was told quality 1.0 (saturated liquid). The latent heat of the unrecognised vapour feed was exactly the missing ~1 MW. At 16 bar the feeds are genuinely liquid ($vf = 0$, consistent with quality 1.0) and the balance closes — which is also why a real depropanizer runs at elevated pressure: so the propane distillate condenses against cooling water (here 321.6 K). A feed’s thermal state must be specified consistently with its actual phase.

*Both feeds are real flowsheet streams (inputs (feed1 feed2), mapped to stages by operation.feeds), so the plant-boundary mass + energy balance sees them (*information follows the streams*). Side draws are outputs in stage order: outputs (distillate bottoms drawC4 drawC5).*

Self-consistently validated (mass + energy close, four cuts in volatility order) on the full-MESH solver that IS paper-validated against Naphtali & Sandholm (1971) in column07_naphtaliSandholm. No external multi-feed+multi-draw single-column benchmark with published profiles is readily available in the open literature.

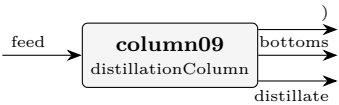
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0125
stream	distillate	F	0.0125
stream	drawC4	F	0.0152777777778
stream	drawC5	F	0.0152777777778
stream	feed1	F	0.0277777777778
stream	feed2	F	0.0277777777778
kpi	radfrac	B	0.0125
kpi	radfrac	D	0.0125
kpi	radfrac	Q_condenser_kW	-804.58622364
kpi	radfrac	Q_reboiler_kW	1698.71150765
kpi	radfrac	R	4
kpi	radfrac	T_bottom	454.498489397
kpi	radfrac	T_top	321.647346922
kpi	radfrac	feedStage	10

... and 5 more reference values in the case’s expected.

column09_tray_hydraulics

tutorials/steady/distillation/column09_tray_hydraulics



What it does: Sieve-tray hydraulics: the column that separates, sized so its trays do not flood

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column09	B	0.0138888888889
kpi	column09	D	0.0138888888889
kpi	column09	L_rect	0.0277777777778
kpi	column09	L_strip	0.0555555555556
kpi	column09	Q_condenser_kW	-1281.04348002
kpi	column09	Q_reboiler_kW	1279.30480005
kpi	column09	R	2
kpi	column09	T_bottom	382.893530951
kpi	column09	T_top	354.21089412
kpi	column09	V_rect	0.0416666666667
kpi	column09	V_strip	0.0416666666667
...and 11 more reference values in the case's <i>expected</i> .			

column10_flooding

tutorials/steady/distillation/column10_flooding



What it does: The same column, built too narrow: every tray floods – and it still converges

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column10	B	0.0138888888889
kpi	column10	D	0.0138888888889
kpi	column10	L_rect	0.0277777777778
kpi	column10	L_strip	0.0555555555556
kpi	column10	Q_condenser_kW	-1281.04348002
kpi	column10	Q_reboiler_kW	1279.30480005
kpi	column10	R	2
kpi	column10	T_bottom	382.893530951
kpi	column10	T_top	354.21089412
kpi	column10	V_rect	0.0416666666667
kpi	column10	V_strip	0.0416666666667
...and 11 more reference values in the case's <i>expected</i> .			

column11_murphree

tutorials/steady/distillation/column11_murphree



What it does: Murphree tray efficiency: 15 REAL trays, not 15 ideal stages – and the purity you actually get

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column11	B	0.0138888888889
kpi	column11	D	0.0138888888889
kpi	column11	L_rect	0.0277777777778
kpi	column11	L_strip	0.0555555555556
kpi	column11	Q_condenser_kW	-1284.29753602
kpi	column11	Q_reboiler_kW	1281.05938025
kpi	column11	R	2
kpi	column11	T_bottom	381.492133108
kpi	column11	T_top	355.136328304
kpi	column11	V_rect	0.0416666666667
kpi	column11	V_strip	0.0416666666667
...and 12 more reference values in the case's <i>expected</i> .			

shortcut01_benzene_toluene

tutorials/steady/distillation/shortcut01_benzene_toluene



What it does: Shortcut (FUG) design of a benzene/toluene column

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column	B	0.0138888888889
kpi	column	D	0.0138888888889
kpi	column	N_min	10.067785029
kpi	column	N_theoretical	21.5880731317
kpi	column	R	1.68248609289
kpi	column	R_min	1.29422007145
kpi	column	alpha_LK_HK	2.49137883567
kpi	column	feed_stage	10.7940365659
kpi	column	theta	1.42716041594
kpi	column	x_B_HK	0.99
kpi	column	x_D_LK	0.99

drying

solidDryer01_sugar

tutorials/steady/drying/solidDryer01_sugar

What it does: Solid dryer: wet sugar powder dried to sorption equilibrium

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	drySolid	F	5.80292648834e-05
stream	hotAir	F	0.222222222222
stream	humidExhaust	F	0.232719748513
stream	wetSolid	F	0.0105555555556
kpi	solidDryer	T_out	347.5126791
kpi	solidDryer	X_equilibrium	0.00109945367945
kpi	solidDryer	X_final	0.00109945367945
kpi	solidDryer	X_initial	0.199991235758
kpi	solidDryer	air_in_kmol_h	800
kpi	solidDryer	drySolid_flow	0.95187873054
kpi	solidDryer	moisture_pct_wb	0.109824620861
kpi	solidDryer	water_activity	0.00431158404453
kpi	solidDryer	water_removed	0.189112936126

sprayDryer01_sugar

tutorials/steady/drying/sprayDryer01_sugar



What it does: Spray dryer: 40 wt% sugar solution → powder (rotary atomiser, Ranz-Marshall)

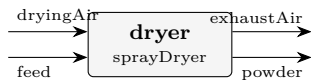
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\}\{\} | H eat-transfer | Version: Choupo-2607 | | \{\}\{\}\{\} | O perations | Website: <https://choupo.org> | | \{\}\{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.01388888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.0009576111111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	d_droplet	6.04023981348e-05
kpi	dryer	d_droplet_micron	60.4023981348
kpi	dryer	d_particle	3.80926671659e-05
... and 12 more reference values in the case's <i>expected</i> .			

sprayDryer02_residence_sweep

tutorials/steady/drying/sprayDryer02_residence_sweep



What it does: Spray dryer: chamber-height sweep → the drying curve (residence = L/v ; constant + falling rate → equilibrium)

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

sprayDryer03_pressure_nozzle

tutorials/steady/drying/sprayDryer03_pressure_nozzle



What it does: Spray dryer: 40 wt% sugar solution → powder (PRESSURE-SWIRL nozzle at 40 bar, $d_{32}=9575 \cdot dP^{-1/3}$; Ranz-Marshall)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24532614778
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203070475508
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.2283799688
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842
...and 30 more reference values in the case's <i>expected</i> .			

sprayDryer04_profiles

tutorials/steady/drying/sprayDryer04_profiles



What it does: Spray dryer DISTRIBUTED: axial profiles of moisture, size and temperature along the chamber

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.01388888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

... and 30 more reference values in the case's *expected*.

sprayDryer05_whey

tutorials/steady/drying/sprayDryer05_whey



What it does: Spray dryer: whey (prob 5.2) – 500 kg/h, X0=5, rotary co-current, sorption-isotherm residual moisture

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.138888888889
stream	exhaustAir	F	0.14529654835
stream	feed	F	0.00649261111111
stream	powder	F	1.73405388823e-05
kpi	dryer	Nu	2.33731434528
kpi	dryer	Pr	0.745230561117
kpi	dryer	Re	0.384509154435
kpi	dryer	Sc	0.602916461126
kpi	dryer	Sh	2.31430911286
kpi	dryer	T_out	382.7774359
kpi	dryer	T_wetbulb	315.824008236
kpi	dryer	X_critical	0.5
kpi	dryer	X_equilibrium	0.0134981947613
kpi	dryer	X_final	0.0134981947613
...and 30 more reference values in the case's <i>expected</i> .			

sprayDryer06_rea

tutorials/steady/drying/sprayDryer06_rea



What it does: Spray dryer REA (model chen): Reaction Engineering Approach kinetics – smooth rate, no critical-moisture break

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842
...and 30 more reference values in the case's <i>expected</i> .			

sprayDryer07_design

tutorials/steady/drying/sprayDryer07_design



What it does: Spray dryer DESIGN: give targets (powder size, residence), the sizing pass inverts to the geometry

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\}\{\} | H eat-transfer | Version: Choupo-2607 | | \{\}\{\}\{\} | O perations | Website: <https://choupo.org> | | \{\}\{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.01388888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

... and 31 more reference values in the case's *expected*.

economics

The theory you need. Sizing (stirred tanks, shell-and-tube areas) feeds costing (Guthrie modules); prices are CASE data (constant/economics, dated and cited) – the engine refuses to invent a price. Expect honest refusal when a price is missing, never a silent default.

economics01_esterification_dcf

tutorials/steady/economics/economics01_esterification_dcf



What it does: Esterification plant: CAPEX (Guthrie) → OPEX (COM_d) → discounted-cash-flow appraisal (NPV/IRR/payback)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

A fixed (designed) ethyl-acetate plant taken all the way from equipment sizing to a discounted-cash-flow economic appraisal:

1. *sizing* — *StirredTank* for the CSTR, *ShellTubeHX* for the heater, sized from each unit's KPIs.
2. *costing* — *Guthrie/Turton* purchased + bare-module + total-module cost, *CEPCI*-updated to the target year, in EUR.
3. *economics* — the new *EconomicsPass*: it builds the CAPEX ladder (FCI, working capital, TCI), the *Turton* cost-of-manufacture *COM_d*, revenue, a straight-line-depreciation cash-flow timeline, then NPV, IRR (bisection bracket + Newton polish), and discounted + simple payback.

Every cost factor and DCF assumption is dict-owned and printed — read the appraisal block and you can re-derive every number by hand.

*Depreciation enters the DCF exactly once (it is **not** in *COM_d*) — the classic textbook double-count is avoided. Working capital is an outflow at startup and is recovered untaxed at end-of-life.*

*Market/time-specific data — product and feed prices, labour rate, FX — lives only in constant/economics, each value dated and primary-cited (an index/bulletin/agency named in a provenance{} block). The *EconomicsPass* refuses to run if a required price is absent, with a remedy message; set *economics.refuseOnMissingPrice* 0; in the *postDict* to switch to defaulted-but-loud. No price ever defaults to a literal in code.*

*The economics pass also publishes its headline scalars into *result.kpis["economics"]* (IRR, NPV, paybackYears, FCI, TCI, *COM_d*), which is what lets a sweep or optimisation read economics as an ordinary response — see *economics02_irr_sweep*.*

economics02_irr_sweep

tutorials/steady/economics/economics02_irr_sweep



What it does: Sweep reactor volume; read economics.IRR/NPV/payback per point (SweepDriver runs the post chain)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

The same esterification plant as economics01, but driven by a sensitivity sweep over the reactor volume that reads economic responses at every point:

*The SweepDriver now runs the full post-processing chain (sizing → costing → economics) on each converged point, so the headline economic scalars the EconomicsPass publishes into result.kpis["economics"] resolve as ordinary unit.kpi responses. A sweep can finally *see cost*:*

A bigger reactor raises conversion (more product, more revenue) but raises CAPEX and feed throughput — the IRR falls monotonically over this range, a trade-off invisible from a single CAPEX number.

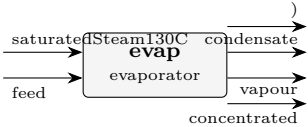
> The range floors at 4 m³ on purpose: below ~3 m³ the lower conversion > drops the mixture bubble point and the fixed-duty heater would cross the > saturation dome (a heater is sensible-only) — the deliberate validity edge.

evaporation

The theory you need. An evaporator boils solvent off a solution. Two pieces of physics beyond a flash: boiling-point elevation, $T_{\text{boil}} = T_{\text{sat}}(P) + \text{BPE}$ with the BPE from the electrolyte activity of water ($\ln a_w = -\varphi \nu m M_w / 1000$); and the heat of dilution/concentration (the relative apparent molar enthalpy $L_\varphi(m, T)$) entering the duty. Expect: the BPE to GROW as the brine concentrates, steam economy below 1 for a single effect, and – for NaCl – the full Silvester–Pitzer T -dependence of $\gamma_\pm$ shaping the answer between 25 and 200°C. *Full derivation: Theory Guide, the electrolyte chapters.*

evaporator01_brine

tutorials/steady/evaporation/evaporator01_brine



What it does: Single-effect evaporator: 35 g/L brine at 1 bar; the fixed steam duty/UA evaporates ~82 % of the feed water ($V/F=0.82$), enriching NaCl ~5.5x ($x\ 0.011 \rightarrow 0.059$) – duty-driven, not a target (BPE included)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0049888708933
stream	condensate	F	0.0277777777778
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0277777777778
stream	vapour	F	0.0227889068845
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	1.79512624904
kpi	evap	F_conc	0.0049888708933
kpi	evap	F_conc_kmol_h	17.9599352159
kpi	evap	F_conc_mass	0.101856029976
kpi	evap	F_cond	0.500416666667
kpi	evap	F_cond_kg_h	1801.5
kpi	evap	F_in	0.0277777777778
... and 19 more reference values in the case's <i>expected</i> .			

evaporator02_triple_effect_sugar

tutorials/steady/evaporation/evaporator02_triple_effect_sugar



What it does: Triple-effect cocurrent evaporator: 22500 kg/h sugar solution 5%→25% (MCFT Ex. 10.4)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1	F	0.24340099744
stream	L2	F	0.150822370581
stream	L3	F	0.0533879416888
stream	V1	F	0.0870984081566
stream	V2	F	0.0925786268589
stream	V3	F	0.0974344288922
stream	cond1	F	0.1375
stream	cond2	F	0.0870984081566
stream	cond3	F	0.0925786268589
stream	feed	F	0.330499405597
stream	utilitySteam_200kPa	F	0.1375
kpi	effect1	A	100
kpi	effect1	A_m2	100
kpi	effect1	BPE	0.107002096415
...and 81 more reference values in the case's expected.			

evaporator03_co_current

tutorials/steady/evaporation/evaporator03_co_current



What it does: Co-current forward-feed triple-effect evaporator, user-defined components (MCFT 10.38, case a)

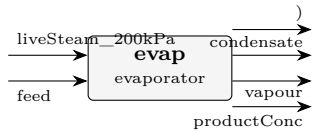
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1to2	F	0.0469536109199
stream	L2to3	F	0.0295647943065
stream	V1	F	0.0161887935578
stream	V2	F	0.0173888166134
stream	V3	F	0.018462595509
stream	cond1	F	0.0262363636123
stream	cond2	F	0.0161887935578
stream	cond3	F	0.0173888166134
stream	feed	F	0.0631424044778
stream	liveSteam	F	0.0262363636123
stream	productConc	F	0.0111021987975
kpi	effect1	A	16.5383245281
kpi	effect1	A_m2	16.5383245281
kpi	effect1	BPE	0
...and 81 more reference values in the case's <i>expected</i> .			

evaporator04_orange_juice_simple

tutorials/steady/evaporation/evaporator04_orange_juice_simple



What it does: Single-effect evaporator concentrating orange juice 10% to 50% solids (MCFT Ex. 10.2 a-c, without BPE)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

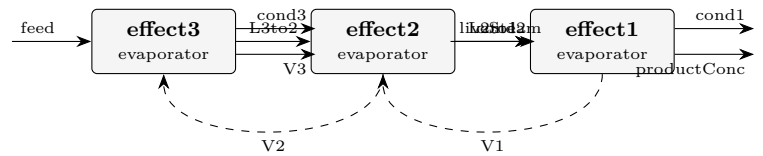
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	0.360686102898
stream	feed	F	0.350791135988
stream	liveSteam_200kPa	F	0.360686102898
stream	productConc	F	0.0424059629635
stream	vapour	F	0.308385173024
kpi	evap	A	75.5133330137
kpi	evap	A_m2	75.5133330137
kpi	evap	BPE	0
kpi	evap	F_conc	0.0424059629635
kpi	evap	F_conc_kmol_h	152.661466669
kpi	evap	F_conc_mass	1.38888555242
kpi	evap	F_cond	6.49776014371
kpi	evap	F_cond_kg_h	23391.9365174
kpi	evap	F_in	0.350791135988

... and 19 more reference values in the case's *expected*.

evaporator05_counter_current

tutorials/steady/evaporation/evaporator05_counter_current



What it does: Counter-current backward-feed triple-effect evaporator (MCFT 10.38 variant)

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

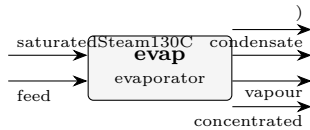
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L2to1	F	0.0319496046345
stream	L3to2	F	0.0496904316429
stream	V1	F	0.020847035226
stream	V2	F	0.0177408270084
stream	V3	F	0.0134519728349
stream	cond1	F	0.0227907734916
stream	cond2	F	0.0208470355984
stream	cond3	F	0.0177408244446
stream	feed	F	0.0631424044778
stream	liveSteam	F	0.0227907734916
stream	productConc	F	0.0111025694085
kpi	effect1	A	17.1719452001
kpi	effect1	A_m2	17.1719452001
kpi	effect1	BPE	0

...and 81 more reference values in the case's *expected*.

evaporator06_nacl_pitzer

tutorials/steady/evaporation/evaporator06_nacl_pitzer



What it does: Single-effect NaCl evaporator with the boiling-point elevation from the PITZER electrolyte model (a_w), not the ideal K_b*m. Concentrated brine → BPE several degrees, the real value.

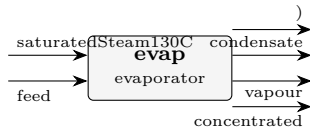
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0203403273098
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00743745046799
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	5.06216808829
kpi	evap	F_conc	0.0203403273098
kpi	evap	F_conc_kmol_h	73.2251783152
kpi	evap	F_conc_mass	0.422576829819
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
... and 19 more reference values in the case's <i>expected</i> .			

evaporator07_nacl_enrtl

tutorials/steady/evaporation/evaporator07_nacl_enrtl



What it does: Single-effect NaCl evaporator with the boiling-point elevation from the eNRTL water activity (electrolyte-NRTL) – the same slot as model pitzer, one word changed. eNRTL is the model for salt in a non-ideal molecular solvent.

The story: - Choupo - * | \{\} | / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | / / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | / / / O perations | Website: <https://choupo.org> | | \{\} \{\} | / / U nits | Licence: GPL-3.0-or-later | | \{\} | / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0200845615144
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00769321626341
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	3.86220786107
kpi	evap	F_conc	0.0200845615144
kpi	evap	F_conc_kmol_h	72.3044214517
kpi	evap	F_conc_mass	0.417969209015
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
...and 19 more reference values in the case's <i>expected</i> .			

evaporator08_naoh_dilution_heat

tutorials/steady/evaporation/evaporator08_naoh_dilution_heat



What it does: Single-effect NaOH evaporator: the heat of dilution IN the duty (fitted L_{ϕ} , Parker-validated)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | \{\} \{\} \{\} / U nits | Licence: GPL-3.0-or-later | \{\} \{\} / P roperties | Tutorial case file – part of Choupo | \{\} \{\} / O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0118365226457
stream	condensate	F	0.0208333333333
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0208333333333
stream	vapour	F	0.0159412551321
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	6.45099255003
kpi	evap	F_conc	0.0118365226457
kpi	evap	F_conc_kmol_h	42.6114815246
kpi	evap	F_conc_mass	0.234606344352
kpi	evap	F_cond	0.3753125
kpi	evap	F_cond_kg_h	1351.125
kpi	evap	F_in	0.0277777777778

... and 20 more reference values in the case's *expected*.

flash

The theory you need. An isothermal flash splits a feed of composition z_i into vapour y_i and liquid x_i at fixed (T, P) . Phase equilibrium is $y_i = K_i x_i$ with $K_i = \gamma_i P_i^{\text{sat}} / P$ (γ - φ formulation), and the vapour fraction β solves the Rachford–Rice equation

$$\sum_i \frac{z_i (K_i - 1)}{1 + \beta (K_i - 1)} = 0,$$

monotone in β – Newton converges in a handful of iterations, all of them visible at verbosity 3. Expect: a bubble point when the equation's root hits $\beta = 0$, a dew point at $\beta = 1$, and non-ideal systems ($\gamma_i \neq 1$: NRTL/Wilson/UNIQUAC/UNIFAC) bending K_i with composition. Liquid–liquid and three-phase cases replace successive-substitution with direct Gibbs-energy minimisation (the symmetric γ -model trap: Newton collapses onto the trivial $K = 1$ root). *Full derivation: Theory Guide, the VLE and flash chapters.*

adiabaticFlash01_benzene_toluene

tutorials/steady/flash/adiabaticFlash01_benzene_toluene



What it does: Adiabatic flash, benzene/toluene 40/60, 380K/3bar → 1atm

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	flashAdiab	T_out	368.661803634
kpi	flashAdiab	V_over_F	0.0518811133842
stream	feed	F	0.02777777777778
stream	liquid	F	0.0263366357393
stream	vapor	F	0.00144114203845

bubbleT01_ethanol_water

tutorials/steady/flash/bubbleT01_ethanol_water



What it does: Bubble-T of 30/70 ethanol/water at 1 atm (NRTL)

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | // H eat-transfer | Version: Choupo-2607 | // O perations | Website: <https://choupo.org> | // U nits | Licence: GPL-3.0-or-later | // P roperties | Tutorial case file – part of Choupo | | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
kpi	bubble01	P	101325
kpi	bubble01	T_bubble	354.424819099
kpi	bubble01	converged	1
kpi	bubble01	y_ethanol	0.581915403358
kpi	bubble01	y_water	0.418084596646

bubbleT02_uniquac_ethanol_water

tutorials/steady/flash/bubbleT02_uniquac_ethanol_water



What it does: Bubble-T of 30/70 ethanol/water at 1 atm (UNIQUAC) – validates the UNIQUAC model against the NRTL/Wilson twins

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
kpi	bubble02	P	101325
kpi	bubble02	T_bubble	353.469206954
kpi	bubble02	converged	1
kpi	bubble02	y_ethanol	0.582711434226
kpi	bubble02	y_water	0.417288565785

dewT01_ethanol_water

tutorials/steady/flash/dewT01_ethanol_water



What it does: Dew-T of 30/70 ethanol/water vapour at 1 atm (NRTL)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778

flash01_benzene_toluene
tutorials/steady/flash/flash01_benzene_toluene



What it does: Isothermal flash benzene/toluene at 370 K, 1 bar (Raoult)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	flash01	V_over_F	0.30398357314
kpi	flash01	T	370
stream	vapor	F	0.00844398814279
stream	liquid	F	0.019333789635

flash02_ethanol_water

tutorials/steady/flash/flash02_ethanol_water



What it does: Isothermal flash 50/50 ethanol/water at 355 K (NRTL)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.00562323865754
stream	vapor	F	0.0221545391202
kpi	flash02	F_alpha	0.00562323865754
kpi	flash02	F_beta	0.0221545391202
kpi	flash02	F_in	0.02777777777778
kpi	flash02	P	101325
kpi	flash02	T	355
kpi	flash02	V_over_F	0.797563408329
kpi	flash02	phaseSet	0

flash03_wilson_ethanol_water

tutorials/steady/flash/flash03_wilson_ethanol_water



What it does: Isothermal flash 50/50 ethanol/water at 355 K (Wilson activity model)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.0063767199082
stream	vapor	F	0.0214010578696
kpi	flash03	F_alpha	0.0063767199082
kpi	flash03	F_beta	0.0214010578696
kpi	flash03	F_in	0.02777777777778
kpi	flash03	P	101325
kpi	flash03	T	355
kpi	flash03	V_over_F	0.770438083305
kpi	flash03	phaseSet	0

flash04_molarFlows_input
tutorials/steady/flash/flash04_molarFlows_input



What it does: Isothermal flash, feed declared via molarFlows (benzene 40 + toluene 60 kmol/h)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.019333789635
stream	vapor	F	0.00844398814279
kpi	flash04	F_alpha	0.019333789635
kpi	flash04	F_beta	0.00844398814279
kpi	flash04	F_in	0.02777777777778
kpi	flash04	P	100000
kpi	flash04	T	370
kpi	flash04	V_over_F	0.30398357314
kpi	flash04	phaseSet	0

flash05_ternary_massFlows

tutorials/steady/flash/flash05_ternary_massFlows



What it does: Ternary flash benzene/toluene/n-hexane, feed in mass flows (5000 kg/h)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0160941453232
stream	liquid	F	0.00814913402453
stream	vapor	F	0.00794501129864
kpi	flash05	F_alpha	0.00814913402453
kpi	flash05	F_beta	0.00794501129864
kpi	flash05	F_in	0.0160941453232
kpi	flash05	P	100000
kpi	flash05	T	365
kpi	flash05	V_over_F	0.493658478851
kpi	flash05	phaseSet	0

flash06_sweep_T

tutorials/steady/flash/flash06_sweep_T



What it does: Flash benzene/toluene at 1 bar: T swept 360→385 K across the bubble→dew window (V/F S-curve + duty)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

flash07_gridsweep_TP

tutorials/steady/flash/flash07_gridsweep_TP



What it does: 2-D grid sweep: flash V/F over T x P (the flash envelope)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

flash08_co2_water_package

tutorials/steady/flash/flash08_co2_water_package

What it does: Carbonated water flash whose thermo is an INLINE property-package MANIFEST: constant/thermoPhysPropDict IS the full record (components, methods per phase, solvent+solute, the CO2-water Henry pair declared+verified). CO2 dissolves by the unsymmetric Henry convention; water stays Raoult. Read the run log: the assembly announces itself.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	flatWater	F	0.000277662593734
stream	gasOff	F	1.15184043824e-07
stream	sodaWater	F	0.000277777777778
kpi	node	F_alpha	0.000277662593734
kpi	node	F_beta	1.15184043824e-07
kpi	node	F_in	0.000277777777778
kpi	node	P	101325
kpi	node	Q	0
kpi	node	Q_kW	0
kpi	node	T	298.15
kpi	node	V_over_F	0.000414662557767
kpi	node	phaseSet	0

flash09_n2ch4_stryjek

tutorials/steady/flash/flash09_n2ch4_stryjek

What it does: phi-phi VLE VALIDATION vs Stryjek, Chappellear & Kobayashi (1974), J. Chem. Eng. Data 19:334 (DOI 10.1021/je60063a023), Table III, -190 F = 149.82 K. The case carries its thermo as an INLINE manifest: liquid+vapour both eos.SRK (one cubic, two roots: $K = \phi_L/\phi_V$) with the N2-CH4 interaction constant $k_{ij} = 0.0289$ (a small measured correction to the mixing rule, from the Knapp/DECHEMA data compilation). Measured at 299 psia: $x_{N2}=0.1503$, $y_{N2}=0.4579$ – compare the Flash Result table in the log with these measured values (agreement ~2%).

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	liq	F	0.000145628387466
stream	vap	F	0.000132149390312
kpi	node	F_alpha	0.000145628387466
kpi	node	F_beta	0.000132149390312
kpi	node	F_in	0.0002777777777778
kpi	node	P	2061600
kpi	node	Q	0
kpi	node	Q_kW	0
kpi	node	T	149.82
kpi	node	V_over_F	0.475737805121
kpi	node	phaseSet	0

flash10_ch4propane_pcsaft

tutorials/steady/flash/flash10_ch4propane_pcsaft

What it does: PC-SAFT phi-phi binary flash, CH₄/propane at 273.15 K / 20 bar, $k_{ij} = 0$ (announced): the PREDICTIVE baseline of the SAFT world – $K = \phi_L/\phi_V$ from ONE a_{res} surface at its two density roots. Sanity anchor carried by the case itself: propane's curated Antoine gives $P_{sat}(273.15\text{ K}) = 4.7\text{ bar}$, so the Raoult-magnitude $K_{propane} \sim 0.24$ – the PC-SAFT $K_{propane} = 0.32$ sits near it (the deviation IS the fugacity correction of a liquid holding ~11% dissolved CH₄, visible); $K_{CH_4} \gg 1$ (supercritical light key). The published- k_{ij} refinement is a NAMED follow-up: it needs the cited pair record (data/standards/parameters/PCSAFT/), never an invented number.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	liq	F	9.97838549688e-05
stream	vap	F	0.000177993922809
kpi	node	F_alpha	9.97838549688e-05
kpi	node	F_beta	0.000177993922809
kpi	node	F_in	0.0002777777777778
kpi	node	P	2000000
kpi	node	Q	632.243350571
kpi	node	Q_kW	0.632243350571
kpi	node	T	273.15
kpi	node	V_over_F	0.640778122112
kpi	node	phaseSet	0

flash_polymer_devolatilisation

tutorials/steady/flash/flash_polymer_devolatilisation

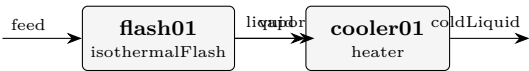


What it does: Devolatilise a polystyrene melt: flash off residual toluene; the polymer (fixed-MW pseudo-component, role nonvolatile) stays liquid

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

flash_pseudoComponent_petCut

tutorials/steady/flash/flash_pseudoComponent_petCut



What it does: Flash of a petroleum pseudo-component (Riazi-Daubert lump) + toluene, with a heater closing the energy balance

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

leaf01_flash

tutorials/steady/flash/leaf01_flash

What it does: Leaf-node form: isothermal flash benzene/toluene (fractal base case)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | Heat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | Operations | Website: <https://choupo.org> | | \{\} \{\} | Units | Licence: GPL-3.0-or-later | | \{\} | Properties | Tutorial case file – part of Choupo | | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	liquid	F	0.019333789635
stream	vapor	F	0.00844398814279
kpi	flash	F_alpha	0.019333789635
kpi	flash	F_beta	0.00844398814279
kpi	flash	F_in	0.0277777777778
kpi	flash	P	100000
kpi	flash	T	370
kpi	flash	V_over_F	0.30398357314
kpi	flash	phaseSet	0

vllc01_waterButanol

tutorials/steady/flash/vllc01_waterButanol



What it does: Liquid-liquid flash of water + 1-butanol at 320 K, 1.013 bar – the real miscibility gap from the cited Winkelman et al. (2009) UNIQUAC LLE fit: water-rich ~98.3 mol% water, butanol-rich ~46 mol% butanol (their Figs 1-2). See README.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

Water + 1-butanol (*n*-butanol) is *the* textbook partially-miscible binary. Below the bubble point it splits into two coexisting liquids: a water-rich phase carrying only a couple of mol % butanol, and a butanol-rich phase holding roughly half water. So a flash of a feed *inside the miscibility gap* should give two liquids — and here it does.

At 320 K, 1.013 bar (well below the bubble point, so no vapour) a feed of 80 mol % water / 20 mol % butanol splits into:

with split fraction $\beta \approx 0.41$ into the butanol-rich phase. The solver reports the Michelsen TPD instability ($tm_min \approx -0.45$, **unstable = YES**) and then a direct Gibbs-energy minimisation that lands the two compositions.

An independent re-implementation of the UNIQUAC γ (fresh code, isoactivity Newton) reproduces the engine's split to 4 decimals across 298–350 K, confirming the parameters are entered correctly. The agreement with the published figures is essentially exact — the UNIQUAC fit is doing its job.

constant/thermoPhysPropDict carries the LLE-specific UNIQUAC for water/1-butanol, cited per value:

- Primary source: J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, *"Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"*, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013. - Binary interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation $A_{ij}(T) = a_{ij} + b_{ij} \cdot T + c_{ij} \cdot T^2$ (their Eq. 10) and $\tau_{ij} = \exp(-A_{ij}/T)$ (their Eq. 7): - $A_{1,3}$ (butanol→water): $a = 155.31$, $b = 1.0822$, $10^3 \cdot c = -43.711$ - $A_{3,1}$ (water→butanol): $a = -579.36$, $b = 2.7517$, $10^3 \cdot c = -6.7700$ - Structural r , q — their Table 1 (p.72) (value Magnussen et al.): 1-butanol $r = 3.9243$, $q = 3.668$; water $r = 0.9200$, $q = 1.400$.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	butanolRich	F	0.0114431411907
stream	feed	F	0.02777777777778
stream	waterRich	F	0.016334636587
kpi	decanter01	F_alpha	0.016334636587
kpi	decanter01	F_beta	0.0114431411907
kpi	decanter01	F_in	0.02777777777778
kpi	decanter01	P	101300
kpi	decanter01	T	320
kpi	decanter01	betaFraction	0.411953082867
kpi	decanter01	phaseSet	1

vlle03_audit_artificial

tutorials/steady/flash/vlle03_audit_artificial



What it does: Algorithm-audit synthetic VLE: 3 components with identical P_{sat} and exaggerated NRTL. Should converge to 3-phase by construction.

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What to expect (golden-locked):

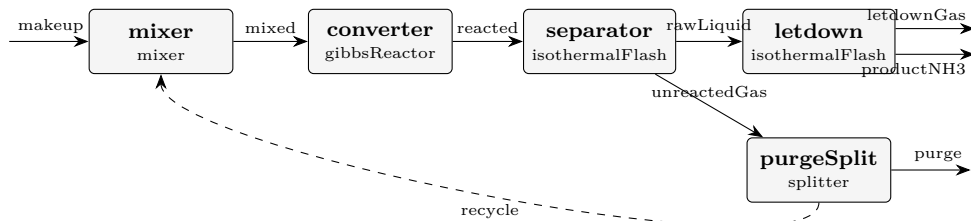
kind	unit/stream	key	expected
stream	aqueous	F	0.011307571384
stream	feed	F	0.0277777777778
stream	organic	F	0.0113091924076
stream	vapor	F	0.00516101398624
kpi	decanter01	F_alpha	0.0164702063938
kpi	decanter01	F_beta	0.011307571384
kpi	decanter01	F_in	0.0277777777778
kpi	decanter01	F_vapor	0.00516101398624
kpi	decanter01	P	100000
kpi	decanter01	T	350
kpi	decanter01	V_over_F	0.407072569823
kpi	decanter01	beta_vapor	0.185796503505
kpi	decanter01	phaseSet	2

flowsheets

The theory you need. Recycle turns a flowsheet into a fixed-point problem: guess the tear stream, march the units, compare. Successive substitution converges linearly (and diverges past unit gain); Wegstein accelerates it; Newton-on-tears solves the tear residuals with a Jacobian. Choupo announces the tear choice, every iteration's residual, and any bound that binds – no silent crutch. Expect: purge streams controlling inert build-up (close the loop with no purge and watch the accumulation), and convergence behaviour that CHANGES with the tear location. *Full derivation: Theory Guide, the flowsheet-convergence chapter.*

ammonia01_synthesis_loop

tutorials/steady/flowsheets/ammonia01_synthesis_loop



What it does: Complete ammonia synthesis loop (Haber-Bosch) with recycle and purge: makeup 1:3 N₂:H₂ (+ Ar inert) → mix with recycle → SRK Gibbs converter at 700 K / 200 bar (real-gas fugacity) → chill-flash at 250 K to condense the NH₃ product → split the unreacted gas into a small Ar-bleeding PURGE and the recycle. The recycle tear + purge are the whole point: high overall N₂/H₂ utilisation despite a modest per-pass conversion, with the inert Ar held at steady state by the purge.

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	letdownGas	F	1.18571362082e-06
stream	makeup	F	0.000277777777778
stream	mixed	F	0.0005681436407
stream	productNH3	F	0.000131387462567
stream	purge	F	1.52824159741e-05
stream	rawLiquid	F	0.000132573176187
stream	reacted	F	0.000438221495669
stream	recycle	F	0.000290365903507
stream	unreactedGas	F	0.000305648319481
kpi	converter	F_in_kmol_h	2.04531710652
kpi	converter	F_out_kmol_h	1.57759738441
kpi	converter	N_in_mol_s	0.5681436407
kpi	converter	N_out_mol_s	0.438221495669
kpi	converter	P	20000000

... and 33 more reference values in the case's *expected*.

ammonia02_full_plant

tutorials/steady/flowsheets/ammonia02_full_plant

What it does: Full-scale ammonia plant (~59 t/h NH₃, 1400 t/day): explicit syngas + recycle COMPRESSORS, feed-effluent HEAT EXCHANGER, ADIABATIC converter (the reaction heats itself – exothermic), water cooler + refrigerated separator, product let-down, and the recycle + purge. SRK fugacity throughout; dissolved H₂/N₂ by measured Henry constants (Wiebe 1934, Larson-Black 1925 via IUPAC SDS 5/6).

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	chilledEffluent	F	10.5918567473
stream	compMakeup	F	2.2222222222
stream	compRecycle	F	9.3296744801
stream	cooledEffluent	F	10.5918567473
stream	cw	F	138.888888889
stream	cwOut	F	138.888888889
stream	hotEffluent	F	10.5918563175
stream	letdownGas	F	0.00792840799315
stream	makeup	F	2.2222222222
stream	mixed	F	11.5518967023
stream	preheatedFeed	F	11.5518967023
stream	productNH3	F	0.965707281277
stream	purge	F	0.28854663174
stream	rawLiquid	F	0.97363568927

...and 108 more reference values in the case's *expected*.

composite01_two_flashes

tutorials/steady/flowsheets/composite01_two_flashes

What it does: Composite node: two flashes in series (fractal step 2)

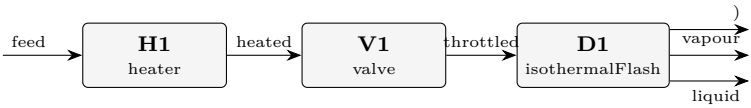
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.00370154472079
stream	feed	F	0.02777777777778
stream	vapor1	F	0.0212163063101
stream	vapor2	F	0.00285992674691
kpi	flash1	F_alpha	0.0065614714677
kpi	flash1	F_beta	0.0212163063101
kpi	flash1	F_in	0.02777777777778
kpi	flash1	P	100000
kpi	flash1	T	370
kpi	flash1	V_over_F	0.763787027163
kpi	flash1	phaseSet	0
kpi	flash2	F_alpha	0.00370154472079
kpi	flash2	F_beta	0.00285992674691
kpi	flash2	F_in	0.0065614714677

*...and 4 more reference values in the case's **expected**.*

credo01_valve_heater_drum
tutorials/steady/flowsheets/credo01_valve_heater_drum



What it does: Info-follows-streams credo: heater (Q) + valve (P) set the state; the flash drum (no knobs) just separates

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	H1	T_out	413.08978161
kpi	V1	T_out	369.30464285
kpi	V1	vf	0.20555173
kpi	V1	dP	900000
stream	liquid	F	0.0220680075
stream	vapour	F	0.0057097703

plant01_two_sectors

tutorials/steady/flowsheets/plant01_two_sectors

What it does: Recursive plant: factory of two composite sectors (fractal step 3)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0
stream	feed	F	0.02777777777778
stream	vaporA	F	0.0212163063101
stream	vaporB	F	0.0065614714677

plant02_basename_twins

tutorials/steady/flowsheets/plant02_basename_twins

What it does: basename twins: a top-level ‘feed’ and a DISTINCT sector-internal ‘A.feed’ coexist

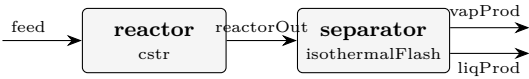
The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001000000000002
stream	hot	F	0.001000000000002
stream	out	F	0.001000000000002
kpi	h	F	0.001000000000002
kpi	h	P	100000
kpi	h	Q	2000
kpi	h	Q_kW	2
kpi	h	Q_per_mol	1999.99999997
kpi	h	T_in	298.15
kpi	h	T_out	366.662038108
kpi	h	dT	68.5120381076

process01_reactor_flash

tutorials/steady/flowsheets/process01_reactor_flash



What it does: Two-unit flowsheet: CSTR → isothermal flash

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	liqProd	F	0.000108248328241
stream	reactorOut	F	0.000277777777778
stream	vapProd	F	0.000169529449537
kpi	reactor	Da_kTau	1.10929132154
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	X_limiting	0.52590711876
kpi	reactor	k	0.00357499942014
kpi	reactor	tau_s	310.291329081
kpi	reactor	xi_mol_per_s	0.0730426553834
kpi	separator	F_alpha	0.000108248328241
... and 6 more reference values in the case's <i>expected</i> .			

process02_reactor_hx_flash

tutorials/steady/flowsheets/process02_reactor_hx_flash



What it does: Three-unit flowsheet: CSTR → heater → flash

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	hotStream	F	0.0002777777777778
stream	liqProd	F	0.000186109936751
stream	reactorOut	F	0.0002777777777778
stream	vapProd	F	9.16678410264e-05
kpi	heater	F	0.0002777777777778
kpi	heater	P	101325
kpi	heater	Q	400
kpi	heater	Q_kW	0.4
kpi	heater	Q_per_mol	1440
kpi	heater	T_in	350
kpi	heater	T_out	361.946251976
kpi	heater	dT	11.9462519765
kpi	reactor	Da_kTau	1.10929132154
... and 15 more reference values in the case's <i>expected</i> .			

process02_with_design

tutorials/steady/flowsheets/process02_with_design



What it does: process02 with full reports: balances + sizing + Guthrie costing

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

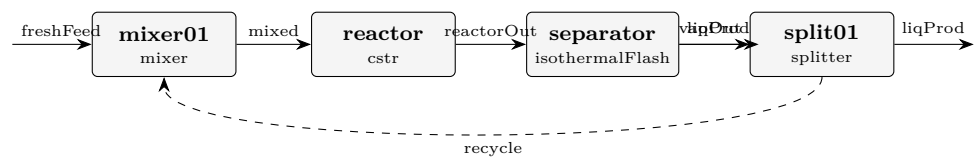
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	hotStream	F	0.0002777777777778
stream	liqProd	F	0.000186109936751
stream	reactorOut	F	0.0002777777777778
stream	vapProd	F	9.16678410264e-05
kpi	heater	F	0.0002777777777778
kpi	heater	P	101325
kpi	heater	Q	400
kpi	heater	Q_kW	0.4
kpi	heater	Q_per_mol	1440
kpi	heater	T_in	350
kpi	heater	T_out	361.946251976
kpi	heater	dT	11.9462519765
kpi	reactor	Da_kTau	1.10929132154

... and 15 more reference values in the case's *expected*.

process03_recycle

tutorials/steady/flowsheets/process03_recycle



What it does: Recycle loop: the tear stream is solved by Newton (5 iters, |r|2~5e-6)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	freshFeed	F	0.000277777777778
stream	liqOut	F	0.000364610626883
stream	liqProd	F	0.000145844250753
stream	mixed	F	0.000496545446603
stream	reactorOut	F	0.000496545446603
stream	recycle	F	0.00021876637613
stream	vapProd	F	0.00013193481972
kpi	reactor	Da_kTau	3.53861103626
kpi	reactor	F_in_kmol_h	1.78756360777
kpi	reactor	F_out_kmol_h	1.78756360777
kpi	reactor	T	354.303008007
kpi	reactor	V_R	0.02
kpi	reactor	X_limiting	0.779668274718
kpi	reactor	k	0.00478796954448
...and 9 more reference values in the case's <i>expected</i> .			

process04_research_workflow

tutorials/steady/flowsheets/process04_research_workflow



What it does: Case-local binary pair file (research workflow pattern)

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The research workflow for managing thermodynamic parameters: instead of declaring NRTL pairs (...) inline in the thermoPhysPropDict, the case declares only the model and lets the simulator load the parameters from a file inside the case directory.

This is the pattern used when you have your own fitted parameters that you do not want to mix into the standards database. The case remains self-contained and reproducible: the parameters travel with the case.

When the simulator builds the NRTL model and discovers there are no inline pairs in thermoPhysPropDict, it falls back to file lookup:

1. Case-local – <case>/constant/parameters/NRTL/ethanol-water.dat ? used here 2. Standards – \$CHOUPO_HOME/data/standards/parameters/NRTL/ethanol-water.dat

In this case the local file exists, so it wins. If you delete the local file, the run still succeeds because the standards file is the same DECHEMA set.

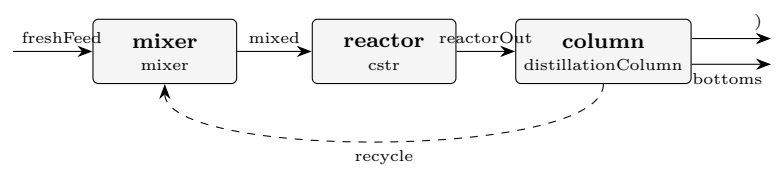
The result is numerically identical to flash02_ethanol_water (same NRTL parameters; just a different way of supplying them).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.00562323865754
stream	vapor	F	0.0221545391202
kpi	flash01	F_alpha	0.00562323865754
kpi	flash01	F_beta	0.0221545391202
kpi	flash01	F_in	0.02777777777778
kpi	flash01	P	101325
kpi	flash01	T	355
kpi	flash01	V_over_F	0.797563408329
kpi	flash01	phaseSet	0

process05_isomerization_recycle

tutorials/steady/flowsheets/process05_isomerization_recycle



What it does: Isomerization neoPentane→nPentane: mixer + CSTR + distillation with recycle

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.000277777777778
stream	freshFeed	F	0.000277777777778
stream	mixed	F	0.000363888888889
stream	reactorOut	F	0.000363888888889
stream	recycle	F	8.61111111111e-05
kpi	column	B	0.000277777777778
kpi	column	D	8.61111111111e-05
kpi	column	L_rect	0.000172222222222
kpi	column	L_strip	0.000536111111111
kpi	column	R	2
kpi	column	T_bottom	364.62708689
kpi	column	T_top	350.366212454
kpi	column	V_rect	0.000258333333333
kpi	column	V_strip	0.000258333333333
... and 13 more reference values in the case's <i>expected</i> .			

proxy01_gas_loop

tutorials/steady/flowsheets/proxy01_gas_loop



What it does: Proxy step 1: gas WGS loop with one recycle (mixer + Gibbs reactor + splitter)

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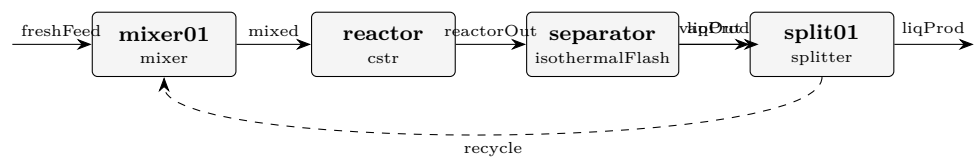
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	CW	F	0.0166666666667
stream	cooledGas	F	0.000375486726088
stream	dryGas	F	0.000325696494167
stream	expandedProduct	F	0.000227987545917
stream	feed	F	0.000277777777778
stream	mixed	F	0.000536409608696
stream	productGas	F	0.000227987545917
stream	productWater	F	4.97902319203e-05
stream	reacted	F	0.000536409608697
stream	recycle1	F	0.000160922882609
stream	recycle2	F	9.77089482502e-05
stream	toFlash	F	0.000375486726088
stream	warmCW	F	0.0166666666667
kpi	F1	F_alpha	4.97902319203e-05

...and 64 more reference values in the case's *expected*.

recycle_autoinit_tear

tutorials/steady/flowsheets/recycle_autoinit_tear



What it does: Recycle with NO authored tear guess – the solver auto-seeds the tear from the feed aggregate (announced), then converges

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	freshFeed	F	0.0002777777777778
stream	liqOut	F	0.000364608639339
stream	liqProd	F	0.000145843455736
stream	mixed	F	0.000496542956892
stream	reactorOut	F	0.000496542956892
stream	recycle	F	0.000218765183603
stream	vapProd	F	0.000131934317553
kpi	mixer01	F_out	0.000496542956892
kpi	mixer01	P_out	100000
kpi	mixer01	T_out	354.302988011
kpi	mixer01	n_in	2
kpi	mixer01	vf	0
kpi	reactor	Da_kTau	3.53861380168
kpi	reactor	F_in_kmol_h	1.78755464481
... and 16 more reference values in the case's <i>expected</i> .			

gas-solid

bagFilter01_dust

tutorials/steady/gas-solid/bagFilter01_dust



What it does: Bag filter (baghouse) on a dusty gas: >99% capture + cake pressure drop

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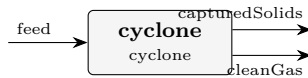
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	baghouse	airToCloth_m_min	1.53857549963
kpi	baghouse	arealDustLoad	0.5
kpi	baghouse	dP_filter	2745.33935569
kpi	baghouse	efficiency	0.999334225794
kpi	baghouse	faceVelocity	0.0256429249938
kpi	baghouse	filterArea	40
kpi	baghouse	mu_gas	1.78433840676e-05
kpi	baghouse	penetration	0.000665774205525
kpi	baghouse	rho_gas	1.13794383208
kpi	baghouse	solidsCaptured_mass	0.0138796420249
kpi	baghouse	solidsEscaped_mass	9.24686396563e-06

...and 1 more reference values in the case's *expected*.

cyclone01_dust_removal

tutorials/steady/gas-solid/cyclone01_dust_removal



What it does: Cyclone dust removal: silica from N2 (Lapple, $D_c = 0.5$ m)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

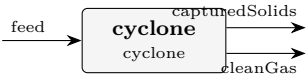
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	2.71094994999e-06
kpi	cyclone	d50_micron	2.71094994999
kpi	cyclone	efficiency	0.820454419117
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
kpi	cyclone	rho_particle	2650
kpi	cyclone	solidsCaptured_mass	0.0113952002655
kpi	cyclone	solidsEscaped_mass	0.00249368862337

... and 1 more reference values in the case's *expected*.

cyclone02_leith_licht

tutorials/steady/gas-solid/cyclone02_leith_licht



What it does: Cyclone dust removal – Leith-Licht back-mixing model

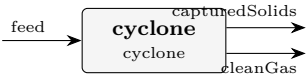
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	1.10984825815e-06
kpi	cyclone	d50_micron	1.10984825815
kpi	cyclone	efficiency	0.880048347849
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
kpi	cyclone	rho_particle	2650
kpi	cyclone	solidsCaptured_mass	0.0122228937201
kpi	cyclone	solidsEscaped_mass	0.00166599516876
...and 1 more reference values in the case's <i>expected</i> .			

cyclone03_iozia_leith

tutorials/steady/gas-solid/cyclone03_iozia_leith



What it does: Cyclone dust removal – Iozia-Leith data-fitted logistic (beta)

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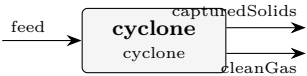
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	Q_gas	1.02571699975
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	9.86636018962e-07
kpi	cyclone	d50_micron	0.986636018962
kpi	cyclone	dP_cyclone	4903.83536027
kpi	cyclone	efficiency	0.96703441953
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	loading	0.0118992372589
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5

...and 5 more reference values in the case's *expected*.

cyclone04_high_loading

tutorials/steady/gas-solid/cyclone04_high_loading



What it does: Cyclone high dust loading – Muschelknautz mass-loading effect

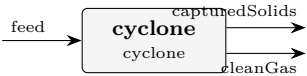
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	1e-09
kpi	cyclone	d50_micron	0.001
kpi	cyclone	dP_cyclone	4903.83536027
kpi	cyclone	efficiency	0.983932479817
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	loading	0.475969490356
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
...and 4 more reference values in the case's <i>expected</i> .			

cyclone05_dirgo_leith_validation

tutorials/steady/gas-solid/cyclone05_dirgo_leith_validation



What it does: VALIDATION vs Dirgo & Leith (1985): a Stairmand high-efficiency cyclone (D=0.305 m, 0.139 m3/s, ~15 m/s inlet, mineral-oil aerosol) run through the flow-pattern Iozia-Leith model. Swap the model slot to see the ~3x d50 spread across correlations on ONE real cyclone – why you validate against data.

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.00563888888889
stream	feed	F	0.00563888888889
kpi	cyclone	Q_gas	0.138813700633
kpi	cyclone	bodyDiameter	0.305
kpi	cyclone	d50	2.24296780859e-06
kpi	cyclone	d50_micron	2.24296780859
kpi	cyclone	dP_cyclone	648.673413872
kpi	cyclone	efficiency	0.702163610002
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	11.9377544216
kpi	cyclone	loading	0.00879253984647
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
...and 5 more reference values in the case's expected.			

gasSolidSplitter01_ideal

tutorials/steady/gas-solid/gasSolidSplitter01_ideal



What it does: Ideal gas-solid splitter: specified 90% solids recovery (no efficiency physics)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	separator	gasCarryover	0
kpi	separator	gasToCleanGas	0.0416666666667
kpi	separator	gasToSolids	0
kpi	separator	solidsCaptured_mass	0.0125
kpi	separator	solidsEscaped_mass	0.00138888888889
kpi	separator	solidsIn_mass	0.0138888888889
kpi	separator	solidsRecovery	0.9

gibbs

The theory you need. Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A} \mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as λ_E , the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~100–250 K when they are not (dissociation is endothermic). Above ~1500 K the catalogue's polynomial c_p fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

Equilibrium maps. Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes T and P in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The **temperatureApproach** ΔT evaluates the *reaction* equilibrium at $T + \Delta T$ while enthalpy and the energy balance stay at the physical T – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on **nh3_equilibrium_map**, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium NH_3 mole fraction crosses 30%; report T to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in T at fixed P . *Full derivation:* *Theory Guide*, the *Gibbs-equilibrium chapter*, *Equilibrium maps*.

gibbs01_water_gas_shift

tutorials/steady/gibbs/gibbs01_water_gas_shift



What it does: Gibbs reactor: water-gas shift $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ at 800 K

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	products	F	0.000277777777778
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	1
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_out_mol_s	0.277777777778
kpi	wgs	P	100000
kpi	wgs	T	800
kpi	wgs	converged	1
kpi	wgs	lambda_C	5549.05961227
kpi	wgs	lambda_H	-60095.3190791
kpi	wgs	lambda_O	-294758.913033
kpi	wgs	y_CO	0.16409231685
kpi	wgs	y_CO2	0.33590768315
... and 2 more reference values in the case's expected.			

gibbs02_steam_reforming

tutorials/steady/gibbs/gibbs02_steam_reforming



What it does: Gibbs reactor: steam reforming of methane at 1000 K

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | // H eat-transfer | Version: Choupo-2607 | // O perations | Website: <https://choupo.org> | // U nits | Licence: GPL-3.0-or-later | // P roperties | Tutorial case file – part of Choupo | | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.00111111111111
stream	syngas	F	0.0016586091548
kpi	reformer	F_in_kmol_h	4
kpi	reformer	F_out_kmol_h	5.97099295728
kpi	reformer	N_in_mol_s	1.1111111111
kpi	reformer	N_out_mol_s	1.6586091548
kpi	reformer	P	100000
kpi	reformer	T	1000
kpi	reformer	converged	1
kpi	reformer	lambda_C	-33527.5812363
kpi	reformer	lambda_H	-75161.3018601
kpi	reformer	lambda_O	-309185.611918
kpi	reformer	y_CH4	0.0024289965616
kpi	reformer	y_CO	0.0979819433622
... and 3 more reference values in the case's expected.			

gibbs03_methane_combustion

tutorials/steady/gibbs/gibbs03_methane_combustion



What it does: Gibbs reactor: methane combustion in stoichiometric air at 1800 K

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.00292222222222
stream	flueGas	F	0.00292404303636
kpi	burner	F_in_kmol_h	10.52
kpi	burner	F_out_kmol_h	10.5265549309
kpi	burner	N_in_mol_s	2.92222222222
kpi	burner	N_out_mol_s	2.92404303636
kpi	burner	P	100000
kpi	burner	T	1800
kpi	burner	converged	1
kpi	burner	lambda_C	-357484.903199
kpi	burner	lambda_H	-201140.892694
kpi	burner	lambda_N	-201371.845813
kpi	burner	lambda_O	-268835.220885
kpi	burner	y_CH4	7.51542148893e-20

...and 6 more reference values in the case's *expected*.

gibbs04_wgs_temperature_sweep

tutorials/steady/gibbs/gibbs04_wgs_temperature_sweep



What it does: Water-gas shift equilibrium swept over $T = 600..1200$ K

The story: - Choupo - * | \{ } / C hemicals | Choupo: open-source, glass-box simulation | | \{ } \{ } // H eat-transfer | Version: Choupo-2607 | | \{ } \{ } \{ } /// O perations | Website: <https://choupo.org> | | \{ } \{ } // U nits | Licence: GPL-3.0-or-later | | \{ } / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

gibbs05_adiabatic_flame
tutorials/steady/gibbs/gibbs05_adiabatic_flame



What it does: Adiabatic flame of lean CH4 in air ($\varphi = 0.5$)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.005566666666667
stream	flueGas	F	0.00556667001791
kpi	burner	F_in_kmol_h	20.04
kpi	burner	F_out_kmol_h	20.0400120645
kpi	burner	N_in_mol_s	5.566666666667
kpi	burner	N_out_mol_s	5.56667001791
kpi	burner	P	100000
kpi	burner	T	1491.58454904
kpi	burner	converged	1
kpi	burner	lambda_C	-432004.832024
kpi	burner	lambda_H	-204860.137156
kpi	burner	lambda_N	-162961.775795
kpi	burner	lambda_O	-186456.945423
kpi	burner	outerIterations	3

...and 7 more reference values in the case's *expected*.

gibbs06_h2_flame_radicals

tutorials/steady/gibbs/gibbs06_h2_flame_radicals



What it does: Stoichiometric H2/air adiabatic flame – the radical pool at equilibrium

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	burnt	F	0.000808562123724
stream	feed	F	0.000938888888889
kpi	flame	F_in_kmol_h	3.38
kpi	flame	F_out_kmol_h	2.91082364541
kpi	flame	N_in_mol_s	0.9388888888889
kpi	flame	N_out_mol_s	0.808562123724
kpi	flame	P	101325
kpi	flame	T	2385.83679448
kpi	flame	converged	1
kpi	flame	lambda_H	-240335.177161
kpi	flame	lambda_N	-277509.500244
kpi	flame	lambda_O	-344071.838893
kpi	flame	outerIterations	3
kpi	flame	y_H	0.00181958387904
...and 9 more reference values in the case's <i>expected</i> .			

gibbs06_thermal_nox
tutorials/steady/gibbs/gibbs06_thermal_nox



What it does: Methane combustion at 1800 K with thermal NOx formation

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.00292222222222
stream	flueGas	F	0.00292416183331
kpi	burner	F_in_kmol_h	10.52
kpi	burner	F_out_kmol_h	10.5269825999
kpi	burner	N_in_mol_s	2.92222222222
kpi	burner	N_out_mol_s	2.92416183331
kpi	burner	P	100000
kpi	burner	T	1800
kpi	burner	converged	1
kpi	burner	lambda_C	-355589.713611
kpi	burner	lambda_H	-200666.346513
kpi	burner	lambda_N	-201373.229482
kpi	burner	lambda_O	-269787.239957
kpi	burner	y_CH4	9.68349314225e-20
... and 7 more reference values in the case's expected.			

gibbs07_wgs_cooled
tutorials/steady/gibbs/gibbs07_wgs_cooled



What it does: Cooled WGS Gibbs reactor (350 K): water condenses, shifting the equilibrium (multi-phase)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.76244802832e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240153299998
kpi	wgs	F_condensate_kmol_h	0.135448129019
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864551879994
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0376244802832
kpi	wgs	N_out_mol_s	0.277777780282
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	50085.9480936
kpi	wgs	lambda_H	-24736.1727481
...and 8 more reference values in the case's expected.			

gibbs08_wgs_cooled_reactiveflash

tutorials/steady/gibbs/gibbs08_wgs_cooled_reactiveflash



What it does: Cooled WGS via reactiveFlash (B1) – cross-checks the elementPotential V+L result

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.76244802827e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240153299999
kpi	wgs	F_condensate_kmol_h	0.135448129018
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864551879995
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0376244802827
kpi	wgs	N_out_mol_s	0.277777780281
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	50085.9480935
kpi	wgs	lambda_H	-24736.1727482
...and 8 more reference values in the case's expected.			

gibbs09_wgs_cooled_directmin

tutorials/steady/gibbs/gibbs09_wgs_cooled_directmin



What it does: Cooled WGS via directMin (B2, Nelder-Mead direct G-min) – cross-checks A/B1 to ~0.3%

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.77165683291e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240061209449
kpi	wgs	F_condensate_kmol_h	0.135779645985
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864220354015
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0377165683291
kpi	wgs	N_out_mol_s	0.277777777778
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	0
kpi	wgs	lambda_H	0
...and 11 more reference values in the case's <i>expected</i> .			

gibbs10_ammonia_fugacity

tutorials/steady/gibbs/gibbs10_ammonia_fugacity



What it does: Ammonia synthesis equilibrium at 200 bar with the REAL-gas fugacity correction: $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$ at 700 K, 200 bar over an SRK equation of state. At this pressure the ideal-gas equilibrium (30.0 % NH_3) is wrong – the Gibbs reactor now folds the SRK fugacity coefficients into the equilibrium ($K_{\text{phi}} \sim 0.75$, favouring the fewer-moles NH_3 side), giving 32.4 % NH_3 , toward the measured ~35-40 %. Switch the equationOfState to idealGas and watch NH_3 drop back to 30.0 % – the fugacity correction is the whole point of a high-pressure reactor.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	products	F	0.000209823048593
kpi	ammonia	F_in_kmol_h	1
kpi	ammonia	F_out_kmol_h	0.755362974933
kpi	ammonia	N_in_mol_s	0.277777777778
kpi	ammonia	N_out_mol_s	0.209823048593
kpi	ammonia	P	20000000
kpi	ammonia	Q_kW	-3.68175302075
kpi	ammonia	T	700
kpi	ammonia	converged	1
kpi	ammonia	lambda_H	-34945.3954982
kpi	ammonia	lambda_N	-59416.3950887
kpi	ammonia	y_H2	0.507099864715
kpi	ammonia	y_N2	0.169033288238

... and 1 more reference values in the case's *expected*.

heat

The theory you need. Heat exchange couples streams without mixing them: $Q = U A \Delta T_{\text{lm}}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

Exchanger DESIGN vs rating. A shell-and-tube poses two inverse questions: RATE it (geometry given → find U , area, ΔP , duty – model geometry;) or DESIGN it (duty given → size the geometry – model design;). U is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` → `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Sheat-exchanger design.*

condenser01_film_nusselt

tutorials/steady/heat/condenser01_film_nusselt



What it does: Horizontal-tube steam condenser: Nusselt film HTC → duty (geometry mode)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	0.02777777777778
stream	steam	F	0.02777777777778
kpi	condenser	F	0.02777777777778
kpi	condenser	P	100000
kpi	condenser	Q	-506700.51677
kpi	condenser	Q_UAdT_kW	506.617311982
kpi	condenser	Q_kW	-506.70051677
kpi	condenser	R_coolant	0.00014880952381
kpi	condenser	R_film	0.000349003587291
kpi	condenser	R_wall	4.35883467862e-05
kpi	condenser	Re_film	15.1166609728
kpi	condenser	T_in	372.76
kpi	condenser	T_out	372.755918611
kpi	condenser	T_wall	325.855275434
... and 8 more reference values in the case's <i>expected</i> .			

heatExchanger01_water_water

tutorials/steady/heat/heatExchanger01_water_water



What it does: Two-stream heat exchanger (eps-NTU): hot/cold water recovery

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cold	F	0.0333333333333
stream	coldOut	F	0.0333333333333
stream	hot	F	0.0277777777778
stream	hotOut	F	0.0277777777778
kpi	hx	C_cold	2516.66666667
kpi	hx	C_hot	2097.22222222
kpi	hx	C_r	0.833333333333
kpi	hx	LMTD	18.7584806629
kpi	hx	NTU	2.38410596026
kpi	hx	Q	93792.4033146
kpi	hx	Q_kW	93.7924033146
kpi	hx	T_cold_in	300
kpi	hx	T_cold_out	337.268504628
kpi	hx	T_hot_in	360
... and 5 more reference values in the case's <i>expected</i> .			

heatExchanger02_geometry_U

tutorials/steady/heat/heatExchanger02_geometry_U



What it does: Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m2.K), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

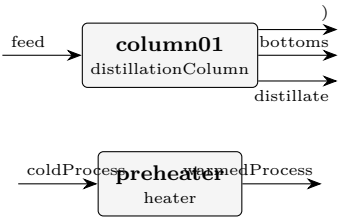
The story: - Choupo - * | \{\} | / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	15.5339177262
kpi	hx	NTU	1.26058061182
kpi	hx	Nu_shell	171.621364606
kpi	hx	Nu_tube	287.732215939
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	2065604.42087
...and 22 more reference values in the case's <i>expected</i> .			

heatlink01_condenser_to_heater

tutorials/steady/heat/heatlink01_condenser_to_heater



What it does: Forward heat-link: column condenser preheats a downstream stream

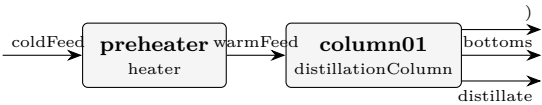
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.01388888888889
stream	coldProcess	F	0.4166666666667
stream	distillate	F	0.01388888888889
stream	feed	F	0.02777777777778
stream	warmedProcess	F	0.4166666666667
kpi	column01	B	0.01388888888889
kpi	column01	D	0.01388888888889
kpi	column01	L_rect	0.02777777777778
kpi	column01	L_strip	0.05555555555556
kpi	column01	Q_condenser_kW	-1281.04348002
kpi	column01	Q_reboiler_kW	1279.30480005
kpi	column01	R	2
kpi	column01	T_bottom	382.893530951
kpi	column01	T_top	354.21089412
...and 14 more reference values in the case's <i>expected</i> .			

heatlink02_condenser_feed_preheat

tutorials/steady/heat/heatlink02_condenser_feed_preheat



What it does: Feedback heat-link (energy tear): column condenser preheats its OWN feed

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.263888888889
stream	coldFeed	F	0.277777777778
stream	distillate	F	0.0138888888889
stream	warmFeed	F	0.277777777778
kpi	column01	B	0.263888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0208333333333
kpi	column01	L_strip	0.298611111111
kpi	column01	Q_condenser_kW	-1067.88365467
kpi	column01	Q_reboiler_kW	1846.42629755
kpi	column01	R	1.5
kpi	column01	T_bottom	365.99791143
kpi	column01	T_top	354.558892551
kpi	column01	V_rect	0.0347222222222
...and 13 more reference values in the case's expected.			

hx01_heater

tutorials/steady/heat/hx01_heater



What it does: Heater: 50/50 ethanol/water from 300 K to ~338 K (sensible)

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	hotStream	F	0.02777777777778
kpi	heater01	F	0.02777777777778
kpi	heater01	P	101325
kpi	heater01	Q	100000
kpi	heater01	Q_kW	100
kpi	heater01	Q_per_mol	3600
kpi	heater01	T_in	300
kpi	heater01	T_out	338.338658147
kpi	heater01	dT	38.338658147

hxWorkflow1_design_from_duty

tutorials/steady/heat/hxWorkflow1_design_from_duty



What it does: PART 1 of 2 – DESIGN: the DUTY TARGET is given (cool the hot shell 330→315 K) and the exchanger GEOMETRY is SOLVED for (model design;). Glass-box Kern sizing: fix the tube choices, then find the tube count (and the shell diameter it implies, Kern/Sinnott bundle) so $U \cdot A = Q_{\text{req}}/\text{LMTD}$; U/dP then rated with Gnielinski (tube) + Kern (shell). The detailed-design (sizing) step, glass-box. Part 2 (hxWorkflow2_rate_designed) rates the bundle this case produces.

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	16.7868881533
kpi	hx	NTU	1.0053658663
kpi	hx	Nu_shell	195.331265319
kpi	hx	Nu_tube	416.603846461
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	1888506.69951
...and 22 more reference values in the case's <i>expected</i> .			

hxWorkflow2_rate_designed

tutorials/steady/heat/hxWorkflow2_rate_designed



What it does: Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m2.K), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	16.7876807682
kpi	hx	NTU	1.00522927337
kpi	hx	Nu_shell	195.283521062
kpi	hx	Nu_tube	416.603846461
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	1888394.47754
...and 22 more reference values in the case's <i>expected</i> .			

mheatx01_two_hot_one_cold

tutorials/steady/heat/mheatx01_two_hot_one_cold



What it does: Multi-stream heat exchanger: 2 hot + 1 cold, balance closes

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cold	F	0.09444444444444
stream	coldOut	F	0.09444444444444
stream	hot1	F	0.02777777777778
stream	hot1Out	F	0.02777777777778
stream	hot2	F	0.01666666666667
stream	hot2Out	F	0.01666666666667
kpi	mhx	Q_cold	142611.111111
kpi	mhx	Q_cold_kW	142.61111111
kpi	mhx	Q_hot	-142611.111111
kpi	mhx	Q_hot1_kW	-104.86111111
kpi	mhx	Q_hot2_kW	-37.75
kpi	mhx	Q_hot_kW	-142.61111111
kpi	mhx	balanceCloses	1
kpi	mhx	dT_min	30

... and 4 more reference values in the case's expected.

phasechange01_partial_condenser

tutorials/steady/heat/phasechange01_partial_condenser



What it does: Partial condenser: benzene/toluene vapour cooled to quality 0.4

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	condenser	vf_out	0.399999998746
kpi	condenser	T_out	367.494289759
kpi	condenser	Q_kW	-630.508868406
kpi	condenser	Tsat	352.826063711
stream	wetOut	F	0.0277777777778
stream	wetOut	vf	0.399999998746

reboiler_water_copper

tutorials/steady/heat/reboiler_water_copper



What it does: Steam-heated copper-surface pool reboiler: Rohsenow nucleate flux → duty, Zuber CHF ceiling (geometry mode)

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

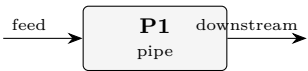
kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	wetVapour	F	0.0277777777778
kpi	reboiler	F	0.0277777777778
kpi	reboiler	P	100000
kpi	reboiler	Q	111804.644945
kpi	reboiler	Q_kW	111.804644945
kpi	reboiler	T_in	372
kpi	reboiler	T_out	372.755918611
kpi	reboiler	T_surface	381.390508143
kpi	reboiler	Tsat	372.755918611
kpi	reboiler	area	1.25663706144
kpi	reboiler	chf_margin	0.0709724846287
kpi	reboiler	dT_excess	8.6345895313
kpi	reboiler	q_CHF	1253960.82217
...and 4 more reference values in the case's <i>expected</i> .			

hydraulics

The theory you need. Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves’ intersection.

pipe01_water_line

tutorials/steady/hydraulics/pipe01_water_line



What it does: Pipe pressure drop: 2000 kmol/h water in a 0.1 m x 50 m commercial-steel line (Churchill friction)

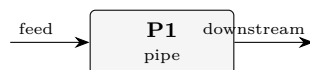
The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	downstream	F	0.555555555556
stream	feed	F	0.555555555556
kpi	P1	F	0.555555555556
kpi	P1	P_in	500000
kpi	P1	P_out	462942.299189
kpi	P1	dP_elevation	17171.7847864
kpi	P1	dP_fittings	10942.8641074
kpi	P1	dP_friction	8943.05191699
kpi	P1	deltaP	37057.7008108
kpi	P1	density	875.517367621
kpi	P1	frictionFactor	0.0192870918591
kpi	P1	head_loss_m	2.31611521712
kpi	P1	regime	2
kpi	P1	reynolds	149291.31749
...and 2 more reference values in the case's expected.			

pipe02_airwater_twophase

tutorials/steady/hydraulics/pipe02_airwater_twophase



What it does: Gas-liquid two-phase pressure drop: air (N₂) + water in a horizontal 0.05 m x 10 m line, Lockhart-Martinelli. The two-phase path activates automatically because the feed flashes two-phase ($0 < V/F < 1$).

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Air (N₂) + water flowing through a horizontal 0.05 m × 10 m commercial-steel line. The pipe's two-phase path activates automatically: the feed flashes two-phase at 2 bar / 25 °C (N₂ is supercritical → all vapour; water → mostly liquid), so $0 < V/F < 1$ and the pipe routes to the gas-liquid models instead of the single-phase liquid path.

The two teaching points: 1. The multiplier $\varphi_{L^2} = 8.6$: two-phase friction is far above either phase alone — the classic Lockhart-Martinelli amplification, visible in the printout via X and the Chisholm C . 2. Slip → holdup: the homogeneous (no-slip) model would put the liquid fraction at $j_L/(j_L+j_G) \approx 0.12$; L-M gives 0.36 because the liquid moves slower than the gas and accumulates. Switch twoPhase { model homogeneous; } to see the difference for yourself.

The Lockhart-Martinelli correlation (Chem. Eng. Prog. 45 (1949) 39) was developed and validated on isothermal air-water (and similar gas-liquid) pipe data — so air-water is exactly the system it is trusted on. The model structure here (X , φ_{L^2} , the $tt/vt/tv/vv$ C-selection, the Butterworth 1975 void fraction) reproduces that framework.

Switch twoPhase { model ...; } and compare — the spread IS a lesson (two-phase ΔP correlations routinely disagree by ~50 %):

Layer 2 (Friedel, BeggsBrill) needs surface tension — supplied here by transport { surfaceTension { model BrockBird; } } (from water's $T_c/P_c/T_b$). Beggs-Brill also classifies the flow pattern (segregated / intermittent / distributed) and corrects holdup for pipe inclination (here horizontal, so $\psi = 1$).

*- Adiabatic: the inlet vapour fraction is held along the pipe (no flashing / condensation in the line). Flashing flow is a future increment. - Liquid density: $\rho_L = 877 \text{ kg/m}^3$ comes from the Rackett branch, ~12 % low for water (a known anomaly, also in pipe01_water_line). It biases the *absolute* ΔP ; the correlation *structure* is unaffected. - Liquid viscosity via operation { viscosity ...; } (water's value): the flash leaves a trace of dissolved N₂ in the liquid and the Vogel model has no N₂ entry.*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	downstream	F	0.0563888888889
stream	feed	F	0.0563888888889
kpi	P1	F	0.0563888888889
kpi	P1	P_in	200000
kpi	P1	P_out	193263.21239
kpi	P1	Re_Gs	27493.3230716
kpi	P1	Re_Ls	28761.0631705
kpi	P1	X_martinelli	2.67141881475
kpi	P1	dP_elevation	0
kpi	P1	dP_friction	6736.78760984
kpi	P1	deltaP	6736.78760984
kpi	P1	holdup_liquid	0.36001183622
kpi	P1	jG	4.29950133573
kpi	P1	jL	0.583736217791

... and 6 more reference values in the case's *expected*.

membranes

The theory you need. Solution-diffusion (Wijmans & Baker 1995): inside a dense membrane the pressure is uniform, so permeation is concentration-driven. Linearised: $J_w = A_w(\Delta P - \Delta\pi)$ and $J_s = B_s(c_m - c_p)$ – water driven by NET pressure, solute blind to pressure; (A_w, B_s) are MATERIAL constants from `data/standards/assets/`. Concentration polarisation (film model) makes the wall concentration $c_m = c_b e^{J_w/k}$ exceed the bulk – the flux–polarisation–osmotic feedback is THE nonlinearity of RO/NF, solved per module segment. Expect: rejection improving with pressure ($c_p = B_s c_m / (J_w + B_s)$), polarisation eating driving force at high flux, and charged-membrane (Donnan) effects on multivalent ions that plain solution-diffusion cannot see – that is what the DSPM-DE sibling model is for. *Full derivation: Theory Guide, the membrane chapter.*

membrane01_RO_NaCl_seawater

tutorials/steady/membranes/membrane01_RO_NaCl_seawater



What it does: Spiral-wound RO of 35 g/L NaCl feed at 55 bar (canonical seawater desalination)

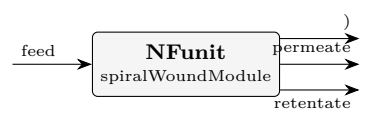
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	ROunit	R_obs_NaCl	0.997859093278
kpi	ROunit	water_recovery	0.157762204919
kpi	ROunit	J_w_avg	1.32725173967e-05

membrane02_NF_sugar

tutorials/steady/membranes/membrane02_NF_sugar



What it does: NF270 partial rejection of glucose at 10 bar (canonical NF: 10 g/L sugar feed)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.05555555555556
stream	permeate	F	0.0149607433177
stream	retentate	F	0.0405951326871
kpi	NFunit	J_w_avg	5.40656892958e-05
kpi	NFunit	R_obs_glucose	0.662821422531
kpi	NFunit	dP_feed	100000
kpi	NFunit	water_recovery	0.267701129953

membrane03_concentration_polarization

tutorials/steady/membranes/membrane03_concentration_polarization



What it does: Concentration polarization study: sweep k_{film} over a SW30HR seawater RO case

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

membrane04_spacer_schock_miquel

tutorials/steady/membranes/membrane04_spacer_schock_miquel



What it does: Seawater RO with k_{film} computed from spacer-channel hydraulics (Schock-Miquel Sherwood); k_{film} falls ~6.6 to ~5.6e-5 m/s along the channel

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0286373914761
stream	retentate	F	0.131086212764
kpi	ROunit	J_w_avg	1.47407124897e-05
kpi	ROunit	R_obs_NaCl	0.99810213206
kpi	ROunit	dP_feed	26065.9064084
kpi	ROunit	water_recovery	0.175213731874

membrane05_train

tutorials/steady/membranes/membrane05_train



What it does: Multi-element membrane train: 3 SW30HR elements in series (seawater RO)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0590874153104
stream	retentate	F	0.100637449271
kpi	ROunit	J_w_avg	1.01384186683e-05
kpi	ROunit	R_obs_NaCl	0.99694467035
kpi	ROunit	c_ret_elem1	0.68349184216
kpi	ROunit	c_ret_elem2	0.793343136295
kpi	ROunit	c_ret_elem3	0.881961644632
kpi	ROunit	dP_feed	101143.55107
kpi	ROunit	elements	3
kpi	ROunit	recovery_elem1	0.175213731874
kpi	ROunit	recovery_elem2	0.289843097471
kpi	ROunit	recovery_elem3	0.361527335552
kpi	ROunit	water_recovery	0.361527335552

membrane06_pitzer

tutorials/steady/membranes/membrane06_pitzer



What it does: Seawater RO with the Pitzer osmotic model (vs membrane01 van't Hoff)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0284588599363
stream	retentate	F	0.13126472432
kpi	ROunit	J_w_avg	1.46488544181e-05
kpi	ROunit	R_obs_NaCl	0.997986472373
kpi	ROunit	dP_feed	200000
kpi	ROunit	water_recovery	0.174121871794

membrane07_scaling_si

tutorials/steady/membranes/membrane07_scaling_si



What it does: Brackish RO, 6 x 8-inch modules: per-module scaling audit (SI at wall vs bulk)

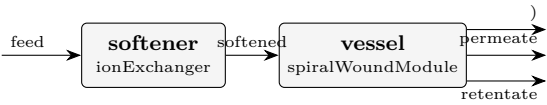
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.107212342596
stream	retentate	F	0.0872324772828
kpi	vessel	J_w_avg	8.653382114e-06
kpi	vessel	R_obs_Ca	0.999474633761
kpi	vessel	R_obs_Cl	0.994770268575
kpi	vessel	R_obs_HCO3	0.996072731843
kpi	vessel	R_obs_K	0.993471097929
kpi	vessel	R_obs_Mg	0.999605925236
kpi	vessel	R_obs_Na	0.994770268575
kpi	vessel	R_obs_SO4	0.999737250092
kpi	vessel	SI0_module_calcite	3
kpi	vessel	SI0_module_gypsum	-1
kpi	vessel	SI0_recovery_calcite	0.305177946028
...and 18 more reference values in the case's <i>expected</i> .			

membrane08_softened_scaling

tutorials/steady/membranes/membrane08_softened_scaling



What it does: Softener + brackish RO: ion-exchange removes the scaling driver upstream of the vessel

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.1054509275
stream	retentate	F	0.0890050837601
stream	softened	F	0.194455667987
kpi	softener	CEC_utilised_pct	100
kpi	softener	Ca_leakage_mgL	0.0098339472306
kpi	softener	I	0.0243975946959
kpi	softener	Mg_leakage_mgL	0.00549119043267
kpi	softener	Na_added_meqL	6.71876194262
kpi	softener	Na_added_mgL	154.464337061
kpi	softener	bedVolume_m3	1.5
kpi	softener	dH_exchange_Jkgw	21.9905304514
kpi	softener	hardness_in_mgL_CaCO3	320.96282036
kpi	softener	hardness_out_mgL_CaCO3	0.0471722805225
...and 34 more reference values in the case's expected.			

membrane09_index_vs_rigorous

tutorials/steady/membranes/membrane09_index_vs_rigorous



What it does: Seawater RO at high recovery: industry calcite indices (LSI / Stiff-Davis / Ryznar) vs the rigorous activity SI at the membrane WALL

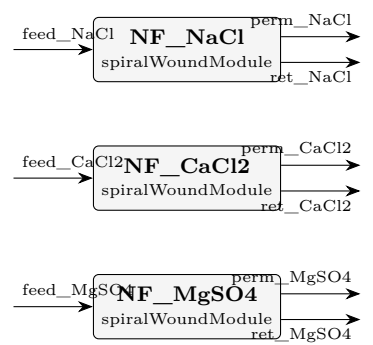
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.25
stream	permeate	F	0.0675548243983
stream	retentate	F	0.182446484545
kpi	vessel	J_w_avg	4.67400706783e-06
kpi	vessel	LSI_calcite_bulk_end	0.76535387412
kpi	vessel	LSI_calcite_wall_end	0.985330125244
kpi	vessel	RSI_calcite_wall_end	5.02933974951
kpi	vessel	R_obs_Ca	0.999255215369
kpi	vessel	R_obs_Cl	0.992602201838
kpi	vessel	R_obs_HCO3	0.994441275819
kpi	vessel	R_obs_K	0.990769979906
kpi	vessel	R_obs_Mg	0.999441306528
kpi	vessel	R_obs_Na	0.992602201838
kpi	vessel	R_obs_SO4	0.99962746766
...and 25 more reference values in the case's expected.			

membrane10_dspmde_divalent

tutorials/steady/membranes/membrane10_dspmde_divalent



What it does: DSPM-DE NF: divalent (MgSO4) > mixed (CaCl2) > monovalent (NaCl) rejection at 10 bar

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | \{\} | Heat-transfer | Version: Choupo-2607 | \{\} | Operations | Website: <https://choupo.org> | \{\} | Units | Licence: GPL-3.0-or-later | \{\} | Properties | Tutorial case file – part of Choupo | | Optimization |

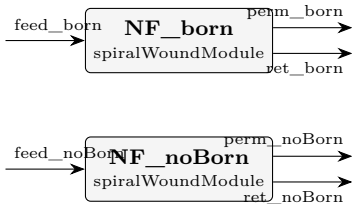
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed_CaCl2	F	0.05555555555556
stream	feed_MgSO4	F	0.05555555555556
stream	feed_NaCl	F	0.05555555555556
stream	perm_CaCl2	F	0.0191775929422
stream	perm_MgSO4	F	0.020040469745
stream	perm_NaCl	F	0.0208297354228
stream	ret_CaCl2	F	0.0363780274704
stream	ret_MgSO4	F	0.0355151987788
stream	ret_NaCl	F	0.034725847724
kpi	NF_CaCl2	J_w_avg	6.91063443929e-05
kpi	NF_CaCl2	R_obs_CaCl2	0.704588151782
kpi	NF_CaCl2	R_obs_MgSO4	1
kpi	NF_CaCl2	R_obs_NaCl	1
kpi	NF_CaCl2	dP_feed	100000

... and 13 more reference values in the case's expected.

membrane11_dspmde_born_toggle

tutorials/steady/membranes/membrane11_dspmde_born_toggle



What it does: DSPM-DE NF: Born dielectric-exclusion toggle (same NaCl feed, poreDielectric on vs off)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed_born	F	0.0555555555556
stream	feed_noBorn	F	0.0555555555556
stream	perm_born	F	0.0103299082671
stream	perm_noBorn	F	0.0112254914132
stream	ret_born	F	0.0452257284146
stream	ret_noBorn	F	0.0443301389741
kpi	NF_born	J_w_avg	3.7229087272e-05
kpi	NF_born	R_obs_NaCl	0.861027895843
kpi	NF_born	dP_feed	100000
kpi	NF_born	water_recovery	0.185615581216
kpi	NF_noBorn	J_w_avg	4.0473218085e-05
kpi	NF_noBorn	R_obs_NaCl	0.65954044001
kpi	NF_noBorn	dP_feed	100000
kpi	NF_noBorn	water_recovery	0.201790063872

optimisation

The theory you need. The outer driver wraps the whole simulation in $f(\mathbf{x})$: sweeps map it, Nelder–Mead descends it, parameter estimation fits it to data (Levenberg–Marquardt, with identifiability diagnostics – χ^2_{red} , correlation, conditioning). Expect every function evaluation to be a full, honest flowsheet pass.

designSpec01_triple_equal_areas

tutorials/steady/optimisation/designSpec01_triple_equal_areas



What it does: DesignSpec: equal-area triple-effect sugar evaporator (manipulate \$A so L3 = 4500 kg/h)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1	F	0.242853688633
stream	L2	F	0.150128936321
stream	L3	F	0.0529527028239
stream	V1	F	0.087645716964
stream	V2	F	0.0927247523111
stream	V3	F	0.0971762334975
stream	cond1	F	0.138858068999
stream	cond2	F	0.087645716964
stream	cond3	F	0.0927247523111
stream	feed	F	0.330499405597
stream	utilitySteam_200kPa	F	0.138858068999
kpi	effect1	A	108.007004701
kpi	effect1	A_m2	108.007004701
kpi	effect1	BPE	0.107244152342
...and 81 more reference values in the case's <i>expected</i> .			

fitNRTL01_ethanol_water

tutorials/steady/optimisation/fitNRTL01_ethanol_water

fit_NRTL_ethanol_water fitParameters
--

What it does: Fit NRTL ethanol/water pair against 1 atm bubble-T data (Levenberg-Marquardt)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

The parameter-estimation workflow: regress the four NRTL binary interaction parameters of the ethanol-water pair (a_{ij} , b_{ij} , a_{ji} , b_{ji} ; α held fixed) against experimental bubble-temperature data at 1.01325 bar, using hand-rolled Levenberg-Marquardt inside the fitParameters engine (successor of the retired fitBinaryPair driver).

** See an outer Marquardt loop (3-up/5-down trust region on λ) wrapped around an inner Newton-1D solving $\Sigma K x = 1$ for each (x_i, T) pair. * Watch λ oscillate as the algorithm switches between Gauss-Newton (λ small) and gradient-descent (λ large) regimes. * Watch some LM steps get rejected — χ^2 gets worse, the step is reversed, λ is multiplied by 5, retry. The whole "trust region" idea visible in one column of the log.*

Note how minimal the forward case is: one stream, one bubbleT unit. The outer driver is what does the work — iterating over the dataset internally and looping LM over the four parameters.

Four keywords (plus optional maxIter, tolerance, fdStep, lambda0). The driver:

1. locates the pair in thermoPhysPropDict.activityModel.pairs by name, 2. reads the dataset (a two-column flat list of x_i T_i), 3. finds the bubbleT unit's input stream from the flowsheetDict, 4. for each LM step: - 8 forward evals to build a central-difference Jacobian over 4 params $\times$ N data points, - solve the 4×4 normal equations with Marquardt damping, - try the step; accept if χ^2 decreased, otherwise raise λ and retry.

After the run header you get (the iteration trace below is illustrative – it shows the SHAPE of an LM run, accept/reject and λ adaptation; the actual run converges to $\chi^2 = 1.0426$, RMS = 0.3079 K with fitted $a_{ij} = 9.328$, $b_{ij} = -3609.18$):

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	fit_NRTL_ethanol_water	chi2	1.04279
diag	fit_NRTL_ethanol_water	chi2_reduced	0.14897
diag	fit_NRTL_ethanol_water	cond_JtJ	3.55369e+11
diag	fit_NRTL_ethanol_water	converged	1
diag	fit_NRTL_ethanol_water	corr_computed	1
diag	fit_NRTL_ethanol_water	dof	7
diag	fit_NRTL_ethanol_water	identifiable	0
diag	fit_NRTL_ethanol_water	invertible	1
diag	fit_NRTL_ethanol_water	iter	24
diag	fit_NRTL_ethanol_water	max_abs_corr	0.999967
diag	fit_NRTL_ethanol_water	max_abs_resid	0.625729
diag	fit_NRTL_ethanol_water	n_data	11
diag	fit_NRTL_ethanol_water	n_params	4
diag	fit_NRTL_ethanol_water	params_at_bound	0

... and 4 more reference values in the case's expected.

optim01_column_reflux

tutorials/steady/optimisation/optim01_column_reflux



What it does: Optimise reflux ratio + feedStage of benzene/toluene column to minimise reboiler vapour V_strip

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0416666666667
kpi	column01	L_strip	0.0694444444444
kpi	column01	R	3
kpi	column01	T_bottom	383.299688974
kpi	column01	T_top	353.770171691
kpi	column01	V_rect	0.0555555555556
kpi	column01	V_strip	0.0555555555556
kpi	column01	feedStage	9
kpi	column01	nStages	15
... and 2 more reference values in the case's expected.			

optim02_process_cost

tutorials/steady/optimisation/optim02_process_cost



What it does: Minimise total module Guthrie cost of reactor + heater by varying reactor V_R and heater Q

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A small reactor $\rightarrow$ heater $\rightarrow$ separator process. The OptimizationDriver (Nelder-Mead) tunes two design variables to minimise the total Guthrie module cost of the two costed units (reactor + heater):

The heater is credo-pure: it specifies Q (the thermal power, hardware), not an outlet T . T_{out} is a reported KPI, not a control variable.

The optimum is not at the lower corner. Nelder-Mead converges (8 moves, simplexInit 0.15) to:

Increasing Q monotonically lowers the objective: the simplex walks the cost from $\sim 2.0e6$ EUR at $Q \approx 170$ W down to $\sim 9.35e5$ EUR at $Q = 500$ W.

The dominant cost is the heat exchanger (~ 916 k of the ~ 935 k total). Its area follows $A = Q / (U \cdot \text{LMTD}) = Q / 6000 \text{ m}^2$, so $Q = 500$ W gives only $A \approx 0.083 \text{ m}^2$. The Turton shell-and-tube correlation used by the Guthrie model is valid for $A \in [10, 1000] \text{ m}^2$; our exchanger is two orders of magnitude below S_{min} . In that extrapolated tail the correlation

*turns *upward* as A shrinks (the quadratic term dominates for $\log_{10} A < 0$), so a larger area is cheaper. The optimiser correctly exploits this: pushing Q to its upper bound grows A toward the correlation's valid range and slides down the cost curve. The reactor volume settles at an interior 0.00807 m^3 — large enough for adequate conversion ($X \approx 0.64$), small enough that its vessel cost (~ 19 k) stays a rounding error next to the exchanger.*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	hotStream	F	0.0002777777777778
stream	liqProd	F	7.65621619403e-05
stream	reactorOut	F	0.0002777777777778
stream	vapProd	F	0.000201215615838
kpi	heater	F	0.0002777777777778
kpi	heater	P	101325
kpi	heater	Q	500
kpi	heater	Q_kW	0.5
kpi	heater	Q_per_mol	1800
kpi	heater	T_in	350
kpi	heater	T_out	364.85579492
kpi	heater	dT	14.8557949202
kpi	reactor	Da_kTau	1.79046552367

*... and 15 more reference values in the case's **expected**.*

optim04_membrane_recovery_scaling

tutorials/steady/optimisation/optim04_membrane_recovery_scaling



What it does: Brackish RO: maximise water recovery up to the scaling limit (SQP, scaling constraints active at the optimum)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.0800476772619
stream	retentate	F	0.11439691471
kpi	vessel	J_w_avg	6.05900819134e-06
kpi	vessel	LSI_calcite_bulk_end	0.376636925312
kpi	vessel	LSI_calcite_wall_end	0.568802059189
kpi	vessel	RSI_calcite_wall_end	5.86239588162
kpi	vessel	R_obs_Ca	0.999415347979
kpi	vessel	R_obs_Cl	0.994183822598
kpi	vessel	R_obs_HCO3	0.995631569104
kpi	vessel	R_obs_K	0.99274024463
kpi	vessel	R_obs_Mg	0.999561447433
kpi	vessel	R_obs_Na	0.994183822598
kpi	vessel	R_obs_SO4	0.999707589243
...and 25 more reference values in the case's <i>expected</i> .			

optim05_reactor_npv

tutorials/steady/optimisation/optim05_reactor_npv



What it does: Esterification CSTR sized & operated for MAX NPV by hand-rolled SQP, subject to a reactor-size cap and a capital-budget cap (active-constraint shadow prices)

The story: - Choupo - * | \{\} | / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | / / O perations | Website: <https://choupo.org> | | \{\} \{\} | / / U nits | Licence: GPL-3.0-or-later | | \{\} | / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

Take the *economics01 ethyl-acetate plant* and stop **fixing** the design — let the engine choose it. The hand-rolled SQP sizes and operates the reactor to maximise the net present value, subject to inequality caps, and prints the shadow prices (Lagrange multipliers) of the binding caps.

The objective *economics.NPV* is a KPI published by the *EconomicsPass* in the *system/postDict* chain (sizing → costing → economics). The optimiser runs that whole post chain on every evaluation, so the *economics* scalars (*economics.NPV*, *economics.FCI*, ...) are reachable as ordinary objective/constraint paths — exactly like a sweep reads them.

The heater runs at a fixed partial duty, so the flash **splits** the reactor effluent into a light ethyl-acetate-rich vapour overhead (the prime product at 1.55 €/kg) and a heavy water/acid-rich liquid bottoms (a low-value byproduct at 0.30 €/kg). Esterification conserves total mass; what conversion buys you is the reshuffling of that mass toward the valuable overhead.

* V_R — a bigger reactor raises the residence time $\tau = V_R/Q$, so the isothermal-CSTR conversion $X = k\tau/(1+k\tau)$ climbs and the priced overhead grows. But a bigger vessel costs more steel — the *StirredTank* sizing → Guthrie costing drives FCI (CAPEX) up. Revenue rises with diminishing returns (X saturates toward 1) while CAPEX keeps climbing → an NPV maximum exists. * T_{feed} — a higher feed temperature raises the Arrhenius rate $k = A \cdot \exp(-E^\circ/RT)$, the cheap conversion **accelerator**. Its bounds stay inside the window where the flash splits cleanly (below the ~353 K edge that boils the bottoms dry).

We deliberately set the size cap tighter than where the NPV-greedy design would otherwise stop, so the reactor-size constraint binds at the optimum.

The optimiser drives the money-losing initial design ($V_R = 4 \text{ m}^3$, $T_{\text{feed}} = 350 \text{ K} \rightarrow \text{NPV} \approx -164 \text{ M}^\circ$) to a profitable, certified optimum:

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.138888888889
stream	hotStream	F	0.138888888889
stream	liqProd	F	0.0017296965168
stream	reactorOut	F	0.138888888889
stream	vapProd	F	0.137159192372
kpi	economics	COM_d	223170524.398
kpi	economics	FCI	522154.982187
kpi	economics	NPV	477589097.525
kpi	economics	TCI	600478.229515
kpi	economics	WC	78323.2473281
kpi	economics	paybackYears	0.00848797903819
kpi	economics	revenue	321660167.134
kpi	heater	F	0.138888888889
kpi	heater	P	101325

... and 25 more reference values in the case's *expected*.

optim06_integer_stages

tutorials/steady/optimisation/optim06_integer_stages



What it does: The classic N-vs-R trade solved in ONE run: nStages declared integer → exhaustive enumeration, one full SQP per stage count, ALL values on the table

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0313732166828
kpi	column01	L_strip	0.0591509944605
kpi	column01	Q_condenser_kW	-1391.66557074
kpi	column01	Q_reboiler_kW	1389.90800504
kpi	column01	R	2.25887160116
kpi	column01	T_bottom	382.835800828
kpi	column01	T_top	354.273383646
kpi	column01	V_rect	0.0452621055716
kpi	column01	V_strip	0.0452621055716
...and 7 more reference values in the case's <i>expected</i> .			

pareto01_purity_energy

tutorials/steady/optimisation/pareto01_purity_energy



What it does: The purity-energy PARETO FRONT by epsilon-constraint: minimise V_strip subject to x_D_LK >= eps, eps swept 0.96..0.994 – each front point one auditable SQP run

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.052352377807
kpi	column01	L_strip	0.0801301555848
kpi	column01	Q_condenser_kW	-2035.41756863
kpi	column01	Q_reboiler_kW	2033.86732819
kpi	column01	R	3.7693712021
kpi	column01	T_bottom	383.489767144
kpi	column01	T_top	353.562992445
kpi	column01	V_rect	0.0662412666959
kpi	column01	V_strip	0.0662412666959
...and 7 more reference values in the case's expected.			

sensitivity01_column_reflux

tutorials/steady/optimisation/sensitivity01_column_reflux



What it does: Sweep reflux ratio $R = 1.5..5.0$ (10 pts) across the column01 case; responses: x_D purity + V_{strip} (reboiler-duty proxy)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

sensitivity02_feasibility_map

tutorials/steady/optimisation/sensitivity02_feasibility_map



What it does: Feasibility MAP: the reflux sweep with declared constraints – every row marked (value, bound, residual, satisfied), nothing filtered; the feasible window between the purity spec and the boilup budget read off the CSV

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0424382716049
kpi	column01	L_strip	0.0702160493827
kpi	column01	Q_condenser_kW	-1730.95769425
kpi	column01	Q_reboiler_kW	1729.37389301
kpi	column01	R	3.05555555556
kpi	column01	T_bottom	383.383107764
kpi	column01	T_top	353.679332066
kpi	column01	V_rect	0.0563271604938
kpi	column01	V_strip	0.0563271604938

...and 7 more reference values in the case's *expected*.

power

The theory you need. Gas (Brayton) and steam (Rankine) cycles, alone and combined. Compressor and turbine hardware is specified by SHAFT WORK and isentropic efficiency – outlet pressure is a RESULT, not an input (the anti-DesignSpec credo). Expect: pressure ratio setting Brayton efficiency, the combined cycle’s gain coming from bottoming the gas turbine’s exhaust exergy, and every energy flow visible in the ledger. *Full derivation: Theory Guide, the power-cycle chapter.*

brayton01_gas_turbine

tutorials/steady/power/brayton01_gas_turbine



What it does: Closed Brayton (gas-turbine) cycle: compress air → combust (gas-aware heater) → expand to feed P

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

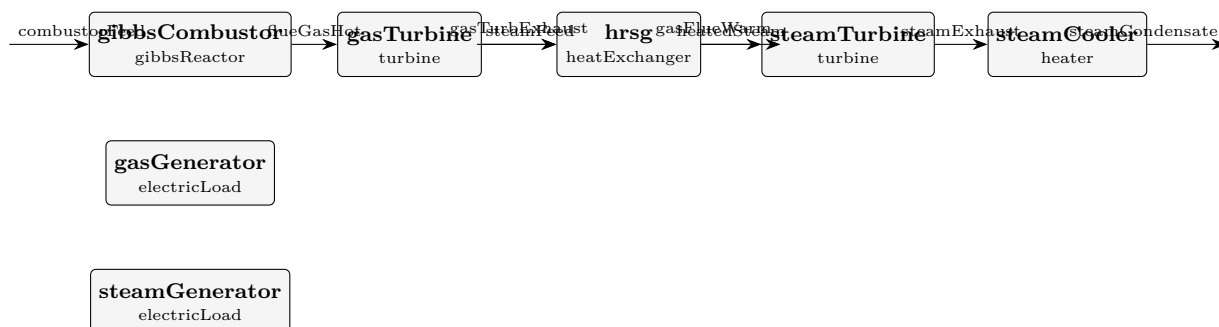
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	compressed	F	0.001
stream	exhaust	F	0.001
stream	feed	F	0.001
stream	hot	F	0.001
kpi	combustor	F	0.001
kpi	combustor	P	991200.306811
kpi	combustor	Q	20000
kpi	combustor	Q_kW	20
kpi	combustor	Q_per_mol	20000
kpi	combustor	T_in	633.374172563
kpi	combustor	T_out	1246.40394079
kpi	combustor	dT	613.029768231
kpi	compressor	F	0.001
kpi	compressor	P_in	100000

...and 27 more reference values in the case's *expected*.

combined01_brayton_rankine

tutorials/steady/power/combined01_brayton_rankine



What it does: Combined-cycle: Brayton (CH₄ + air, Gibbs combustor) + HRSG + Rankine + 2 generators

The story: - Choupo -*—————* combined01_brayton_rankine – combined-cycle power plant

Two thermodynamic cycles linked by a HRSG (heat-recovery steam generator). Top cycle: gas turbine on combustion products of CH₄ in air, made with a Gibbs equilibrium reactor. Bottom cycle: simplified Rankine on steam (gas-only, see rankine01_water). HRSG transfers heat from the gas-turbine exhaust to the Rankine working fluid. Both turbines drive their own ElectricLoad generators.

Pre-mixed combustion air (N₂/O₂/CH₄, 8 bar, 600 K) → gibbsCombustor (T 1300 K, P 8 bar) → flueGasHot (8 bar, 1300 K) → gasTurbine (-600 kW, eta 0.85) → gasTurbExhaust (~4.0 bar, ~1131 K) → HRSG hot side — counter — HRSG cold side (steam) → gasFlueWarm (~4.0 bar, ~1051 K) ^

v heatedSteam (5 bar, ~1131 K) v steamTurbine (-50 kW, eta 0.85) v steamExhaust (~2.6 bar, ~1011 K) → steamCooler → condensate

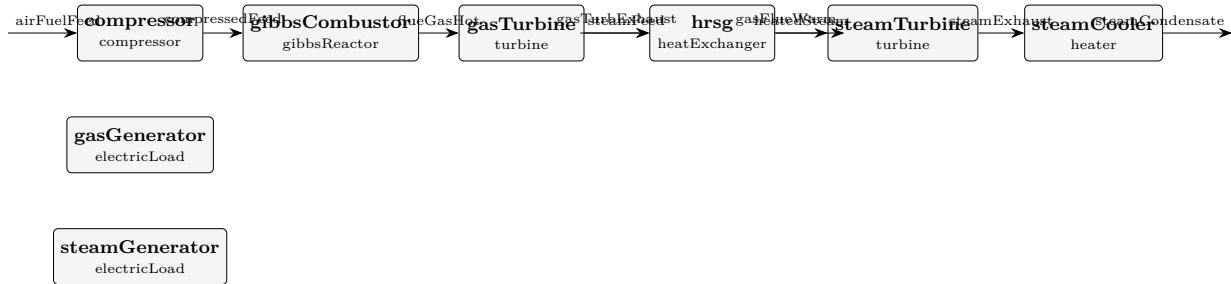
(T/P annotations above are this run's results; gas-only Rankine has no phase change, so the steam side heats to nearly the gas-exhaust T.)

Shaft routing (the v0.0.44 electricLoad unit): gasTurbine.shaft –wire→ gasGenerator (electricLoad, eta 0.97) steamTurbine.shaft –wire→ steamGenerator (electricLoad, eta 0.97)

Simplifications (deliberate, documented for v0.0.44): Pre-mixed combustion feed: a real Brayton has a separate compressor + air/fuel mixer. The current Choupo Mixer is liquid-only; gas-aware Mixer + Compressor closing this loop ship in v0.0.45. Rankine bottom cycle is gas-only (no liquid leg, no phase change). A multi-phase Boiler unit op is also on the v0.0.45 roadmap.

combined02_brayton_rankine_shaft

tutorials/steady/power/combined02_brayton_rankine_shaft



What it does: Combined cycle with SHAFT-SPLIT: gas turbine drives the compressor + net to the generator

The story: - Choupo -*—————* combined02_brayton_rankine_shaft – combined cycle, FULL Brayton with a SHAFT-SPLIT node.

Upgrades combined01: a real COMPRESSOR now sits on the gas-turbine shaft. The gas turbine’s gross work drives the compressor AND the net goes to the generator – a SHAFT NODE that SUMS work inputs:

$\text{gasGenerator.W} = \text{gasTurbine.shaft (+gross)} + \text{compressor.shaft (-load)}$

(Both rotating units publish a ‘shaft’ port with expression “-W_shaft”; the turbine’s $W_{\text{shaft}} < 0 \rightarrow +$, the compressor’s $W_{\text{shaft}} > 0 \rightarrow -$, so the generator nets to the delivered power.)

airFuelFeed (1 bar) $\rightarrow$ compressor ($\rightarrow \sim 8$ bar) $\rightarrow$ gibbsCombustor (1300 K) $\rightarrow$ gasTurbine $\rightarrow$ exhaust $\rightarrow$ HRSG hot side $\rightarrow$ flue | \{\} | (shaft) (shaft) (counter) | \{\} | compressor gasGenerator(net) steamFeed $\rightarrow$ heatedSteam $\rightarrow$ steamTurbine $\rightarrow$ steamGenerator $\rightarrow$ steamCooler $\rightarrow$ condensate

Bottom Rankine stays gas-only (phase-change boiler is the next step).

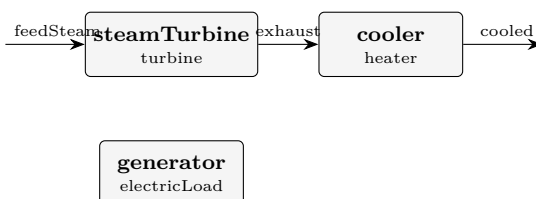
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	airFuelFeed	F	0.1025
stream	compressedFeed	F	0.1025
stream	flueGasHot	F	0.1025
stream	gasFlueWarm	F	0.1025
stream	gasTurbExhaust	F	0.1025
stream	heatedSteam	F	0.01
stream	steamCondensate	F	0.01
stream	steamExhaust	F	0.01
stream	steamFeed	F	0.01
kpi	compressor	F	0.1025
kpi	compressor	P_in	100000
kpi	compressor	P_out	828817.033759
kpi	compressor	T_in	300
kpi	compressor	T_out	592.090873473

... and 89 more reference values in the case’s *expected*.

rankine01_water

tutorials/steady/power/rankine01_water



What it does: Simplified gas-phase Rankine: superheated steam → turbine (-10 kW) → cooler (gas-only, no phase change)

The story: - Choupo -*—————* rankine01_water – simplified Rankine (gas-phase only)

A pedagogical Rankine cycle that demonstrates: a turbine extracting work from superheated steam, an ElectricLoad consuming the turbine's shaft via an energy wire, a thermal efficiency derived from the unit KPIs.

Simplification (documented, deliberate): the cycle runs entirely in the gas phase — there is no liquid leg and no phase change. A real Rankine has a pump + boiler that crosses the saturation line; that needs a multi-phase Boiler unit op which is on the v0.0.45 roadmap.

For the closed Rankine demonstration here:

feedSteam (superheated, 5 bar, 800 K, 1 mol/s) → steamTurbine (W_{shaft} -10 kW, eta 0.85) → exhaust (vapour, ~0.53 bar, ~531 K) → cooler (heater, Q -5 kW) → cooled (vapour, ~388 K)

Shaft routing (the new electricLoad unit, v0.0.44): steamTurbine.shaft —wire—→ generator (electricLoad, eta 0.97)

What the turbine produces (this run): W_{generated} = 10 kW (the shaft delivers this; W_{isen} would be 11.765 kW at eta = 1) P_{out} = 0.528 bar (a RESULT — the gas expands here) T_{out} = 530.9 K (258 degC) T_{out_isen} = 480.9 K (ideal isentropic discharge) expansion ratio = 9.47

W_{shaft} = -10 kW is the realistic specification for 1 mol/s of steam at this 5 → ~0.5 bar expansion (an early draft over-specified -300 kW, which would have demanded a physically absurd expansion).

The cooler duty is deliberately bounded: the turbine exhaust at 0.528 bar has T_{sat} = 355.5 K, so the sensible-only 'heater' must stop ABOVE the saturation dome. Q = -5 kW removes the desuperheat down to ~388 K (~33 K of margin above T_{sat}) and the cooled stream stays vapour (vf = 1) – a legitimate gas-phase desuperheater. Push the duty past ~6 kW and the outlet would cross the dome; a sensible-only heater correctly refuses (it asks for a phaseChanger/condenser). Fully closing the cycle (condensing back to a liquid leg + pump) needs the multi-phase Boiler/Condenser unit op on the v0.0.45 roadmap; this gas-phase demonstration stops at the desuperheater by design.

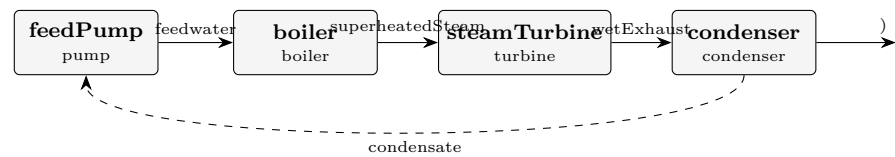
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cooled	F	0.001
stream	exhaust	F	0.001
stream	feedSteam	F	0.001
kpi	cooler	F	0.001
kpi	cooler	P	52827.3976295
kpi	cooler	Q	-5000
kpi	cooler	Q_kW	-5
kpi	cooler	Q_per_mol	-5000
kpi	cooler	T_in	530.878445241
kpi	cooler	T_out	388.087132345
kpi	cooler	dT	-142.791312896
kpi	generator	W_electric	9700
kpi	generator	W_electric_kW	9.7
kpi	generator	W_losses	300

...and 17 more reference values in the case's *expected*.

rankine02_water

tutorials/steady/power/rankine02_water



What it does: Closed IF97 Rankine cycle: pump → boiler → turbine → condenser → recycle

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

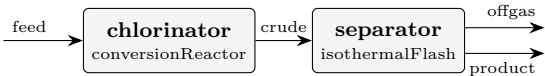
kind	unit/stream	key	expected
kpi	boiler	T_out	773.15
kpi	boiler	vf_out	1.0
kpi	boiler	Tsat	568.159121229
kpi	boiler	Q_kW	47.8559413747
kpi	condenser	vf_out	0.0
kpi	condenser	Tsat	446.515894355
kpi	condenser	Q_kW	-39.017165456
kpi	steamTurbine	P_out	858590.397228
kpi	steamTurbine	W_gen_kW	9.0
stream	superheatedSteam	vf	1.0
stream	condensate	vf	0.0

reactors

The theory you need. Batch reactors integrate $dn_i/dt = \nu_i r V$ from a charged initial state – no in/out flows; time replaces residence time. Adiabatic runs couple the energy balance $\sum_i n_i \bar{c}_{p,i} dT/dt = -\Delta H_{rxn} r V$, so composition and temperature evolve together. Stiffness appears the moment fast and slow reactions coexist (radical chains: μs radical lifetimes vs ms induction) – the explicit RK4 integrator then needs Δt at the FASTEST scale or blows up, while Rosenbrock23 (L-stable, linearly implicit) steps at the scale of the SOLUTION. The stiffness ratio is printed; watch it. *Full derivation: Theory Guide, the batch/ODE chapter.*

chlorophenol01_chlorination

tutorials/steady/reactors/chlorophenol01_chlorination



What it does: Chlorophenol: classic phenol chlorination ($\text{PhOH} + \text{Cl}_2 \rightarrow 4\text{-CP} + \text{HCl}$) + HCl flash

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	crude	F	0.0005833333333333
stream	feed	F	0.0005833333333333
stream	offgas	F	0.000274077911624
stream	product	F	0.00030925542171
kpi	chlorinator	F_in_kmol_h	2.1
kpi	chlorinator	F_out_kmol_h	2.1
kpi	chlorinator	T	353
kpi	chlorinator	conversion	0.9
kpi	chlorinator	extent_kmol_h	0.9
kpi	separator	F_alpha	0.00030925542171
kpi	separator	F_beta	0.000274077911624
kpi	separator	F_in	0.0005833333333333
kpi	separator	P	101325
kpi	separator	T	353

... and 2 more reference values in the case's *expected*.

conversion01_hda

tutorials/steady/reactors/conversion01_hda



What it does: Conversion reactor: HDA toluene+H2 → benzene+CH4 at 75%/pass (gas-basis, no kinetics, no Vliq)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.111111111111
stream	product	F	0.111111111111
kpi	reactor	F_in_kmol_h	400
kpi	reactor	F_out_kmol_h	400
kpi	reactor	Q_kW	-1007.67392146
kpi	reactor	T	900
kpi	reactor	conversion	0.75
kpi	reactor	dHrxn_kJ_per_mol	-48.3683482303
kpi	reactor	extent_kmol_h	75

conversion02_parallel_selectivity

tutorials/steady/reactors/conversion02_parallel_selectivity



What it does: Conversion reactor, PARALLEL network $A \rightarrow B$ / $A \rightarrow C$: selectivity as a SPEC (75 %)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	product	F	0.0001888888888889
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.68
kpi	reactor	T	350
kpi	reactor	extent_a_to_b_kmol_h	0.12
kpi	reactor	extent_a_to_c_kmol_h	0.04
kpi	reactor	nReactions	2

cstr01_first_order

tutorials/steady/reactors/cstr01_first_order



What it does: CSTR with pseudo-first-order Arrhenius esterification ($\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$), ~53% conversion of ethanol

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	X_limiting	0.52590711876
kpi	reactor	T	350
kpi	reactor	k	0.00357499942014

cstr02_reversible_wgs

tutorials/steady/reactors/cstr02_reversible_wgs



What it does: Reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) in a CSTR

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	feed	vf	1
stream	products	F	0.000277777777778
stream	products	vf	1
kpi	wgs	Da_kTau	145.857299066
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	1
kpi	wgs	T	600
kpi	wgs	V_R	0.001
kpi	wgs	X_limiting	0.841431955275
kpi	wgs	k	1.08521882096
kpi	wgs	tau_s	134.403584096
kpi	wgs	xi_mol_per_s	0.116865549344

cstr03_series_selectivity

tutorials/steady/reactors/cstr03_series_selectivity



What it does: CSTR with a SERIES network $A \rightarrow B \rightarrow C$: multi-reaction, the selectivity problem

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	product	F	0.000199218293117
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.717185855222
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	k_a_to_b	0.00214522777035
kpi	reactor	k_b_to_c	0.000536306942589
kpi	reactor	nReactions	2
kpi	reactor	newtonResidual	6.93889390391e-18
kpi	reactor	tau_s	225
kpi	reactor	xi_a_to_b_mol_per_s	0.0392797423303
kpi	reactor	xi_b_to_c_mol_per_s	0.0126884461066

cstr04_adiabatic

tutorials/steady/reactors/cstr04_adiabatic



What it does: ADIABATIC CSTR: T_{out} solved WITH the extents; Q=0 by construction (elements datum)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	reactorOut	F	0.0002777777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	4.70379291073e-12
kpi	reactor	T	350
kpi	reactor	T_guess	350
kpi	reactor	T_out	454.152948902
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	104.152948902
kpi	reactor	k_esterification_etac	0.88934531923
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	3.5527136788e-15
kpi	reactor	tau_s	310.291329081
... and 1 more reference values in the case's expected.			

cstr05_multiplicity

tutorials/steady/reactors/cstr05_multiplicity



What it does: CSTR multiplicity: three steady states – the reactor prints them all, T_guess picks the branch

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	8.68283223099e-12
kpi	reactor	T	300
kpi	reactor	T_guess	380
kpi	reactor	T_out	405.903598573
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	105.903598573
kpi	reactor	k_esterification_etac	0.406108484365
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	8.32667268469e-16
kpi	reactor	steadyStates	3
...and 2 more reference values in the case's expected.			

cstr06_jacketed

tutorials/steady/reactors/cstr06_jacketed



What it does: JACKETED CSTR: the coolant carries the reaction heat away – $Q = UA (T_c - T_{out})$

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-1.85832125175
kpi	reactor	T	350
kpi	reactor	T_guess	400
kpi	reactor	T_out	394.33285007
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	44.3328500695
kpi	reactor	k_esterification_etac	0.0534254733127
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	1.38777878078e-16
kpi	reactor	steadyStates	1

...and 2 more reference values in the case's expected.

cstr07_lhhw_methylAcetate

tutorials/steady/reactors/cstr07_lhhw_methylAcetate



What it does: LHHW on activities: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	reactorOut	F	0.0002777777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-2.09513270004
kpi	reactor	T	333.15
kpi	reactor	T_out	333.15
kpi	reactor	V_R	0.005
kpi	reactor	k_esterification_meoac	0.00280769603355
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	1.2490009027e-16
kpi	reactor	tau_s	366.524129505
kpi	reactor	xi_esterification_meoac_mol_per_s	0.0680930737737

equil01_reforming

tutorials/steady/reactors/equil01_reforming



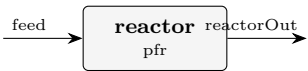
What it does: Equilibrium reactor (REquil): SMR + WGS to SIMULTANEOUS equilibrium at 1100 K

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001111111111111
stream	syngas	F	0.00166593958576
kpi	reformer	F_in_kmol_h	4
kpi	reformer	F_out_kmol_h	5.99738250873
kpi	reformer	Kp_smr	312.740523794
kpi	reformer	Kp_wgs	0.978425617548
kpi	reformer	Q_kW	59.6424648629
kpi	reformer	T	1100
kpi	reformer	conversion_smr	0.998691254363
kpi	reformer	conversion_wgs	0.109768673641
kpi	reformer	extent_smr_kmol_h	0.998691254363
kpi	reformer	extent_wgs_kmol_h	0.329306020923
kpi	reformer	nReactions	2
kpi	reformer	newtonResidual	3.2596148003e-13

pfr01_first_order
tutorials/steady/reactors/pfr01_first_order



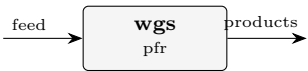
What it does: PFR with the same kinetics as cstr01 (RK4 integration)

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778

pfr02_reversible_wgs
tutorials/steady/reactors/pfr02_reversible_wgs



What it does: Reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) in a PFR; approaches the same X_{eq} as the CSTR

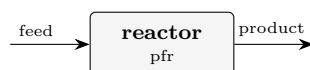
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	feed	vf	1
stream	products	F	0.0002777777777778
stream	products	vf	1

pfr03_series_selectivity

tutorials/steady/reactors/pfr03_series_selectivity



What it does: PFR with the SAME series network $A \rightarrow B \rightarrow C$: no back-mixing $\rightarrow$ MORE of the intermediate B

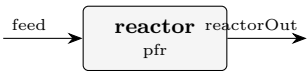
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | // O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	product	F	0.000176612457804
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.635804848095
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	k_a_to_b	0.00214522777035
kpi	reactor	k_b_to_c	0.000536306942589
kpi	reactor	nReactions	2
kpi	reactor	nSteps	200
kpi	reactor	tau_s	225

pfr04_adiabatic

tutorials/steady/reactors/pfr04_adiabatic



What it does: ADIABATIC PFR: the exothermic esterification heats its own stream – T(V) is a RESULT

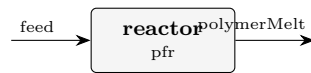
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-1.42108547152e-14
kpi	reactor	T	350
kpi	reactor	T_max	454.513818186
kpi	reactor	T_min	350
kpi	reactor	T_out	454.513818186
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	104.513818186
kpi	reactor	hotSpot_dT	104.513818186
kpi	reactor	k_esterification_etac	0.00357499942014
kpi	reactor	nReactions	1
...and 2 more reference values in the case's expected.			

pfr_polyesterification

tutorials/steady/reactors/pfr_polyesterification



What it does: PFR step-growth polyesterification — Carothers/Flory chain statistics

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A molten AB-type hydroxy-acid monomer (6-hydroxyhexanoic acid) self- condenses to polycaprolactone in a plug-flow reactor, releasing one water per ester bond:

As the monomer conversion p climbs along the reactor, the Carothers relation makes the average chain length rise steeply and the polydispersity approach 2 — the most-probable (Flory–Schulz) distribution.

and emits the Flory–Schulz distribution as a profile (chainLength vs n_x , w_x) so the GUI plots it: $n(x)=(1-p)p^{x-1}$ decays monotonically, $w(x)=x(1-p)^2 p^{x-1}$ peaks near $x \approx X_n$.

Run a sweep over operation. V_R (reactor volume $\rightarrow$ residence time $\rightarrow p$) and watch PDI climb toward 2, X_n and M_n shoot up, and the weight distribution broaden. The whole Carothers/Flory story in one glass-box run.

Lives inside the reaction. Absent ? the reactor behaves as an ordinary PFR. p is taken from the reactor's own conversion of limitingReactant (single source of truth). The distribution + PDI = $1+p$ are valid because a PFR has a uniform residence time; on a CSTR the residence-time distribution broadens both, so the distribution is gated to PFR/batch only.

- The chain-statistics math is exact and standard: P.J. Flory, **Principles of Polymer Chemistry**, Cornell Univ. Press (1953), Ch. VIII–IX (the most-probable distribution and $PDI = 1+p$); G. Odian, **Principles of Polymerization**, 4th ed., Ch. 2; H.S. Fogler, **Elements of Chemical Reaction Engineering**, 5th ed., ?9. - The monomer/repeat component properties are ILLUSTRATIVE (clearly labelled in each .dat) — plausible round numbers for a teaching case, not curated literature values. The Arrhenius rate is illustrative, tuned so $Da = k\tau \approx 4$ gives $p \approx 0.98$. Curate these before any quantitative use.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	polymerMelt	F	0.000550395491925
kpi	reactor	Da_kTau	3.98587252785
kpi	reactor	F_out_kmol_h	1.98142377093
kpi	reactor	Mn_kg_kmol	6144.41174045
kpi	reactor	Mw_kg_kmol	12174.6834809
kpi	reactor	PDI	1.98142377093
kpi	reactor	T	450
kpi	reactor	V_R	0.2
kpi	reactor	X_limiting	0.981423770929
kpi	reactor	Xn	53.8322388335
kpi	reactor	Xw	106.664477667
kpi	reactor	p_conversion	0.981423770929
kpi	reactor	tau_s	5829.95951417

rotating

The theory you need. Compressors, turbines, pumps: specified by shaft work W_s and isentropic efficiency η_s ; the outlet state follows from $h_{out} = h_{in} + W_s/\dot{n}$ with the isentropic reference path. Outlet pressure is a RESULT (hardware, not a spec).

compressor01_air

tutorials/steady/rotating/compressor01_air



What it does: Adiabatic air compressor: 10 kW shaft, eta 0.75 — P_out and T_out are RESULTS

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	out	F	0.001
kpi	compressor01	F	0.001
kpi	compressor01	P_in	100000
kpi	compressor01	P_out	890211.642995
kpi	compressor01	T_in	298.15
kpi	compressor01	T_out	633.374206884
kpi	compressor01	T_out_isen	551.37943898
kpi	compressor01	W_isen	7500
kpi	compressor01	W_isen_kW	7.5
kpi	compressor01	W_shaft	10000
kpi	compressor01	W_shaft_kW	10
kpi	compressor01	dS_gen	4.22615345389
kpi	compressor01	dT	335.224206884
... and 3 more reference values in the case's <i>expected</i> .			

compressor02_designspec_pout

tutorials/steady/rotating/compressor02_designspec_pout



What it does: Air compressor sized to hit P_out = 8 bar exactly, via DesignSpec on \$W_shaft

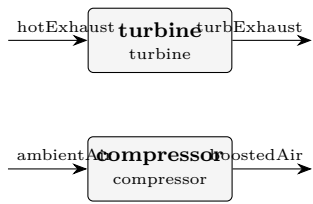
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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	out	F	0.001
kpi	compressor	F	0.001
kpi	compressor	P_in	100000
kpi	compressor	P_out	800004.793643
kpi	compressor	T_in	298.15
kpi	compressor	T_out	612.41049696
kpi	compressor	T_out_isen	535.384868081
kpi	compressor	W_isen	7017.30522226
kpi	compressor	W_isen_kW	7.01730522226
kpi	compressor	W_shaft	9356.40696301
kpi	compressor	W_shaft_kW	9.35640696301
kpi	compressor	dS_gen	4.08116534481
kpi	compressor	dT	314.26049696
...and 3 more reference values in the case's <i>expected</i> .			

compressorTurbine01_turbo_booster

tutorials/steady/rotating/compressorTurbine01_turbo_booster



What it does: Turbo-booster: turbine shaft drives a compressor (energy wire)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	ambientAir	F	0.0005
stream	boostedAir	F	0.0005
stream	hotExhaust	F	0.001
stream	turbExhaust	F	0.001
kpi	compressor	F	0.0005
kpi	compressor	P_in	100000
kpi	compressor	P_out	890211.642995
kpi	compressor	T_in	298.15
kpi	compressor	T_out	633.374206884
kpi	compressor	T_out_isen	551.37943898
kpi	compressor	W_isen	3750
kpi	compressor	W_isen_kW	3.75
kpi	compressor	W_shaft	5000
kpi	compressor	W_shaft_kW	5
...and 19 more reference values in the case's <i>expected</i> .			

pump01_water

tutorials/steady/rotating/pump01_water



What it does: Water pump: 2 kW shaft, eta 0.65 — incompressible, P_out / T_out are RESULTS

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.01
stream	out	F	0.01
kpi	pump01	F	0.01
kpi	pump01	P_in	100000
kpi	pump01	P_out	7294244.60432
kpi	pump01	T_in	298.15
kpi	pump01	T_out	299.077152318
kpi	pump01	W_hydraulic	1300
kpi	pump01	W_shaft	2000
kpi	pump01	W_shaft_kW	2
kpi	pump01	dP	7194244.60432
kpi	pump01	dT	0.927152317881
kpi	pump01	eta	0.65
kpi	pump01	head_m	735.848502288
...and 1 more reference values in the case's <i>expected</i> .			

pump02_pressure_spec

tutorials/steady/rotating/pump02_pressure_spec



What it does: Water pump specified by DISCHARGE PRESSURE (P_out 30 bar) — the shaft power W_shaft is the RESULT (closed-form inverse of pump01)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.01
stream	out	F	0.01
kpi	pump02	F	0.01
kpi	pump02	P_in	100000
kpi	pump02	P_out	3000000
kpi	pump02	T_in	298.15
kpi	pump02	T_out	298.523735099
kpi	pump02	W_hydraulic	524.03
kpi	pump02	W_shaft	806.2
kpi	pump02	W_shaft_kW	0.8062
kpi	pump02	dP	2900000
kpi	pump02	dT	0.373735099338
kpi	pump02	eta	0.65
kpi	pump02	head_m	296.620531272
...and 1 more reference values in the case's expected.			

turbine01_air

tutorials/steady/rotating/turbine01_air



What it does: Adiabatic gas turbine: extract 10 kW from hot air, eta 0.85 — P_out / T_out are RESULTS

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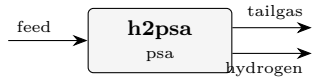
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	exhaust	F	0.001
stream	hotAir	F	0.001
kpi	turbine01	F	0.001
kpi	turbine01	P_in	1000000
kpi	turbine01	P_out	131729.126973
kpi	turbine01	T_in	900
kpi	turbine01	T_out	582.935206477
kpi	turbine01	T_out_isen	524.463853734
kpi	turbine01	W_gen_kW	10
kpi	turbine01	W_generated	10000
kpi	turbine01	W_shaft	-10000
kpi	turbine01	dS_gen	3.19034018436
kpi	turbine01	dT	-317.064793523
kpi	turbine01	dT_isen	-375.536146266
... and 2 more reference values in the case's expected.			

separation

psa01_h2_psa

tutorials/steady/separation/psa01_h2_psa



What it does: H2-PSA: 20-bar syngas (H2/CO2/CH4/CO/N2) over zeolite 5A → high-purity H2 raffinate + CO2-rich tail gas (equilibrium cyclic-steady-state shortcut)

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.2777777777778
stream	hydrogen	F	0.145276064457
stream	tailgas	F	0.13250171332
kpi	h2psa	Dq_CH4	0.0260273101613
kpi	h2psa	Dq_CO	0.0367785976251
kpi	h2psa	Dq_CO2	1.74280135323
kpi	h2psa	Dq_H2	0.0186248522203
kpi	h2psa	Dq_N2	0.00987823175641
kpi	h2psa	eta_used	0.8
kpi	h2psa	pressureRatio	16.6666666667
kpi	h2psa	productivity	0.0715324198173
kpi	h2psa	purgeRatio	0.15
kpi	h2psa	purity_H2	0.984779152498
kpi	h2psa	recovery_H2	0.735762032407
...and 5 more reference values in the case's <i>expected</i> .			

tsa01_co2_twin_bed
tutorials/steady/separation/tsa01_co2_twin_bed



What it does: Twin-bed TSA shortcut: continuous CO2 removal from helium by alternating adsorption at 298 K and regeneration at 398 K

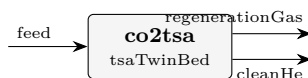
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	2.77777777778e-4
stream	cleanHe	F	2.475e-4
stream	regenerationGas	F	3.02777777778e-5
kpi	co2tsa	recovery_He	0.9
kpi	co2tsa	purity_He	1.0
kpi	co2tsa	Dq_CO2	0.408156018947
kpi	co2tsa	capacity_utilisation_CO2	0.680567638068
kpi	co2tsa	Q_sensible_per_cycle_kJ	276.0
kpi	co2tsa	Q_desorption_per_cycle_kJ	37.5
kpi	co2tsa	Q_regeneration_kW	1.045
kpi	co2tsa	Q_adsorption_cooling_kW	-1.045

tsa02_refused_capacity

tutorials/steady/separation/tsa02_refused_capacity



What it does: TSA refusal lesson: one undersized column cannot hold the CO₂ loaded during the declared switch interval

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solids

The theory you need. Solids carry mass and enthalpy without VLE: crystallisers, dryers, cyclones, gas–solid splitters. A dryer’s air is a REAL stream (two inlets, adiabatic mixing, outlet humidity from the psychrometrics); a cyclone’s cut size comes from the selected model (Lapple...Muschelknautz). Expect solid streams flagged separately (F_{solid}) and energy balances that close across phase change.

pneumaticConveyor01_sand

tutorials/steady/solids/pneumaticConveyor01_sand

What it does: Dilute-phase pneumatic conveying of sand in nitrogen: the glass-box dP breakdown + the saltation-margin safety check. Particle velocity from a mechanistic FORCE BALANCE (drag = wall friction + gravity); solids friction from Yang (1974, AIChE J 20:605); Rizk (1973) saltation. Loading ~2 kg/kg (in the Yang voidage range); solids friction DOMINATES the dP – conveying solids IS a pressure-drop problem. $u_g/u_{salt} = 1.72$ (safe margin above plugging).

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A single pneumaticConveyor unit: silica sand (~200 μm) carried by nitrogen up a 100 mm line, 50 m long, with a 10 m vertical rise.

*The total conveying-line pressure drop as a sum of physical contributions (glass-box) — Rhodes, *Introduction to Particle Technology*, Ch. 8:*

- superficial gas velocity $u_g = 21.9$ m/s, - particle velocity $u_p = 21.1$ m/s (Hinkle slip), - single-particle terminal velocity $u_t = 1.03$ m/s, - saltation velocity $u_{salt} = 3.6$ m/s (Rizk) — u_g is well above it, so transport is stable. The conveyor warns aloud (verbosity ≥ 1) if u_g ever falls below u_{salt} (risk of solids settling and blocking the line) — a first-class, announced design check, never a silent clip.

Solids loading = 0.20 kg solid / kg gas (dilute phase), suspended-solids density $\rho_s = 0.47$ kg/m³.

- particle velocity: Hinkle $u_p = u_g(1 - 0.68 d_p^{0.92} \rho_p^{0.5} \rho_g^{-0.2} D^{-0.54})$, with an honest $u_g - u_t$ fallback if the correlation leaves the dilute regime; - gas friction: Fanning (Blasius turbulent / 16/Re laminar); - solids friction: Hinkle $f_p = (3/8)(\rho_g/\rho_p) C_D (D/d_p)((u_g - u_p)/u_p)^2$, C_D from Schiller–Naumann at the slip Reynolds number; - saltation: Rizk (1973).

Operation block is hardware only (the credo): pipe diameter D , length L , vertical rise dz . Every velocity, the suspended density and the pressure drop are results. Particle density comes from the solid component's solid { ρ_p ; } (silica = 2650 kg/m³); the size from the inlet PSD (Sauter mean for the drag). The carrier-gas viscosity needs a transport { model Chung; } block in the thermoPhysPropDict (for the friction Reynolds numbers).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.0138888888889
stream	conveyed	F	0.0138888888889
kpi	node	P_out	234394.335968
kpi	node	accelerationLength	16.8207521299
kpi	node	deltaP	65605.6640315
kpi	node	deltaP_bends	0
kpi	node	deltaP_gasAccel	361.754574941
kpi	node	deltaP_gasFriction	1492.10736304
kpi	node	deltaP_solidsAccel	1383.86777275
kpi	node	deltaP_solidsFriction	61751.3873786
kpi	node	deltaP_static	616.546942159
kpi	node	nBends	0
kpi	node	particleWallFriction	0.4
kpi	node	rho_gas	3.39180174393

...and 10 more reference values in the case's *expected*.

pneumaticConveyor02_bends

tutorials/steady/solids/pneumaticConveyor02_bends

What it does: A SHORT (8 m) conveying line routed through 6 bends – the lesson: A FITTING ADDS PRESSURE DROP WITHOUT ADDING LENGTH. With long-radius sweeps the 6 bends cost ~10% of the total dP; swap them for blind tees (type blindTee;) and their share jumps to ~25% – the erosion-vs-pressure trade made visible. (On this short line the straight-pipe solids friction still carries most of the dP; the bends’ share GROWS as lines get shorter and bendier.) Same gas/solids as pneumaticConveyor01 (which runs 30 m / 6 m rise, bend-free) – compare the PER-METRE dP, not the totals.

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.0138888888889
stream	conveyed	F	0.0138888888889
kpi	node	P_out	279210.165228
kpi	node	accelerationLength	15.5453372639
kpi	node	deltaP	20789.8347721
kpi	node	deltaP_bends	2070.63576081
kpi	node	deltaP_gasAccel	361.754574941
kpi	node	deltaP_gasFriction	397.89529681
kpi	node	deltaP_solidsAccel	1380.42384054
kpi	node	deltaP_solidsFriction	16373.2628912
kpi	node	deltaP_static	205.862407804
kpi	node	nBends	6
kpi	node	particleWallFriction	0.4
kpi	node	rho_gas	3.39180174393

... and 10 more reference values in the case’s *expected*.

thermo

The theory you need. Consistency tests interrogate DATA, not models: the van Ness point test checks measured (x, y, T, P) sets against the Gibbs–Duhem constraint (residuals in $\ln(\gamma_1/\gamma_2)$); Herington’s area test is its integral cousin. A dataset that fails is not “bad” – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

air01_heater_expand

tutorials/steady/thermo/air01_heater_expand



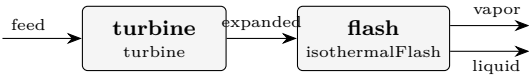
What it does: Predefined mixture: heat a stream fed with ‘components (air)’, which expands to N2/O2/Ar at load time

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	hot	F	0.001
kpi	airHeater	F	0.001
kpi	airHeater	P	100000
kpi	airHeater	Q	5000
kpi	airHeater	Q_kW	5
kpi	airHeater	Q_per_mol	5000
kpi	airHeater	T_in	298.15
kpi	airHeater	T_out	468.511847495
kpi	airHeater	dT	170.361847495

perUnitThermo01_srk_nrtl
tutorials/steady/thermo/perUnitThermo01_srk_nrtl



What it does: Per-unit thermo: turbine SRK (real gas) → flash NRTL (non-ideal liquid)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

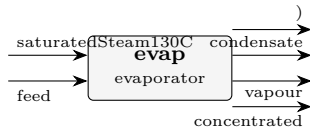
kind	unit/stream	key	expected
stream	expanded	F	0.0138888888889
stream	feed	F	0.0138888888889
stream	liquid	F	0.00854830793289
stream	vapor	F	0.005340580956
kpi	flash	F_alpha	0.00854830793289
kpi	flash	F_beta	0.005340580956
kpi	flash	F_in	0.0138888888889
kpi	flash	P	100000
kpi	flash	T	353
kpi	flash	V_over_F	0.384521828832
kpi	flash	phaseSet	0
kpi	turbine	F	0.0138888888889
kpi	turbine	P_in	600000
kpi	turbine	P_out	553956.972451

...and 11 more reference values in the case's *expected*.

thermoTest

model1_lumped_evaporator

tutorials/steady/thermoTest/model1_lumped_evaporator



What it does: thermoTest model 1 – NaCl as a LUMPED molecular component: the boiling-point elevation is the IDEAL $K_b \cdot m$ (van’t Hoff), BPE ~ 2.11 K. Compare model2 (same evaporator, Pitzer $a_w \rightarrow$ BPE ~ 5.07 K): the REPRESENTATION axis, lumped vs dissociated, on one operation.

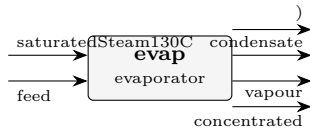
The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0200845615144
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00769321626341
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	2.11135909531
kpi	evap	F_conc	0.0200845615144
kpi	evap	F_conc_kmol_h	72.3044214517
kpi	evap	F_conc_mass	0.417969209015
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
...and 19 more reference values in the case's <i>expected</i> .			

model2_pitzer_evaporator

tutorials/steady/thermoTest/model2_pitzer_evaporator



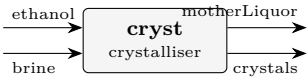
What it does: thermoTest model 2 – NaCl as a PITZER electrolyte (Na+ + Cl-, aqueous-inf-dilution reference): the boiling-point elevation comes from the real water activity a_w , BPE ~ 5.07 K vs the ideal 2.11 K of model1.

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0203403273092
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00743745046862
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	5.0621680882
kpi	evap	F_conc	0.0203403273092
kpi	evap	F_conc_kmol_h	73.225178313
kpi	evap	F_conc_mass	0.422576829808
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
... and 19 more reference values in the case's <i>expected</i> .			

model3_enrtl_antisolvent_crystalliser
tutorials/steady/thermoTest/model3_enrtl_antisolvent_crystalliser



What it does: Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises γ_{pm} , and DROPS the saturation molality m_{sat} – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent γ_{pm} .

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.02777777777778
stream	crystals	F	0.000329661969914
stream	ethanol	F	0.01666666666667
stream	motherLiquor	F	0.0424240933568
kpi	cryst	Ksp_activity	34.8439846653
kpi	cryst	Q_kW	-6.55987377671
kpi	cryst	Q_removed	6559.87377671
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	cakeLiquidFlow	0.000329661969914
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0988038720389
...and 9 more reference values in the case's expected.			

model5_nrtl_flash

tutorials/steady/thermoTest/model5_nrtl_flash



What it does: thermoTest model 5 – water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model. This is the antisolvent (ethanol) and the solvent (water) in their own role – the same substances that dissolve/precipitate NaCl in models 2-4, here separated by distillation.

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | /// O perations | Website: <https://choupo.org> | | \{\} \{\} | /// U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	liquid	F	0.0170966158658
stream	vapor	F	0.010681161912
kpi	flash	F_alpha	0.0170966158658
kpi	flash	F_beta	0.010681161912
kpi	flash	F_in	0.0277777777778
kpi	flash	P	100000
kpi	flash	Q	0
kpi	flash	Q_kW	0
kpi	flash	T	353
kpi	flash	V_over_F	0.384521828832
kpi	flash	phaseSet	0

userops

The theory you need. User-defined operations: the extension seam. A custom property op or unit compiled into the binary via the explicit factory – the tutorial shows the registration pattern (`registerBuiltins`) end to end.

userOp01_yield_reactor

tutorials/steady/userops/userOp01_yield_reactor



What it does: User-defined yieldReactor (case-local code/): neoPentane → nPentane, X=0.8

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What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	out	F	0.0002777777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	T	320
kpi	reactor	X	0.8
kpi	reactor	converted_kmol_h	0.8

userOp02_component_splitter

tutorials/steady/userops/userOp02_component_splitter



What it does: User-defined componentSplitter (case-local code/): shortcut benzene/toluene split

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0134722222222
stream	feed	F	0.0277777777778
stream	overhead	F	0.0143055555556
kpi	splitter	F_bottoms_kmol_h	48.5
kpi	splitter	F_overhead_kmol_h	51.5
kpi	splitter	overhead_fraction	0.515

utilities

The theory you need. Utility streams (steam levels, cooling water, refrigerants, electricity) carry duty into and out of the plant at declared temperature levels and prices; allocation is by *T*-level feasibility, announced.

utility01_dowtherm_preheat

tutorials/steady/utilities/utility01_dowtherm_preheat



What it does: Dowtherm A loop preheats a hydrocarbon feed (eps-NTU)

The story: - Choupo -*—————* utility01_dowtherm_preheat

A hydrocarbon feed (50/50 n-hexane / n-octane) needs to be heated to ~530 K before downstream conversion. HP steam can deliver at most ~525 K (T_{sat} at 40 barg) – not enough; the next escalation up the temperature ladder is a hot-oil loop, the textbook choice in the 250-400 C range being Dowtherm A (delivered here at 623 K).

NOTE: the 25 m² / $U=600$ exchanger is deliberately generous – $NTU=13.8$, $C_r=0.06$ give effectiveness ~ 1.00 , so the cold outlet is PINNED to the 623 K oil inlet, OVERSHOOTING the 530 K target by ~93 K. The case shows a pinned-effectiveness exchanger; shrink the area to land near 530 K.

Hardware (the credo): area + U + flow direction. The duty Q , the outlet temperatures, and the Dowtherm flowrate fall out of the effectiveness-NTU solution.

Process : 40 kmol/h at 320 K, 5 bar (mixed C6 + C8 hydrocarbon) Utility : Dowtherm A loop, delivery 623 K (350 C), ‘utility dowthermA;’ pulls the rest of the conditions from the catalogue Hardware: $A = 25 \text{ m}^2$, $U = 600 \text{ W}/(\text{m}^2\cdot\text{K})$, counter-current

The ‘utilities’ report aggregates the hot-side mass flow and emits reports/utilities/consumption.csv with MW and EUR/h delivered. The inspection question for the student: how much Dowtherm A do we actually consume? At what monetary cost per hour?

utility02_hitec_csp_heater

tutorials/steady/utilities/utility02_hitec_csp_heater



What it does: HITEC molten salt loop sensibly heats pressurised (40 bar) boiler feedwater to ~244 C, just below T_{sat} , for a downstream steam generator (eps-NTU, no phase change), ~208 kW duty

The story: - Choupo -*—————* utility02_hitec_csp_heater

When the duty needs a hot side above Dowtherm A's ceiling (~370 C), the next escalation up the ladder is a molten salt loop – HITEC is the textbook ternary ($\text{NaNO}_3\text{-NaNO}_2\text{-KNO}_3$) used for solar-thermal storage and high-T process heat down to its 142 C melting point.

Process here: pressurised water at 40 bar pre-heated towards its T_{sat} (~250 C) for a downstream HP steam generation step. Hot side: HITEC at 773 K (500 C) delivery, 'utility hitecSalt;' pulls the loop conditions from data/standards/utilities/hitecSalt.dat.

Hardware: $A = 0.85 \text{ m}^2$, $U = 800 \text{ W}/(\text{m}^2\cdot\text{K})$, counter-current. The Q and T_{out} fall out of eps-NTU. The area is deliberately sized so the cold outlet lands at ~517 K (244 C) – a sensible pre-heat that stops JUST below $T_{\text{sat}}(40 \text{ bar}) \approx 523.5 \text{ K}$ (250 C): the water leaves as a hot liquid ($vf = 0$), no phase change, exactly as a feedwater pre-heater ahead of the steam generator should. Oversize the area and the cold outlet would try to cross the saturation dome – a separate boiler's job, not this sensible-only eps-NTU unit's.

Inspection: how much HITEC do we circulate to deliver the duty? At what hourly cost? Compare to the equivalent steam-generated route via fired heater – the salt loop trades higher Capex for cleaner T-control and the absence of decomposition byproducts in the process side.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	boilerFeed	F	0.0222222222222
stream	boilerOut	F	0.0222222222222
stream	boilerOut	T	517.400926398
stream	saltReturn	F	0.0833333333333
stream	saltSupply	F	0.0833333333333
kpi	boilerHX	C_cold	1677.77777778
kpi	boilerHX	C_hot	12795.4797468
kpi	boilerHX	C_r	0.131122694184
kpi	boilerHX	LMTD	306.566828203
kpi	boilerHX	NTU	0.405298013245
kpi	boilerHX	Q	208465.443178
kpi	boilerHX	Q_kW	208.465443178
kpi	boilerHX	T_cold_in	393.15
kpi	boilerHX	T_cold_out	517.400926398

...and 6 more reference values in the case's *expected*.

utility03_glycol_brine_chiller

tutorials/steady/utilities/utility03_glycol_brine_chiller



What it does: PG-30 brine sensibly cools an ethanol vapour stream from 330 K to ~273 K (eps-NTU, no phase change), ~20.8 kW duty

The story: - Choupo -*—————* utility03_glycol_brine_chiller

When the process T sits below the freezing margin of plain chilled water (~5 C), the next escalation DOWN the temperature ladder is a secondary brine loop – propylene-glycol-30 protects sub-zero coils in food, pharma and freeze-cryst applications.

Process: a stream of ethanol vapour from the top of a fermentation column, to be sensible-cooled towards 285 K (~12 C). Utility: PG-30 brine, ‘utility refrigerationPG;’ pulls the loop conditions from data/standards/utilities/refrigerationPG.dat (delivery 0 C, return 5 C).

Hardware: $A = 8 \text{ m}^2$, $U = 350 \text{ W}/(\text{m}^2\cdot\text{K})$ (lower U on the vapour side is realistic for a film condenser-style geometry), counter-current.

Inspection: how much brine flow is required? At 6 EUR/GJ the refrigeration cost is sharply higher per GJ than cooling water – is it justified by the process T? Compare with what cooling water alone could deliver.

Category II: The property bench (choupoProps)

Questions asked of the thermodynamics BEFORE a simulation uses it – CREATE, SEE, TEST, FIT, PROMOTE.

adsorption

The theory you need. The isotherm bench: evaluating a curated adsorption-equilibrium record (Henry-limit, saturation, pin and unit-invariance gates run aloud) and REGRESSING one from data – the multi-temperature van't Hoff Langmuir fit with identifiability verdicts: single- T data refuses ΔH_{ads} , Henry-regime data yields only the product $q_{\text{max}}b$, and a non-converged fit refuses its verdict by name. *Full derivation: Theory Guide, the adsorption chapter.*

isotherm01_eval_13x_co2

tutorials/props/adsorption/isotherm01_eval_13x_co2

eval_13x_co2 isothermEval

What it does: Adsorption isotherm bench: CO2/zeolite13X q(T,p) grid + Henry/saturation/pin/unit-invariance gates

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	eval_13x_co2	T0	2.931e+02
diag	eval_13x_co2	T1	3.131e+02
diag	eval_13x_co2	T2	3.331e+02
diag	eval_13x_co2	b_T0	1.227e+01
diag	eval_13x_co2	b_T1	3.772e+00
diag	eval_13x_co2	b_T2	1.337e+00
diag	eval_13x_co2	henry_T0	6.133e-04
diag	eval_13x_co2	henry_T1	1.886e-04
diag	eval_13x_co2	henry_T2	6.683e-05
diag	eval_13x_co2	qsat	5.000e+00
diag	eval_13x_co2	tRef	2.981e+02
diag	eval_13x_co2	gate_henry	1.000e+00
diag	eval_13x_co2	gate_henry_maxdev	1.227e-07
diag	eval_13x_co2	gate_saturation	1.000e+00

... and 7 more reference values in the case's *expected*.

isotherm02_fit_jaschik

tutorials/props/adsorption/isotherm02_fit_jaschik

fit_jaschik
fitParametersrecovery_multiT
fitParametersgate_singleT
fitParametersgate_henry
fitParameters

What it does: Isotherm regression: CO₂/13X Jaschik 2020 (CC BY) multi-T Langmuir fit + recovery/single-T/Henry identifiability gates

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	fit_jaschik	converged	1.00000e+00
diag	fit_jaschik	identifiable	1.00000e+00
diag	fit_jaschik	n_data	1.32000e+02
diag	fit_jaschik	n_params	3.00000e+00
diag	fit_jaschik	dof	1.29000e+02
diag	fit_jaschik	single_T	0.00000e+00
diag	fit_jaschik	henry_regime	0.00000e+00
diag	fit_jaschik	endothermic_suspicious	0.00000e+00
diag	fit_jaschik	fit.q_max.value	4.31448e+00
diag	fit_jaschik	fit.q_max.stderr	5.17246e-02
diag	fit_jaschik	fit.b_ref.value	4.06758e-04
diag	fit_jaschik	fit.b_ref.stderr	2.90989e-05
diag	fit_jaschik	fit.dH_ads.value	-3.73835e+04
diag	fit_jaschik	fit.dH_ads.stderr	1.90453e+03

... and 32 more reference values in the case's *expected*.

isotherm03_fit_refusals
tutorials/props/adsorption/isotherm03_fit_refusals

gate_henry_nonconverged fitParameters	gate_nonconverged fitParameters	gate_langfull_nonconverged fitParameters
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What it does: Fit refusal gates: non-converged fits refuse their verdicts by name

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

compare

The theory you need. Method comparison runs the SAME property through different models side-by-side – UNIFAC vs NRTL, Joback vs measured, ideal vs EOS. The gap between curves IS the modelling uncertainty; picking a method is a decision the case author owns, made with eyes open. *Full derivation: Theory Guide, the Props Guide.*

compare_activity_etoh_water

tutorials/props/compare/compare_activity_etoh_water

g_ideal

propertyScan1D

g_nrtl

propertyScan1D

g_wilson

propertyScan1D

What it does: Compare 3 activity models (ideal / NRTL / Wilson) for γ_{ethanol} in equimolar mixture with water vs T

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

compare_eos_co2

tutorials/props/compare/compare_eos_co2

z_ideal
propertyScan1D

z_srk
propertyScan1D

z_pr
propertyScan1D

What it does: Compare 3 EoS (idealGas / SRK / PR) for Z of CO2 near critical (T=310 K, P from 1 to 100 bar)

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compare_kinetics_order

tutorials/props/compare/compare_kinetics_order

rate_order1

kinetics1D

rate_order2

kinetics1D

What it does: Rate-law order from DATA: 1st- vs 2nd-order fits overlaid on measured $c(t)$ — the data picks the order

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compare_psat_ambrose_walton

tutorials/props/compare/compare_psat_ambrose_walton

psat_aw_vs_antoine
propertyScan1D

What it does: Psat: Ambrose-Walton corresponding-states PREDICTION vs fitted Antoine DATA, for benzene (non-polar, good) and ethanol (associating, degrades) — see the error, then decide

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization |

compare_transport_water

tutorials/props/compare/compare_transport_water

mu_iapws

propertyScan1D

mu_generic

propertyScan1D

k_iapws

propertyScan1D

k_generic

propertyScan1D

What it does: Compare water transport: IAPWS (IF97 R12-08/R15-11) vs generic correlations (Andrade / SatoRiedel), 280-370 K

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	mu_iapws	n_failures	0
diag	mu_iapws	n_points	46
diag	mu_generic	n_failures	0
diag	mu_generic	n_points	46
diag	k_iapws	n_failures	0
diag	k_iapws	n_points	46
diag	k_generic	n_failures	0
diag	k_generic	n_points	46

compare_visc_liquid_ammonia

tutorials/props/compare/compare_visc_liquid_ammonia

mu_andrade
 propertyScan1D

What it does: Liquid-viscosity Andrade for ammonia, 235-300 K (cross-check $\mu@239.85\text{K} \sim 0.255 \text{ mPa.s}$)

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compare_visc_liquid_water

tutorials/props/compare/compare_visc_liquid_water

mu_andrade

propertyScan1D

mu_vogel

propertyScan1D

What it does: Compare 2 liquid viscosity correlations (Andrade vs Vogel) for water, 280-360 K

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

compare_vle_etoh_water

tutorials/props/compare/compare_vle_etoh_water

txy_ideal
propertyScan1Dtxy_nrtl
propertyScan1Dtxy_unifac
propertyScan1Dconsistency_etoh_water
vleConsistency

What it does: Ethanol/water T-x-y at 1 atm: ideal vs NRTL model curves OVERLAID with measured VLE data — see the data, pick the model

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

electrolyte

The theory you need. Ions interact at long range (Debye–Hückel, $\sqrt{I}$ law) and short range (ion–ion, ion–solvent). Choupo’s ladder: Davies (charge-only, $I \leq 0.5$), single-salt Pitzer ($\beta_0, \beta_1, C^\varphi$; to $I \sim 6$, and with the Silvester–Pitzer T -functions to 300°C), multi-ion Pitzer–HMW (per-ion, mixing terms θ, ψ – seawater-class brines), and eNRTL (local-composition; single-salt, mixed-solvent, and the 2009 symmetric multi-salt with both published and Gibbs–Duhem-consistent formulations). Speciation solves mass action + charge balance by Newton in log-molality space; saturation indices $SI = \log(IAP/K_{sp})$ flag scaling. Expect: models to DISAGREE above their validity (γ divergence is the lesson, e.g. Davies vs Pitzer at RO recoveries), and every activity model to announce itself. *Full derivation: Theory Guide, the electrolyte + speciation chapters.*

composition01_nacl

tutorials/props/electrolyte/composition01_nacl

naclWater
 speciate

What it does: Apparent-salt input: the analysis is given as composition { NaCl 2.0 mol/kg }, expanded to Na + Cl ion totals through NaCl's dissociatesTo, with electroneutrality validated (Sum nu*charge = 0). Byte-identical to writing totals { Na 2.0; Cl 2.0 } by hand.

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The speciate op expands each salt to ion totals through its dissociatesTo ($\text{NaCl} \rightarrow \text{Na } 1, \text{Cl } 1$), and validates electroneutrality ($\text{Sum nu} \times \text{charge} = 0$) so a formulated-salts input can never silently unbalance charge. This is byte-identical to writing the ions by hand (totals { Na 2.0; Cl 2.0 }): SI_halite -0.953, feed charge imbalance 0.00%.

composition{\} is the apparent-salt input mode; analyticalTotals{\} / totals{\} stay as the mode for waters measured directly in ions/masters (roadmap Phase B).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	naclWater	I	2
diag	naclWater	SI_halite	-0.953442
diag	naclWater	aw	0.930475
diag	naclWater	gamma_Cl	1.01683
diag	naclWater	gamma_Na	1.01683
diag	naclWater	m_Cl	2
diag	naclWater	m_Na	2
diag	naclWater	pH	7.13303

enrtl_mixed_nacl_ethanol_esteso

tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso

esteso

enrtlMixedSolvent

What it does: VALIDATION of the generalized segment-based eNRTL (Chen & Song 2004, AIChE J 50:1928, DOI 10.1002/aic.10151) against the PRIMARY measured $\gamma_{\pm}$ of NaCl in ethanol-water at 25 C (Esteso et al., J. Solution Chem. 18 (1989) 277, emf cell, 0-90 wt% ethanol). ZERO refit: every $\tau_{X:Z}$ is the published Table 2/4/7 value; A_{ϕ} per composition is Esteso’s measured Table II. Headline AAD ~4.3% over 47 points (paper quotes RMS 0.026 on γ); ~1% in pure water, ~1.2% at 70 wt%. The +8% at 80/90 wt% is DECLARED physics the complete-dissociation model cannot carry: Esteso measures ion association $K_A = 30\text{-}46$ there (his Table V).

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	esteso	AAD_total_pct	4.3
diag	esteso	AAD_wt0_pct	1
diag	esteso	AAD_wt20_pct	5.3
diag	esteso	AAD_wt40_pct	5.3
diag	esteso	AAD_wt60_pct	2.7
diag	esteso	AAD_wt70_pct	1.2
diag	esteso	AAD_wt80_pct	7.8
diag	esteso	AAD_wt90_pct	7.8
diag	esteso	n_points	47

enrtl_multisalt_nacl_kcl
tutorials/props/electrolyte/enrtl_multisalt_nacl_kcl

harned_nacl_kcl
enrtlMultiSalt

harned_nacl_kcl_refined
enrtlMultiSalt

What it does: Multi-salt symmetric eNRTL (Song & Chen 2009) – NaCl+KCl Harned sweep at 4 mol/kg

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	harned_nacl_kcl	consistent	0.00e+00
diag	harned_nacl_kcl	formulation_gap_mid	4.13e-02
diag	harned_nacl_kcl	gamma1_pure	8.01e-01
diag	harned_nacl_kcl	gamma1_trace	6.17e-01
diag	harned_nacl_kcl	gamma2_pure	5.76e-01
diag	harned_nacl_kcl	gamma2_trace	7.29e-01
diag	harned_nacl_kcl	gibbsDuhem_dev_mid	4.72e-01
diag	harned_nacl_kcl	harned_R2_min	9.99e-01
diag	harned_nacl_kcl	harned_alpha_1	2.85e-02
diag	harned_nacl_kcl	harned_alpha_2	-2.56e-02
diag	harned_nacl_kcl	harned_slope_1	-2.62e-01
diag	harned_nacl_kcl	harned_slope_2	-2.36e-01
diag	harned_nacl_kcl	oracle_pure_dev	4.03e-09
diag	harned_nacl_kcl	reduction_pin_dev	1.11e-15
... and 18 more reference values in the case's expected.			

enthalpy_naoh_water

tutorials/props/electrolyte/enthalpy_naoh_water

gamma_naoh

pitzerActivity

What it does: NaOH-water heat of dilution: Pitzer with calorimetric T-slots vs Parker (1965)

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_naoh	Lphi_AAD_abs_Jmol	22.9776
diag	gamma_naoh	Lphi_AAD_rel_pct	4.7712
diag	gamma_naoh	Lphi_extrap_maxdev_Jmol	15774.0028
diag	gamma_naoh	Lphi_nMeas	29.0000
diag	gamma_naoh	aw_1m	0.9665
diag	gamma_naoh	gamma_pm_1m	0.6679
diag	gamma_naoh	gamma_pm_3m	0.7838
diag	gamma_naoh	phi_1m	0.9471

farelo_licl_range

tutorials/props/electrolyte/farelo_licl_range

licl_1.0
speciatelicl_6.0
speciatelicl_13.0
speciatelicl_19.7
speciatefarelo_LiCl_sat
speciate

What it does: LiCl Pitzer REFITTED to Hamer-Wu + Farelo saturation: $\gamma_{\pm}$ to ~11% over 0.5-19 mol/kg (was 774 at 19.7m), and the LiCl.H₂O hydrate Ksp reproduces Farelo's 19.7 mol/kg via the HMW osmotic water activity ($a_w=0.11$).

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*Prof. Fátima Farelo measured LiCl solubility at 19.7 → 23.5 mol/kg (solid: LiCl·H₂O) — *J. Chem. Eng. Data* 50 (2005) 1470, DOI 10.1021/je050111j. That is one of the most concentrated aqueous electrolytes there is.*

*The catalogue's earlier Li-Cl Pitzer parameters (Appelo, Parkhurst & Post, *Geochim. Cosmochim. Acta* 125 (2014) 49), fitted below ~6 mol/kg, blow up at high concentration — $\gamma_{\pm} = 774$ at 19.7 mol/kg vs ~45 measured. The two-parameter B-term is linear in m , so γ grows exponentially.*

Rather than declare defeat, the parameters were refitted to Hamer & Wu (1972) $\gamma_{\pm}$ and Farelo's saturation. A negative C_{φ} (-0.0026) tames the high- m blow-up, and the three-parameter form then reaches saturation:

*carries a water-activity leg. At 19.7 mol/kg the model gives $a_w = 0.11$ (not 1!) — which only the rigorous Pitzer-HMW osmotic coefficient delivers, not the $\varphi = 1$ dilute shortcut. The lithiumChlorideH₂O mineral Ksp ($\log K_{25} = 4.984$) is calibrated to Farelo's saturation *through* that a_w . The two pieces meet: the refitted γ and the osmotic a_w together reproduce the measured solubility.*

- electrolyte/pairs.dat: the Li-Cl Pitzer pair, refitted to Hamer-Wu + Farelo, valid to ~19 mol/kg (supersedes the Appelo low- m fit for concentrated LiCl). - electrolyte/minerals.dat: the LiCl·H₂O hydrate Ksp, calibrated to Farelo. - constant/species/Li.dat: Li⁺ aqueous formation datum ($h_f A_q = -278.49$ kJ/mol, CODATA).

*- The full measured dataset (this and the other five ternaries) is tabulated, cited, in constant/experimental/farelo_solubility.csv. - The equilibrate round-trip (precipitate LiCl·H₂O from a supersaturated feed) diverges at $I \approx 20$ — a known active-set / γ fixed-point numerical limit at extreme ionic strength, *independent of the activity model*. The $SI = 0$ calibration at the saturation composition is exact; reaching it by precipitation is the convergence problem, not the thermodynamics. - The AlCl₃ systems from the same paper remain deferred: Al³⁺ hydrolyses (the paper notes $pH < 2.75$) and needs an Al speciation model Choupo does not yet carry.*

farelo_nacl_nh4cl

tutorials/props/electrolyte/farelo_nacl_nh4cl

farelo_NaCl_pure
speciatefarelo_NaCl_nh4_093
speciatefarelo_NaCl_nh4_187
speciatefarelo_NaCl_nh4_280
speciatefarelo_NH4Cl_pure
speciate

What it does: Pitzer-HMW vs Farelo (JCED 2005): NaCl+NH₄Cl common-ion salting-out – model SI ~ 0 at each measured saturation point (NaCl 6.15→5.5→5.0; NH₄Cl 7.35 mol/kg, 298 K).

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A glass-box validation of Choupo's multi-ion Pitzer electrolyte model against the solubility measurements of Prof. F?tima Farelo (Chemical Engineering Dept., IST – T?cnico Lisboa).

> M. Farelo, C. Fernandes & A. Avelino, * "Solubilities for Six Ternary Systems: > NaCl + NH₄Cl + H₂O, KCl + NH₄Cl + H₂O, NaCl + LiCl + H₂O, KCl + LiCl + H₂O, > NaCl + AlCl₃ + H₂O, and KCl + AlCl₃ + H₂O at T = (298 to 333) K", > J. Chem. Eng. Data 50 (2005) 1470–1477. DOI [10.1021/je050111j](<https://doi.org/10.1021/je050111j>).

Saturation molalities measured to ±0.0007 mol/kg (visual polythermal / isothermal methods). The pure-salt anchors at 298.15 K: NaCl 6.15, NH₄Cl 7.35 mol/kg; the salting-out series (Table 1): NaCl solubility falls 6.15 → 5.5 → 5.0 → 4.95 as NH₄Cl rises 0 → 0.93 → 1.87 → 2.80 mol/kg.

The model is calibrated to only the two pure-salt solubilities — halite's K_{sp} from PHREEQC, sal-ammoniac's K_{sp} pinned to Farelo's NH₄Cl point. Everything else is a prediction. The NH₄–Cl Pitzer interaction parameters come from the primary single-electrolyte fit (Pitzer & Mayorga 1973); the catalogue already carried Na–Cl and K–Cl.

At each of Farelo's measured saturation compositions the model's saturation index should be ≈ 0 if the model reproduces the measurement. It is, to ±0.03 log units:

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	farelo_NaCl_pure	I	6.15
diag	farelo_NaCl_pure	SI_halite	0.0168246
diag	farelo_NaCl_pure	aw	0.751868
diag	farelo_NaCl_pure	gamma_Cl	1.0105
diag	farelo_NaCl_pure	gamma_Na	1.0105
diag	farelo_NaCl_pure	m_Cl	6.15
diag	farelo_NaCl_pure	m_Na	6.15
diag	farelo_NaCl_pure	pH	7.11704
diag	farelo_NaCl_nh4_093	I	6.43
diag	farelo_NaCl_nh4_093	SI_halite	-0.022236
diag	farelo_NaCl_nh4_093	SI_salammoniac	-0.705926
diag	farelo_NaCl_nh4_093	aw	0.746426
diag	farelo_NaCl_nh4_093	gamma_NH4	0.59147
diag	farelo_NaCl_nh4_093	gamma_Na	1.04074

...and 29 more reference values in the case's expected.

ksp_temperature

tutorials/props/electrolyte/ksp_temperature

T10
speciate

T25
speciate

T40
speciate

T60
speciate

T80
speciate

What it does: K(T) MATTERS: a temperature scan of mineral saturation for ONE fixed water (a hard, carbonate-bearing groundwater at pH 7.6), speciated at 10, 25, 40, 60 and 80 C. The point is the SIGN of dK/dT. CALCITE dissolution is EXOTHERMIC, so its equilibrium constant FALLS as temperature rises and its saturation index SI_calcite RISES – RETROGRADE SOLUBILITY: calcite is LESS soluble hot. This is why hot heat-exchanger surfaces and warm RO concentrate scale with calcium carbonate even when the cold feed looks safe. Contrast HALITE (NaCl): no analytic K(T) in the database, so it falls back to constant-dH van’t Hoff with a small POSITIVE dH (endothermic) – normal solubility, SI_halite drifts gently DOWN with T. At 25 C nothing in the K(T) machinery moves (every form returns logK25 exactly, byte-stable); off 25 C the run announces which K(T) form each entry uses (analytic / van’t Hoff / flat) and warns if any analytic is used beyond its fitted validity range. K(T): analytic = PHREEQC -analytical_expression anchored on logK25 (logK(T)=logK25+[ana(T)-ana(298.15)]); USGS PHREEQC, public domain.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	T10	I	0.079772
diag	T10	SI_calcite	0.586526
diag	T10	SI_gypsum	-0.329589
diag	T10	SI_halite	-4.56692
diag	T10	aw	0.998112
diag	T10	pH	7.6
diag	T25	I	0.0775557
diag	T25	SI_calcite	0.813492
diag	T25	SI_gypsum	-0.358481
diag	T25	SI_halite	-4.60676
diag	T25	aw	0.998123
diag	T25	pH	7.6
diag	T40	I	0.0749471
diag	T40	SI_calcite	1.00956

...and 16 more reference values in the case's *expected*.

overlay01_nacl_ksp
tutorials/props/electrolyte/overlay01_nacl_ksp

naclWater
speciate

What it does: PARTIAL deep-merge overlay: a case recalibrates ONLY the halite equilibrium logK25 via a case-local overlayOf NaCl; every other datum (dH, dissolutionReaction, calorimetric, crystal) still comes from the standard components/NaCl.dat record. The [overlay] line names exactly which leaf came from the case tier.

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A case recalibrates ONLY the halite equilibrium logK25 (1.60 here) to its own value, via a case-local constant/components/NaCl.dat declaring overlayOf NaCl;.

The reader deep-merges this delta over the standard components/NaCl.dat: only solidPhases.halite.equilibrium.logK25 is replaced; dH, the dissolutionReaction, the calorimetric anchor and the crystal block all still resolve from the standard record. The run prints the provenance:

This is the roadmap Phase A acceptance test: a case ships ONLY its deltas, never a second competing complete record, and per-value provenance (tier + file) is preserved.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	naclWater	I	2
diag	naclWater	SI_halite	-0.983442
diag	naclWater	aw	0.930475
diag	naclWater	gamma_Cl	1.01683
diag	naclWater	gamma_Na	1.01683
diag	naclWater	m_Cl	2
diag	naclWater	m_Na	2
diag	naclWater	pH	7.13303

pitzer01_nacl
tutorials/props/electrolyte/pitzer01_nacl

gamma_nacl
pitzerActivity

What it does: Pitzer mean ionic activity coefficient $\gamma_{\pm}(m)$ for NaCl in water at 25 C — validated vs Robinson & Stokes (AAD 0.17%). The glass-box electrolyte foundation: see the famous $\gamma_{\pm}$ dip (~0.66 near 1 m) and the rise above 1 at high concentration.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.9668
diag	gamma_nacl	gamma_pm_1m	0.6572
diag	gamma_nacl	gamma_pm_3m	0.7140
diag	gamma_nacl	phi_1m	0.9363

pitzer02_cacl2
tutorials/props/electrolyte/pitzer02_cacl2

gamma_cacl2
pitzerActivity

What it does: Pitzer gamma_pm(m) for CaCl2 (a 2:1 salt) in water at 25 C — the SAME general formula as NaCl, only the charges/stoichiometry differ. Validated vs Robinson & Stokes (AAD 0.49%, 0.1-2 m).

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_cacl2	aw_1m	0.9449
diag	gamma_cacl2	gamma_pm_1m	0.4985
diag	gamma_cacl2	gamma_pm_3m	1.4771
diag	gamma_cacl2	phi_1m	1.0483

pitzer02_nacl_package
tutorials/props/electrolyte/pitzer02_nacl_package

gamma_nacl
electrolyteActivity

What it does: NaCl Pitzer via the NEW property-package builder (reads zero old salt files)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.966826
diag	gamma_nacl	gamma_pm_1m	0.657172
diag	gamma_nacl	gamma_pm_3m	0.714026
diag	gamma_nacl	phi_1m	0.93634

pitzer_calcite_brine

tutorials/props/electrolyte/pitzer_calcite_brine

seawater_pitzer
speciateseawater_davies
speciate

What it does: Pitzer-HMW carbonate system on surface seawater – pins SI_calcite / SI_gypsum against the published oceanographic saturation state, captures the neutral CO₂ salting-out ($\gamma_{\text{CO2aq}} > 1$ via λ), and the Davies-vs-Pitzer SI divergence.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	seawater_pitzer	I	0.6828
diag	seawater_pitzer	SI_anhydrite	-1.24594
diag	seawater_pitzer	SI_aragonite	0.553044
diag	seawater_pitzer	SI_calcite	0.667044
diag	seawater_pitzer	SI_gypsum	-0.962237
diag	seawater_pitzer	SI_halite	-2.48856
diag	seawater_pitzer	SI_sylvite	-3.51959
diag	seawater_pitzer	aw	0.981411
diag	seawater_pitzer	γ_{CO2aq}	1.10587
diag	seawater_pitzer	γ_{CO3}	0.105425
diag	seawater_pitzer	γ_{Ca}	0.194354
diag	seawater_pitzer	γ_{HCO3}	0.659536
diag	seawater_pitzer	m_{CO2aq}	1.46581e-05
diag	seawater_pitzer	m_{CO3}	7.90109e-05

... and 20 more reference values in the case's *expected*.

pitzer_gamma_hamer_wu
tutorials/props/electrolyte/pitzer_gamma_hamer_wu

gamma_nacl
pitzerActivity

gamma_kcl
pitzerActivity

gamma_kbr
pitzerActivity

gamma_lcl_consistency
pitzerActivity

gamma_hcl
pitzerActivity

gamma_naoh
pitzerActivity

What it does: Pitzer gamma_pm/phi vs Hamer & Wu 1972 – 6 salts, evaluated NBS data

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.9668
diag	gamma_nacl	gamma_AAD_pct	0.1117
diag	gamma_nacl	gamma_nMeas	8.0000
diag	gamma_nacl	gamma_pm_1m	0.6572
diag	gamma_nacl	gamma_pm_3m	0.7140
diag	gamma_nacl	phi_1m	0.9363
diag	gamma_nacl	phi_AAD_pct	0.0870
diag	gamma_nacl	phi_nMeas	8.0000
diag	gamma_kcl	aw_1m	0.9681
diag	gamma_kcl	gamma_AAD_pct	0.1340
diag	gamma_kcl	gamma_nMeas	7.0000
diag	gamma_kcl	gamma_pm_1m	0.6043
diag	gamma_kcl	gamma_pm_3m	0.5698
diag	gamma_kcl	phi_1m	0.8987

... and 34 more reference values in the case's *expected*.

pitzer_hmw_hot_agreement
tutorials/props/electrolyte/pitzer_hmw_hot_agreement

hmw_nacl_100C

speciate

kernel_nacl_100C

pitzerActivity

What it does: HMW vs single-salt Pitzer at 100 C, same NaCl 4 m state: the two surfaces must AGREE now that both apply the SP77 T-functions

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	hmw_nacl_100C	I	4.0000
diag	hmw_nacl_100C	SI_halite	-0.7734
diag	hmw_nacl_100C	aw	0.8533
diag	hmw_nacl_100C	gamma_Cl	0.7309
diag	hmw_nacl_100C	gamma_Na	0.7309
diag	hmw_nacl_100C	m_Cl	4.0000
diag	hmw_nacl_100C	m_Na	4.0000
diag	hmw_nacl_100C	pH	7.0000
diag	kernel_nacl_100C	aw_1m	0.9670
diag	kernel_nacl_100C	gamma_pm_1m	0.6221
diag	kernel_nacl_100C	gamma_pm_3m	0.6760
diag	kernel_nacl_100C	phi_1m	0.9327

pitzer_nacl_sp77_hot

tutorials/props/electrolyte/pitzer_nacl_sp77_hot

nacl_100C
pitzerActivity

nacl_150C
pitzerActivity

nacl_200C
pitzerActivity

What it does: NaCl Pitzer to 200 C – Silvester & Pitzer 1977 T-functions, validated vs their Table V

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	nacl_100C	aw_1m	0.9670
diag	nacl_100C	gamma_AAD_pct	0.2851
diag	nacl_100C	gamma_nMeas	1.0000
diag	nacl_100C	gamma_pm_1m	0.6221
diag	nacl_100C	gamma_pm_3m	0.6760
diag	nacl_100C	phi_1m	0.9327
diag	nacl_150C	aw_1m	0.9679
diag	nacl_150C	gamma_AAD_pct	0.0425
diag	nacl_150C	gamma_nMeas	1.0000
diag	nacl_150C	gamma_pm_1m	0.5593
diag	nacl_150C	gamma_pm_3m	0.5736
diag	nacl_150C	phi_1m	0.9054
diag	nacl_200C	aw_1m	0.9693
diag	nacl_200C	gamma_AAD_pct	0.4317
... and 4 more reference values in the case's <i>expected</i> .			

pitzer_seawater_verify

tutorials/props/electrolyte/pitzer_seawater_verify

reduction_nacl

speciate

seawater

speciate

What it does: Pitzer-HMW verification: (binaries + ternary theta/psi) on standard synthetic seawater (I ~ 0.7). Pins per-ion + mean activity coefficients against PUBLISHED HMW-1984 / seawater-Pitzer values, and the single-salt reduction (ternaries → 0).

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	reduction_nacl	I	1
diag	reduction_nacl	SI_halite	-1.93464
diag	reduction_nacl	aw	0.966826
diag	reduction_nacl	gamma_Cl	0.657172
diag	reduction_nacl	gamma_Na	0.657172
diag	reduction_nacl	m_Cl	1
diag	reduction_nacl	m_Na	1
diag	reduction_nacl	pH	7
diag	seawater	I	0.720461
diag	seawater	SI_anhydrite	-0.970682
diag	seawater	SI_gypsum	-0.687047
diag	seawater	SI_halite	-2.48552
diag	seawater	SI_sylvite	-3.51313
diag	seawater	aw	0.981335

... and 13 more reference values in the case's *expected*.

pitzer_vs_davies_ro_concentrate

tutorials/props/electrolyte/pitzer_vs_davies_ro_concentrate

scan_davies

scalingScan

scan_pitzer

scalingScan

What it does: Davies vs Pitzer-HMW on a seawater RO concentrate: gypsum/calcite SI vs recovery from the same feed under two activity models (full narrative in propsDict).

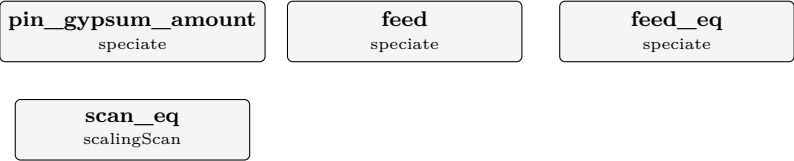
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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	scan_davies	I_r0	0.657383
diag	scan_davies	I_rmax	3.03644
diag	scan_davies	SI_anhydrite_r0	-0.887801
diag	scan_davies	SI_anhydrite_rmax	0.280924
diag	scan_davies	SI_aragonite_r0	1.26415
diag	scan_davies	SI_aragonite_rmax	2.10558
diag	scan_davies	SI_calcite_r0	1.37815
diag	scan_davies	SI_calcite_rmax	2.21958
diag	scan_davies	SI_gypsum_r0	-0.605599
diag	scan_davies	SI_gypsum_rmax	0.492667
diag	scan_davies	SI_halite_r0	-2.39559
diag	scan_davies	SI_halite_rmax	-0.454866
diag	scan_davies	SI_sylvite_r0	-3.39459
diag	scan_davies	SI_sylvite_rmax	-1.44932
...and 33 more reference values in the case's <i>expected</i> .			

precipitation_ro_brackish

tutorials/props/electrolyte/precipitation_ro_brackish



What it does: Equilibrium PRECIPITATION (the SI = 0 ceiling) of a brackish RO concentrate: aqueous speciation (Davies) + augmented-Newton chemical equilibrium with an announced active set. Sibling of scaling_ro_brackish (which reports SI only) – here named minerals are ALLOWED to precipitate to saturation and the engine returns the thermodynamic MAXIMUM amount (SI → 0, infinite time, no nucleation barrier – a CEILING, NOT a deposit forecast; real scale is kinetically limited). Story: (1) pin_gypsum_amount – a CaSO4 water with Ca and SO4 DOUBLED past gypsum saturation; the machine hands back EXACTLY the over-added excess as solid (n_gypsum). (2) feed / feed_eq – the brackish water raw and equilibrated side by side: calcite reaches its ceiling, releases H+ (pH drops, solved closure), and consuming shared Ca pushes gypsum further from saturation. (3) scan_eq – calcite + gypsum allowed across recovery to r = 0.85 with a feedFlow, giving a kg/day calcite curve; gypsum stays n = 0 throughout while its SIeq drifts BELOW the raw free-water SI once calcite consumes the shared Ca – the coupling, visible. Cross-check tool: PHREEQC EQUILIBRIUM_PHASES.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_gypsum_amount	I	0.0455084
diag	pin_gypsum_amount	SI_anhydrite	-0.299583
diag	pin_gypsum_amount	SI_gypsum	1.33618e-10
diag	pin_gypsum_amount	aw	0.99952
diag	pin_gypsum_amount	n_gypsum	0.0152707
diag	pin_gypsum_amount	pH	7
diag	feed	I	0.0266631
diag	feed	SI_anhydrite	-1.67859
diag	feed	SI_aragonite	0.126445
diag	feed	SI_calcite	0.240445
diag	feed	SI_gypsum	-1.3792
diag	feed	SI_halite	-5.35336
diag	feed	SI_sylvite	-6.39791
diag	feed	aw	0.999302
...and 36 more reference values in the case's <i>expected</i> .			

rainwater_air

tutorials/props/electrolyte/rainwater_air

air25
speciate

air10
speciate

What it does: WATER OPEN TO AIR: the generic multi-gas Henry pin. Pure water equilibrated with the atmosphere (pCO2 4.2e-4, pO2 0.2095, pN2 0.7808 atm) at 25 C and at 10 C. Each listed gas PINS its dissolved species by Henry’s law and that master’s mole balance is REPLACED by the pin – three pins at once: CO2 chains through the carbonate family (CO2aq is a computed species, the HCO3 row is swapped), while O2 and N2 are inert neutral masters pinned directly. TWO CLASSIC NUMBERS FALL OUT: (1) unpolluted RAINWATER pH ~ 5.6 – carbonic acid from 420 ppm CO2 alone, no pollutant needed; (2) DISSOLVED-OXYGEN saturation ~ 8.5 mg/L at 25 C rising to ~ 11 mg/L at 10 C – gas solubility is RETROGRADE with temperature (K_H analytic from USGS PHREEQC PHASES, public domain), which is why cold streams hold more oxygen than warm ones. The ‘totals’ entries for HCO3 / O2 / N2 are initial guesses only (announced): in an OPEN system each pinned total is a SOLVED OUTCOME of gas-liquid equilibrium.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	air25	I	2.52807e-06
diag	air25	aw	0.999986
diag	air25	gamma_CO2aq	1
diag	air25	gamma_OH	0.998138
diag	air25	m_CO2aq	1.42971e-05
diag	air25	m_OH	3.97038e-09
diag	air25	pH	5.59803
diag	air10	I	2.79349e-06
diag	air10	aw	0.999981
diag	air10	gamma_CO2aq	1
diag	air10	gamma_OH	0.998089
diag	air10	m_CO2aq	2.2591e-05
diag	air10	m_OH	1.04436e-09
diag	air10	pH	5.55469

scaling_ro_brackish

tutorials/props/electrolyte/scaling_ro_brackish

pin_kw
speciatepin_gypsum
speciatepin_neutral
speciatefeed
speciatescan
scalingScanscan_open
scalingScan

What it does: RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices $SI = \log_{10}(IAP/K)$, scanned over water recovery to $r = 0.85$ (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere (a(CO2aq) pinned by Henry at $pCO_2 = 4.2e-4$ atm, DIC a solved outcome – this CO2-rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces Kw ($m_{OH} \sim 1e-7$), a back-computed gypsum-saturated CaSO4 solution gives $SI_{gypsum} \sim 0.00$, and pure water with pH solved lands at exactly 7.000.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_kw	I	1.00037e-07
diag	pin_kw	aw	1
diag	pin_kw	m_OH	1.00037e-07
diag	pin_kw	pH	7
diag	pin_gypsum	I	0.0455076
diag	pin_gypsum	SI_anhydrite	-0.299593
diag	pin_gypsum	SI_gypsum	-1.04349e-05
diag	pin_gypsum	aw	0.99952
diag	pin_gypsum	m_CaSO4aq	0.0038941
diag	pin_gypsum	pH	7
diag	pin_neutral	I	1.00037e-07
diag	pin_neutral	aw	1
diag	pin_neutral	pH	7
diag	feed	I	0.0266631

... and 40 more reference values in the case's *expected*.

softener_brackish
tutorials/props/electrolyte/softener_brackish

raw

scalingScan

softened

exchange

softened_scaling

scalingScan

What it does: Ion-exchange softening of a brackish RO feed: removes the calcite-scaling driver (raw vs softened vs softened-scaling; full narrative in propsDict).

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	raw	I_r0	0.0266631
diag	raw	I_rmax	0.166892
diag	raw	SI_anhydrite_r0	-1.67859
diag	raw	SI_anhydrite_rmax	-0.623608
diag	raw	SI_aragonite_r0	0.126445
diag	raw	SI_aragonite_rmax	1.31552
diag	raw	SI_calcite_r0	0.240445
diag	raw	SI_calcite_rmax	1.42952
diag	raw	SI_gypsum_r0	-1.3792
diag	raw	SI_gypsum_rmax	-0.327587
diag	raw	SI_halite_r0	-5.35336
diag	raw	SI_halite_rmax	-3.82513
diag	raw	SI_sylvite_r0	-6.39791
diag	raw	SI_sylvite_rmax	-4.87602
...and 35 more reference values in the case's <i>expected</i> .			

tartaricAcid_acidulation

tutorials/props/electrolyte/tartaricAcid_acidulation

acidulation_pitzer
speciateacidulation_davies
speciategypsum_ceiling
speciatepKa2_davies
speciatepKa2_pitzer
speciate

What it does: Tartaric-acid acidulation liquor: rigorous Pitzer brine chemistry (Ca-SO4-K-Cl) + the gypsum precipitation ceiling, with tartrate honestly on the Debye-Hueckel limit (no published Pitzer params). Davies vs Pitzer shown side by side.

The story: - Choupo - *—————* Solution chemistry of the tartaric-acid acidulation liquor. Tartaric acid is a REAL weak diprotic acid: neutral H2Tart is the speciation master, dissociating $\text{H2Tart} \rightleftharpoons \text{HTart}^- \rightleftharpoons \text{Tart}^{2-}$ (pKa1 2.98, pKa2 4.34, dH from Kochergina 2006). Neutral master => an acid liquor is electroneutral at 0.0% (the H+ is from dissociation). Every gamma below is Pitzer-HMW – NOT ideal.

(1) acidulation_pitzer – the acid brine, Pitzer-HMW (I=1.8): activity coeffs + tartaric speciation; tartrate announced Debye-Hueckel-only (no Pitzer pair). (2) acidulation_davies – the SAME brine, Davies (self-flags beyond I~0.5). (3) gypsum_ceiling – gypsum to SI=0 (thermodynamic MAXIMUM, a ceiling). (4) pKa2_davies / (5) pKa2_pitzer – the SAME cut (H2Tart in 1.5 m KCl, at the thermodynamic pKa2) under the two models: at this REAL ionic strength Pitzer drives $[\text{Tart}^{2-}]/[\text{HTart}^-]$ to ~17 (gamma_Tart 0.02), NOT the ideal ratio 1 – the activity-coefficient shift of the apparent pKa, made quantitative.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	acidulation_pitzer	I	1.6414
diag	acidulation_pitzer	SI_anhydrite	1.07386
diag	acidulation_pitzer	SI_arcanite	-2.80366
diag	acidulation_pitzer	SI_gypsum	1.36221
diag	acidulation_pitzer	SI_sylvite	-3.52403
diag	acidulation_pitzer	aw	0.986675
diag	acidulation_pitzer	gamma_Ca	0.0716778
diag	acidulation_pitzer	gamma_H2Tart	1
diag	acidulation_pitzer	gamma_HTart	0.409404
diag	acidulation_pitzer	gamma_SO4	0.0662673
diag	acidulation_pitzer	gamma_Tart	0.036756
diag	acidulation_pitzer	m_Ca	0.397714
diag	acidulation_pitzer	m_H2Tart	0.0656066
diag	acidulation_pitzer	m_HTart	0.0314016

... and 52 more reference values in the case's *expected*.

estimate

The theory you need. When a compound is missing, group contribution builds it: Joback sums first-order group increments into $T_b, T_c, P_c, V_c, \Delta H_f^\circ, \Delta H_{\text{vap}}, c_p^{\text{ig}}(T)$; Lee–Kesler closes ω from (T_b, T_c, P_c) ; Ambrose–Walton gives $P^{\text{sat}}(T)$ from corresponding states; Rackett the liquid volume. The estimate is a PROPOSAL: a .dat you review against any reference data and promote by hand – the engine never estimates at runtime. Molecular structure is IDENTITY: components carry `jobackGroups{}` in their own record, and `bin/estimate <name>` is the one-command curation act. Expect few-percent errors on criticals (and the guide's worked example shows them), weaker T_b for hydrogen-bonded species, and polymers via van Krevelen / Yang on the repeat unit. *Full derivation: Theory Guide, the estimation chapter of the Props Guide.*

estimate_acetone
tutorials/props/estimate/estimate_acetone

estimate_acetone

estimateComponent

What it does: CREATE a component from its groups: Joback estimate of acetone (2 CH3 + 1 ketone), validated against NIST — see the estimate built up from groups

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_acetone	Cp298	74.97
diag	estimate_acetone	Hvap_kJmol	27.53
diag	estimate_acetone	MW	58.08
diag	estimate_acetone	Pc_bar	48.02
diag	estimate_acetone	Psat_298_bar	0.39
diag	estimate_acetone	Psat_Tb_bar	0.98
diag	estimate_acetone	Tb_K	322.11
diag	estimate_acetone	Tc_K	500.56
diag	estimate_acetone	Vc_cm3mol	209.50
diag	estimate_acetone	Vliq298_cm3mol	82.11
diag	estimate_acetone	dGf_kJmol	-154.54
diag	estimate_acetone	dHf_kJmol	-217.83
diag	estimate_acetone	omega	0.30

estimate_ethanol_benzene
tutorials/props/estimate/estimate_ethanol_benzene

estimate_ethanol

estimateComponent

estimate_benzene

estimateComponent

What it does: Joback estimates: ethanol (H-bonding → Tb weak, the honest limit) and benzene (aromatic groups → good) — see where group contribution wins and loses

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_ethanol	Cp298	64.62
diag	estimate_ethanol	Hvap_kJmol	36.73
diag	estimate_ethanol	MW	46.07
diag	estimate_ethanol	Pc_bar	57.57
diag	estimate_ethanol	Psat_298_bar	0.16
diag	estimate_ethanol	Psat_Tb_bar	0.97
diag	estimate_ethanol	Tb_K	337.54
diag	estimate_ethanol	Tc_K	499.41
diag	estimate_ethanol	Vc_cm3mol	166.50
diag	estimate_ethanol	Vliq298_cm3mol	57.70
diag	estimate_ethanol	dGf_kJmol	-170.86
diag	estimate_ethanol	dHf_kJmol	-236.84
diag	estimate_ethanol	omega	0.57
diag	estimate_benzene	Cp298	81.82
...and 12 more reference values in the case's <i>expected</i> .			

estimate_petroleum_cut
tutorials/props/estimate/estimate_petroleum_cut

estimate_petroleum_cut
estimateComponent

What it does: ESTIMATE a petroleum PSEUDO-COMPONENT by name from (Tb, SG) – Riazi-Daubert; see the SCREAMING ESTIMATE provenance and that the lump FLASHES

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_petroleum_cut	Cp298	248.06
diag	estimate_petroleum_cut	MW	151.51
diag	estimate_petroleum_cut	Pc_bar	19.78
diag	estimate_petroleum_cut	Psat_Tb_bar	0.97
diag	estimate_petroleum_cut	SG	0.73
diag	estimate_petroleum_cut	Tb_K	447.30
diag	estimate_petroleum_cut	Tc_K	617.80
diag	estimate_petroleum_cut	Vc_cm3mol	627.46
diag	estimate_petroleum_cut	Vliq298_cm3mol	203.43
diag	estimate_petroleum_cut	omega	0.48

userEstimate01_linear_tb
tutorials/props/estimate/userEstimate01_linear_tb

tb_acetone
linearTbEstimate

What it does: User-code property estimation: a student’s OWN linear group-contribution Tb estimator (linearTbEstimate), compiled from code/ and linked into choupoprops.

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	tb_acetone	Tb_K	322.11

vanKrevelen01_polystyrene_density

tutorials/props/estimate/vanKrevelen01_polystyrene_density

vanKrevelen01_polystyrene_density

estimateComponent

What it does: ESTIMATE the density of polystyrene by Van Krevelen group contribution on the repeat unit – glass-box additive sum, Vw from Bondi 1964 (licence-clean via UNIFAC R)

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	vanKrevelen01_polystyrene_density	M0_g_per_mol	104
diag	vanKrevelen01_polystyrene_density	V_cm3_per_mol	101
diag	vanKrevelen01_polystyrene_density	Vw_cm3_per_mol	62.9
diag	vanKrevelen01_polystyrene_density	density_g_cm3	1.04
diag	vanKrevelen01_polystyrene_density	packing_k	1.6

yang2020_Tg_polystyrene

tutorials/props/estimate/yang2020_Tg_polystyrene

yang2020_Tg_polystyrene

estimateComponent

What it does: ESTIMATE the glass-transition Tg of polystyrene by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH2-CH(phenyl)- gives Tg(inf)=444 K, an HONEST +19% over-estimate of the measured ~373 K (a teaching point about group-contribution limits)

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	yang2020_Tg_polystyrene	MO_g_per_mol	104

yang2020_Tg_pvc

tutorials/props/estimate/yang2020_Tg_pvc

yang2020_Tg_pvc
 estimateComponent

What it does: ESTIMATE the glass-transition Tg of poly(vinyl chloride) (PVC) by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH₂-CHCl- gives Tg(inf)=351 K, EXACT vs the paper (Fig 6)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	yang2020_Tg_pvc	M0_g_per_mol	62.5
diag	yang2020_Tg_pvc	Tg_K	351
diag	yang2020_Tg_pvc	YgSum_1e3gKmol	21.937

gibbs

The theory you need. Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A} \mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as λ_E , the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~100–250 K when they are not (dissociation is endothermic). Above ~1500 K the catalogue's polynomial c_p fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

Equilibrium maps. Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes T and P in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach` ΔT evaluates the *reaction* equilibrium at $T + \Delta T$ while enthalpy and the energy balance stay at the physical T – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium NH₃ mole fraction crosses 30%; report T to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in T at fixed P . *Full derivation:* Theory Guide, the Gibbs-equilibrium chapter, *Equilibrium maps*.

nh3_equilibrium_map
tutorials/props/gibbs/nh3_equilibrium_map

haber_map
gibbsMap

What it does: Ammonia equilibrium map (T x P): why Haber runs at 400-500 C and 150-300 bar

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	haber_map	metric_T400C_P10atm	0.0409234
diag	haber_map	metric_T400C_P50atm	0.157785
diag	haber_map	metric_T450C_P10atm	0.0221983
diag	haber_map	metric_T500C_P100atm	0.104969
diag	haber_map	n_cells	625
diag	haber_map	n_unconverged	0

heat

The theory you need. Heat exchange couples streams without mixing them: $Q = U A \Delta T_{lm}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

Exchanger DESIGN vs rating. A shell-and-tube poses two inverse questions: RATE it (geometry given → find U , area, ΔP , duty – **model geometry**;) or DESIGN it (duty given → size the geometry – **model design**;) . U is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` → `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Sheat-exchanger design.*

htbench01_correlations

tutorials/props/heat/htbench01_correlations

bench
 heatTransferBench

What it does: Heat-transfer correlation bench – each convective correlation (Dittus-Boelter, Gnielinski, Kern) verified against its published Nusselt anchor. Dittus-Boelter and Gnielinski are pinned to Incropera & DeWitt 6th ed. Example 8.4; Kern’s closed j_H form is self-consistency-checked (its original pin is a chart). AAD tolerance 5% honours textbook rounding + the correlations’ own scatter.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	bench	Nu_choupo_DittusBoelter	200.475
diag	bench	Nu_choupo_Gnielinski	223.734
diag	bench	Nu_choupo_Kern	97.5647
diag	bench	Nu_published_DittusBoelter	205
diag	bench	Nu_published_Gnielinski	224
diag	bench	Nu_published_Kern	97.5647
diag	bench	allPass	1
diag	bench	dev_DittusBoelter	0.0220718
diag	bench	dev_Gnielinski	0.00118956
diag	bench	dev_Kern	0

molecular

The theory you need. Molecular activity models on display: NRTL/Wilson/UNIQUAC correlate binary γ ’s from pair parameters; UNIFAC PREDICTS them from group counts. Expect UNIFAC within a few percent of fitted models for families it was trained on – and to be honestly worse outside them. *Full derivation: Theory Guide, the activity-model chapter.*

cosmoSAC01_water_ethanol

tutorials/props/molecular/cosmoSAC01_water_ethanol

pure_water
propertyScan1D

bin_wet
propertyScan1D

ternary
propertyScan1D

water_hexane
propertyScan1D

What it does: COSMO-SAC 2002 (NIST benchmark, VT-2005 profiles): pure→lngamma=0, binary/ternary multicomponent gammas, no NaN.

The story: - Choupo -*—————* COSMO-SAC 2002 validation ops. T = 298.15 K. 1. pure water → gamma ~ 1 (lnGamma = 0 by construction) 2. water/ethanol binary → finite, no NaN 3. water/ethanol/acetone ternary → multicomponent 4. water/nHexane → strongly non-ideal (gamma_hexane » 1)

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pure_water	n_failures	0
diag	pure_water	n_points	2
diag	bin_wet	n_failures	0
diag	bin_wet	n_points	2
diag	ternary	n_failures	0
diag	ternary	n_points	2
diag	water_hexane	n_failures	0
diag	water_hexane	n_points	2

eval_nrtl_catalogue

tutorials/props/molecular/eval_nrtl_catalogue

catalogue_as_assembled fitParameters
--

What it does: EVALUATE mode: score the CATALOGUE NRTL pair against your own data – the scale before any fit

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	catalogue_as_assembled	aad	2.0400
diag	catalogue_as_assembled	chi2	60.1945
diag	catalogue_as_assembled	max_abs_resid	3.4379
diag	catalogue_as_assembled	max_resid_x1	0.5000
diag	catalogue_as_assembled	mode_evaluate	1.0000
diag	catalogue_as_assembled	n_data	11.0000
diag	catalogue_as_assembled	n_feasible	11.0000
diag	catalogue_as_assembled	partial	0.0000
diag	catalogue_as_assembled	penalised_points	0.0000
diag	catalogue_as_assembled	rms	2.3393
diag	catalogue_as_assembled	sse	60.1945
diag	catalogue_as_assembled	validity_declared	1.0000
diag	catalogue_as_assembled	n_outside_validity	0.0000

eval_nrtl_pinned

tutorials/props/molecular/eval_nrtl_pinned

scaffold_as_is

fitParameters

rough_paper_values

fitParameters

scaffold_again

fitParameters

What it does: EVALUATE a pinned what-if: a pinned run scores the package but mutates NOTHING (the leakage sentinel)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	scaffold_as_is	aad	2.0400
diag	scaffold_as_is	chi2	60.1945
diag	scaffold_as_is	max_abs_resid	3.4379
diag	scaffold_as_is	max_resid_x1	0.5000
diag	scaffold_as_is	mode_evaluate	1.0000
diag	scaffold_as_is	n_data	11.0000
diag	scaffold_as_is	n_feasible	11.0000
diag	scaffold_as_is	partial	0.0000
diag	scaffold_as_is	penalised_points	0.0000
diag	scaffold_as_is	rms	2.3393
diag	scaffold_as_is	sse	60.1945
diag	rough_paper_values	aad	1.5942
diag	rough_paper_values	chi2	61.7621
diag	rough_paper_values	max_abs_resid	5.5866
... and 19 more reference values in the case's <i>expected</i> .			

nrtl01_ethanol_water_package
tutorials/props/molecular/nrtl01_ethanol_water_package

g_nrtl
activityCoefficients

What it does: ethanol-water NRTL via the SAME property-package builder (molecular U4)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	g_nrtl	gamma_ethanol	1.26245
diag	g_nrtl	gamma_water	1.4349

pcsaft01_pure_nhexane

tutorials/props/molecular/pcsaft01_pure_nhexane

liquid_298K_1bar
 propertyPoint

What it does: PC-SAFT pure n-hexane: Z + liquid molar volume at 298 K, 1 bar

The story: - Choupo - * PC-SAFT (Gross & Sadowski 2001, non-associating) – pure n-hexane at 298.15 K, 1 bar: BOTH density roots of the SAME a_res, labelled.

The stable phase here is LIQUID ($P_{\text{sat}}(298 \text{ K}) \sim 0.2 \text{ bar} < 1 \text{ bar}$), and the point publishes one coherent VAPOUR-root family (Z, v_molar, H_R, S_R – the EoS-interface contract shared with the cubics) PLUS the explicitly labelled liquid root:

$v_{\text{molar_liq}} = 1.3265 \times 10^{-4} \text{ m}^3/\text{mol} \rightarrow \rho_{\text{liq}} = \text{MW}/v = 649.6 \text{ kg/m}^3$ vs the experimental 654.8 kg/m³ at 25 degC (CRC Handbook of Chemistry and Physics, d₂₅ = 0.6548 g/mL): -0.8%, the magnitude of the liquid-density AAD Gross & Sadowski 2001 (Table 2) report across the n-alkane series – PREDICTED here from (m, sigma, epsilon/k) alone. $Z_{\text{liquid}} = P v_{\text{liq}} / RT = 0.00535$ (a liquid: $Pv \ll RT$) Z (vapour) = 0.9425 (the metastable vapour root)

What the student sees (verbosity 3): the construction-time verify() gate – ideal-gas limit, $\sum x \ln \phi_i = g_{\text{res}}/RT$ (Gibbs-Duhem integral), and the lnphi numeric-derivative STEP-CONVERGENCE sweep ($h_{\text{rel}} 1\text{e-}7..1\text{e-}5$, plateau $\sim 1\text{e-}8$) – the declared price of the numeric μ_{res} derivative.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	liquid_298K_1bar	Cp_ig	1.43106e+02
diag	liquid_298K_1bar	H_R	-3.91460e+02
diag	liquid_298K_1bar	H_ig	-1.67200e+05
diag	liquid_298K_1bar	H_real	-1.67591e+05
diag	liquid_298K_1bar	P	1.00000e+05
diag	liquid_298K_1bar	S_R	-3.54309e-01
diag	liquid_298K_1bar	S_ig	3.88510e+02
diag	liquid_298K_1bar	S_real	3.88156e+02
diag	liquid_298K_1bar	T	2.98150e+02
diag	liquid_298K_1bar	Z	9.42530e-01
diag	liquid_298K_1bar	Z_liquid	5.35107e-03
diag	liquid_298K_1bar	gamma	1.06168e+00
diag	liquid_298K_1bar	v_molar	2.33649e-02
diag	liquid_298K_1bar	v_molar_liq	1.32651e-04

pcsaft02_states_crosscheck

tutorials/props/molecular/pcsaft02_states_crosscheck

CH4_vapour_pcsaft
propertyPointCH4_vapour_srk
propertyPointpropane_liquid
propertyPointnButane_liquid
propertyPoint**What it does:** PC-SAFT across states: CH4 Z vs SRK cross-check + propane/n-butane liquid density vs handbook**The story:** - Choupo -*—————* PC-SAFT across states + an independent-model cross-check (all goldened).

1. CH4_vapour_pcsaft CH4 at 300 K / 50 bar, PC-SAFT vapour root. 2. CH4_vapour_srk the SAME state through SRK (per-op thermo{} override): two independently-published models, fitted to different data, agreeing on Z for a simple near-spherical vapour is real evidence – the student compares the two printed numbers. 3. propane_liquid 298.15 K / 15 bar (above Psat ~ 9.5 bar, so the stable phase is liquid): v_molar_liq → rho_liq, vs the 0.493 g/mL saturated-liquid handbook value at 25 degC (CRC). PREDICTED, not fitted. 4. nButane_liquid 298.15 K / 5 bar (Psat ~ 2.4 bar): rho_liq vs 0.573 g/mL at 25 degC (CRC).

Measured here (goldened): CH4 Z 0.9121 (PC-SAFT) vs 0.9243 (SRK) – 1.3 % apart, two independent models; rho_liq(propane) = 494.2 kg/m3 (+0.24 % vs 0.493 g/mL), rho_liq(nButane) = 571.4 kg/m3 (-0.28 % vs 0.573 g/mL) – inside the 1-2 % liquid-density AAD Gross & Sadowski 2001 (Table 2) report for the n-alkane series. No kij anywhere: pure states.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	CH4_vapour_pcsaft	Cp_ig	3.56607e+01
diag	CH4_vapour_pcsaft	H_R	-8.28601e+02
diag	CH4_vapour_pcsaft	H_ig	-7.44541e+04
diag	CH4_vapour_pcsaft	H_real	-7.52827e+04
diag	CH4_vapour_pcsaft	P	5.00000e+06
diag	CH4_vapour_pcsaft	S_R	-1.24811e+00
diag	CH4_vapour_pcsaft	S_ig	1.53964e+02
diag	CH4_vapour_pcsaft	S_real	1.52716e+02
diag	CH4_vapour_pcsaft	T	3.00000e+02
diag	CH4_vapour_pcsaft	Z	9.12141e-01
diag	CH4_vapour_pcsaft	gamma	1.30404e+00
diag	CH4_vapour_pcsaft	v_molar	4.55038e-04
diag	CH4_vapour_srk	Cp_ig	3.56607e+01
diag	CH4_vapour_srk	H_R	-8.31158e+02

...and 38 more reference values in the case's *expected*.**scan**

The theory you need. Property scans sweep a model across T , P or composition and plot what the thermodynamics DOES: $P^{\text{sat}}(T)$ curves, $\gamma_{\pm}(m)$, VLE envelopes, ternary diagrams, phase boundaries. Nothing is fitted – these ops answer “how does the selected model behave?”, the SEE step of the props loop (CREATE → SEE → TEST → FIT → PROMOTE). Expect model disagreements to be VISIBLE (overlay two methods and look), and every curve exportable as CSV next to the case. *Full derivation: Theory Guide, the Props Guide.*

binary01_lle_water_nbutanol

tutorials/props/scan/binary01_lle_water_nbutanol

lle

propertyScanBinary

What it does: Binary LLE g_{mix} + common-tangent (water/1-butanol) at 298.15 K via propertyScanBinary, cited Winkelman 2009 UNIQUAC

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | // O perations | Website: <https://choupo.org> | | \{\} \{\} | // U nits | Licence: GPL-3.0-or-later | | \{\} | // P roperties | Tutorial case file – part of Choupo | | O ptimization | |

*Water + 1-butanol (n-butanol) is *the* classic partially-miscible binary. At 25 °C it splits into a water-rich liquid (~2 mol % butanol) and a butanol-rich liquid (~50 mol % butanol). A $g_{\text{mix}}(x_{\text{water}})$ scan should therefore show a clearly non-convex region: two binodal points sharing a common tangent, which a student reads the split straight off.*

and binaryLle.csv is a $g_{\text{mix}}(x_1)$ curve that goes non-convex, plus the two binodal points the liquid-liquid flash locates on it:

The agreement is essentially exact: the cited UNIQUAC fit reproduces the published mutual solubilities.

- J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, * "Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"*, *Fluid Phase Equilibria* 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013.

- Interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation $A_{ij}(T) = a + b \cdot T + c \cdot T^2$ (Eq. 10), $\tau_{ij} = \exp(-A_{ij}/T)$ (Eq. 7): - $A_{1,3}$ (butanol→water): $a = 155.31$, $b = 1.0822$, $10^3 \cdot c = -43.711$ - $A_{3,1}$ (water→butanol): $a = -579.36$, $b = 2.7517$, $10^3 \cdot c = -6.7700$ - Structural r , q — their Table 1 (p.72): 1-butanol $r = 3.9243$, $q = 3.668$; water $r = 0.9200$, $q = 1.400$.

The same numbers live, fully provenanced, in data/standards/parameters/UNIQUAC/nButanol-water.dat. The engine evaluates the full $a + b \cdot T + c \cdot T^2$ temperature dependence (the quadratic $c \cdot T^2$ term is essential — at 320 K it is ~450 K, comparable to a).

phase01_water_full

tutorials/props/scan/phase01_water_full

pd

purePhaseDiagram

What it does: Complete P-T phase diagram of water (purePhaseDiagram, solid+liquid+vapour)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

ternary01_audit_vlle

tutorials/props/scan/ternary01_audit_vlle

lle
propertyScanTernary

What it does: Ternary VLLE phase map (compA/compB/compC) at 350 K, 1 bar via propertyScanTernary

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

ternary02_bubbleT_benzene_toluene_water

tutorials/props/scan/ternary02_bubbleT_benzene_toluene_water

bt
propertyScanTernary

What it does: Ternary T_bubble surface (benzene/toluene/water) at 1 bar via propertyScanTernary mode bubbleT

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

ternary03_lle_water_ethanol_benzene

tutorials/props/scan/ternary03_lle_water_ethanol_benzene

lle
propertyScanTernary

What it does: Ternary LLE solubility (water/ethanol/benzene) at 298 K via UNIFAC + propertyScanTernary mode lle

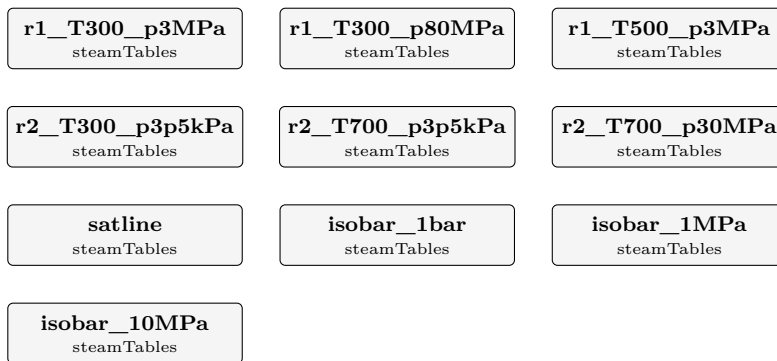
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

steam

The theory you need. Water/steam properties from IAPWS-IF97: regions (liquid, vapour, supercritical), $T_{\text{sat}}(P)$, and $h(T, P)/s(T, P)$ from the dimensionless Gibbs functions. Power cycles are enthalpy bookkeeping around them: pump work $\approx v \Delta P$, turbine work from isentropic expansion $\times$ efficiency, cycle efficiency $\eta = W_{\text{net}}/Q_{\text{in}}$ bounded by Carnot. *Full derivation: Theory Guide, the steam/power chapter.*

steam01_if97_verification

tutorials/props/steam/steam01_if97_verification



What it does: IAPWS-IF97 steam tables (R7-97(2012)) – the verification case. The six computer-program verification states of the release (Table 5 region 1, Table 15 region 2) evaluated as points, the saturation line 0.01-350 degC, and three isobars whose Tsat crossings pin the release’s Table 36 backward values. The published numbers ARE the golden master (reltol 1e-8).

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	r1_T300_p3MPa	region	1
diag	r1_T300_p3MPa	v	0.100215168e-2
diag	r1_T300_p3MPa	h	0.115331273e3
diag	r1_T300_p3MPa	u	0.112324818e3
diag	r1_T300_p3MPa	s	0.392294792
diag	r1_T300_p3MPa	cp	0.417301218e1
diag	r1_T300_p3MPa	w	0.150773921e4
diag	r1_T300_p3MPa	psat_MPa	0.353658941e-2
diag	r1_T300_p80MPa	region	1
diag	r1_T300_p80MPa	v	0.971180894e-3
diag	r1_T300_p80MPa	h	0.184142828e3
diag	r1_T300_p80MPa	u	0.106448356e3
diag	r1_T300_p80MPa	s	0.368563852
diag	r1_T300_p80MPa	cp	0.401008987e1

...and 39 more reference values in the case’s *expected*.

steam02_water_transport

tutorials/props/steam/steam02_water_transport

pin_25C
propertyPoint

pin_50C
propertyPoint

pin_75C
propertyPoint

pin_100C
propertyPoint

transport_scan
propertyScan1D

What it does: IAPWS water transport properties – viscosity (R12-08), thermal conductivity (R15-11) and surface tension (R1-76). Water flagged as a pure fluid (method IF97) routes the transport accessors to the matching IAPWS releases: the density comes from the IF97 thermodynamic kernel at the saturation state, fed to the (rho,T) transport kernels; surface tension is the R1-76 saturation-line form. A coarse 4-point scan (25/50/75/100 degC) pinned against the published IAPWS values, plus a fine sweep for the plot.

The story: - Choupo - * | \{\} C hemicals | Choupo: open-source, glass-box simulation | | \{\}\{\} H eat-transfer | Version: Choupo-2607 | | \{\}\{\}\{\} O perations | Website: <https://choupo.org> | | \{\}\{\} U nits | Licence: GPL-3.0-or-later | | \{\} P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_25C	surface_tension	0.0720
diag	pin_25C	thermal_conductivity	0.6065
diag	pin_25C	viscosity	0.0009
diag	pin_50C	surface_tension	0.0679
diag	pin_50C	thermal_conductivity	0.6406
diag	pin_75C	surface_tension	0.0636
diag	pin_75C	thermal_conductivity	0.6635
diag	pin_100C	surface_tension	0.0589
diag	pin_100C	thermal_conductivity	0.6772
diag	transport_scan	n_points	46
diag	transport_scan	n_failures	0

thermoTest

h_surface_identity

tutorials/props/thermoTest/h_surface_identity

h_identity
 hConsistency

What it does: The state identities of the canonical enthalpy surface: $dh/dT == C_{p_phase}$, the 298 K vaporisation anchor, and Kirchhoff on H_{vap_state} – pinned to round-off

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	h_identity	benzene_anchor298_rel	0.000e+00
diag	h_identity	benzene_dhgas_vs_cpig_rel	7.289e-11
diag	h_identity	benzene_dhliq_vs_cpliq_rel	4.729e-13
diag	h_identity	benzene_kirchhoff_rel	2.058e-10
diag	h_identity	ethanol_anchor298_rel	1.694e-16
diag	h_identity	ethanol_dhgas_vs_cpig_rel	3.826e-11
diag	h_identity	ethanol_dhliq_vs_cpliq_rel	1.907e-11
diag	h_identity	ethanol_kirchhoff_rel	1.287e-10
diag	h_identity	toluene_anchor298_rel	0.000e+00
diag	h_identity	toluene_dhgas_vs_cpig_rel	6.085e-11
diag	h_identity	toluene_dhliq_vs_cpliq_rel	1.854e-13
diag	h_identity	toluene_kirchhoff_rel	2.214e-10
diag	h_identity	water_anchor298_rel	6.529e-16
diag	h_identity	water_dhgas_vs_cpig_rel	3.627e-11

... and 3 more reference values in the case's *expected*.

model4_speciation_scaling

tutorials/props/thermoTest/model4_speciation_scaling

brine

speciate

concentrate

scalingScan

What it does: RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices $SI = \log_{10}(IAP/K)$, scanned over water recovery to $r = 0.85$ (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere ($a(\text{CO}_2\text{aq})$ pinned by Henry at $p\text{CO}_2 = 4.2\text{e-}4$ atm, DIC a solved outcome – this CO_2 -rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces K_w ($m_{\text{OH}} \sim 1\text{e-}7$), a back-computed gypsum-saturated CaSO_4 solution gives $SI_{\text{gypsum}} \sim 0.00$, and pure water with pH solved lands at exactly 7.000.

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What to expect (golden-locked):

kind	unit/stream	key	expected
diag	brine	I	3.26782
diag	brine	SI_anhydrite	-0.22134
diag	brine	SI_gypsum	-0.0221674
diag	brine	SI_halite	-0.910511
diag	brine	SI_mirabilite	-2.13799
diag	brine	SI_thenardite	-2.98985
diag	brine	aw	0.890402
diag	brine	gamma_CaSO4aq	1
diag	brine	m_CaSO4aq	0.00466294
diag	brine	pH	7
diag	concentrate	I_r0	3.26782
diag	concentrate	I_rmax	15.6285
diag	concentrate	SI_anhydrite_r0	-0.22134
diag	concentrate	SI_anhydrite_rmax	1.39807
... and 13 more reference values in the case's expected.			

Category III: Batch and time-dependent (choupoBatch)

$d\mathbf{Y}/dt = f(\mathbf{Y})$ integrated in real time, with recipes for discrete events; trajectories, not steady states.

adsorber

The theory you need. Adsorption in a vessel or a bed: the isotherm $q^*(T, p)$ (Langmuir $q = q_{\max}bp/(1+bp)$, van't Hoff $b(T)$) is the curated PAIR datum; the LDF rate $dq/dt = k(q^* - q)$ carries the EQUIPMENT datum k (this particle, this bed – it lives in the case, never the catalogue). Work the cases as a staircase. **batch09:** a closed vessel relaxes to the inventory equilibrium – one algebraic root, pinned independently of the run. **batch10:** two species compete for the same sites (extended Langmuir) – the selectivity is an equilibrium statement, and the model's Gibbs–Duhem inconsistency for unequal $q_{\max}$ is declared, not hidden. **batch11:** in the Henry limit the whole vessel is LINEAR and the integrator must match $q_{\infty}(1 - e^{-k_{\text{eff}}t})$ to 10^{-8} – the analytic gate. **batch12:** declaring the headspace pressure AND the inventory over-specifies a closed ideal-gas vessel – the contradiction FATALs; P is a result. **batch13:** the fixed bed – the stoichiometric time $t_{\text{st}} = (L/u)(\varepsilon + \rho_b q^*/c_{\text{in}})$ is announced BEFORE integration, breakthrough spreads around it, and the constant- u closure's fabricated carrier is printed, pinned and discounted: the DECLARED model residual is itself the lesson. **batch14/batch15** are the numerical hygiene: pure transport vs the analytic advection–dispersion solution, and the mesh study that earns $N = 100$. *Full derivation: Theory Guide, the adsorption chapter.*

batch09_adsorber_co2

tutorials/batch/adsorber/batch09_adsorber_co2

adsorber
 batchAdsorber

What it does: Closed isothermal CO₂ uptake on zeolite 13X (LDF, solid film): the vessel relaxes to the inventory equilibrium $p_{\text{inf}} = 7229.13$ Pa, $q_{\text{inf}} = 1.9708$ mol/kg

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	7229.128209
kpi	adsorber	Q_ads_isosteric_kJ	-88.68771114
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	2.916197467e-05
kpi	adsorber	q_CO2_final	1.970838025
kpi	adsorber	t_end	600
kpi	adsorber	totalMoles_final	2.916197467e-05
kpi	adsorber	uptake_CO2_mol	1.970838025
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0

... and 9 more reference values in the case's *expected*.

batch10_adsorber_co2_n2

tutorials/batch/adsorber/batch10_adsorber_co2_n2

adsorber
 batchAdsorber

What it does: Closed isothermal competitive CO₂/N₂ uptake on zeolite 13X: extended-Langmuir site competition drives the equilibrium selectivity $S = 71.43$

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	181876.7854
kpi	adsorber	Q_ads_isosteric_kJ	-36.1181845604
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	6.500995276e-06
kpi	adsorber	n_N2_final	7.271816914e-04
kpi	adsorber	q_CO2_final	0.4934990047
kpi	adsorber	q_N2_final	0.7728183086
kpi	adsorber	t_end	600
kpi	adsorber	totalMoles_final	0.000733682686703
kpi	adsorber	uptake_CO2_mol	0.4934990047
kpi	adsorber	uptake_N2_mol	0.7728183086
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_N_closure_rel	0

... and 13 more reference values in the case's *expected*.

batch11_adsorber_henry

tutorials/batch/adsorber/batch11_adsorber_henry

adsorber
 batchAdsorber

What it does: Closed isothermal Henry-limit uptake (linear isotherm): the coupled gas+solid system is LINEAR and the run must match $q(t) = q_{\text{inf}}(1 - \exp(-k_{\text{eff}} t))$ to $1e-8$

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	124785.4636
kpi	adsorber	Q_ads_isosteric_kJ	-9.93242221329
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	5.033788893e-04
kpi	adsorber	q_CO2_final	0.4966211107
kpi	adsorber	t_end	60
kpi	adsorber	totalMoles_final	5.033788893e-04
kpi	adsorber	uptake_CO2_mol	0.4966211107
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0

...and 9 more reference values in the case's *expected*.

batch12_adsorber_refused_P

tutorials/batch/adsorber/batch12_adsorber_refused_P

adsorber
 batchAdsorber

What it does: REFUSAL lesson: declared initial.P (10 bar) contradicts nRT/V (4.958 bar) – the honest cross-check FATALs; P of a closed ideal-gas headspace is a result, not an input

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batch13_breakthrough_co2

tutorials/batch/adsorber/batch13_breakthrough_co2

bed
 fixedBedAdsorber

What it does: CO₂ 15%/He breakthrough on zeolite 13X – 1-D isothermal fixed bed (A3): $t_{st} = 3040.5$ s announced pre-run, $t_b(5\%) = 2890.7$ s, ledger closure at machine level

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_CO2	0.999999989455
kpi	bed	carrier_fabricated_mol	9.19148935961
kpi	bed	integral_anchor_CO2	3040.51245606
kpi	bed	mass_closure_CO2	8.94231742202e-15
kpi	bed	n_CO2_final	1.21079552014e-05
kpi	bed	n_He_final	6.86117461711e-05
kpi	bed	physical_mass_closure_rel	0.108446875314
kpi	bed	qbar_CO2_final	2.87234042488
kpi	bed	retention_factor_CO2	304.051250948
kpi	bed	t_50_CO2	3036.44839047
kpi	bed	t_breakthrough_5pct_CO2	2890.66919752

... and 20 more reference values in the case's *expected*.

batch14_transport_only

tutorials/batch/adsorber/batch14_transport_only

bed

fixedBedAdsorber

What it does: Pure transport ($k = 0$) on the batch13 bed, $N = 200$: t_{50} vs the Ogata-Banks analytic ADE solution (gate G1) under fixed RK4 with the Gershgorin bound printed term by term

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_C02	0.999999999893
kpi	bed	carrier_fabricated_mol	0
kpi	bed	integral_anchor_C02	3.99999999999
kpi	bed	mass_closure_C02	4.09039795713e-14
kpi	bed	n_C02_final	1.21079552058e-05
kpi	bed	n_He_final	6.86117461666e-05
kpi	bed	physical_mass_closure_rel	0
kpi	bed	qbar_C02_final	0
kpi	bed	t_50_C02	3.97561121931
kpi	bed	t_breakthrough_5pct_C02	3.29598772214
kpi	bed	t_breakthrough_95pct_C02	4.78720615773
...and 18 more reference values in the case's expected.			

batch15_mesh_study

tutorials/batch/adsorber/batch15_mesh_study

bed
 fixedBedAdsorber

What it does: Mesh-study member $N = 25$ of the batch13 breakthrough: $t_b(5\%)$ converges monotonically $2732 \rightarrow 2839 \rightarrow 2891 \rightarrow 2916$ s with observed order 1.03-1.05; full table in reports/meshStudy.md

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_C02	0.999973635802
kpi	bed	carrier_fabricated_mol	9.19148014522
kpi	bed	integral_anchor_C02	3040.50945345
kpi	bed	mass_closure_C02	7.40535661511e-15
kpi	bed	n_C02_final	1.21079316034e-05
kpi	bed	n_He_final	6.86117697691e-05
kpi	bed	physical_mass_closure_rel	0.108446766597
kpi	bed	qbar_C02_final	2.87233754538
kpi	bed	retention_factor_C02	304.051250948
kpi	bed	t_50_C02	3041.40641271
kpi	bed	t_breakthrough_5pct_C02	2732.02788103

... and 20 more reference values in the case's *expected*.

batch16_ergun_profile
tutorials/batch/adsorber/batch16_ergun_profile

bed

fixedBedAdsorber

What it does: A4 F1: pure-He Ergun profile against the closed-form P-squared solution

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_first_cell_Pa	99999.4670134
kpi	bed	P_mid_cell_Pa	99946.1540012
kpi	bed	P_last_cell_Pa	99893.8796441
kpi	bed	P_out_boundary_Pa	99893.3459257
kpi	bed	carrier_fabricated_mol	0
kpi	bed	physical_mass_closure_rel	0
kpi	campaign	mass_closure_rel	0

batch17_dilute_ergun_limit
tutorials/batch/adsorber/batch17_dilute_ergun_limit

legacy
fixedBedAdsorber

ergun
fixedBedAdsorber

What it does: A4 F2: 400 ppm uptake, Ergun must approach the A3 constant-velocity limit

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	legacy	t_breakthrough_5pct_CO2	373.706923746
kpi	ergun	t_breakthrough_5pct_CO2	373.819840914
kpi	legacy	t_50_CO2	691.597682330
kpi	ergun	t_50_CO2	691.598159617
kpi	ergun	carrier_fabricated_mol	0
kpi	ergun	physical_mass_closure_rel	0
kpi	ergun	mass_closure_CO2	0

batch18_ergun_conservation

tutorials/batch/adsorber/batch18_ergun_conservation

bed

fixedBedAdsorber

What it does: A4 F3: 15 percent CO2 with explicit carrier continuity and Ergun velocity

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	t_breakthrough_5pct_CO2	2767.86955912
kpi	bed	t_50_CO2	3030.27628270
kpi	bed	t_breakthrough_95pct_CO2	3282.90182463
kpi	bed	carrier_fabricated_mol	0
kpi	bed	physical_mass_closure_rel	0
kpi	bed	mass_closure_CO2	0
kpi	campaign	mass_closure_rel	0
kpi	campaign	element_worst_closure_rel	0

combustion

The theory you need. Radical chain chemistry: initiation creates the first radicals (slow – $\text{H}_2 + \text{M}$, $E_a \sim 105 \text{ kcal}$), branching multiplies them ($\text{H} + \text{O}_2 \rightarrow \text{O} + \text{OH}$: one in, two out), termination removes them. Ignition is the race between branching and termination; the induction time is set by the SLOW build-up, then heat release runs away. Rate constants $k = AT^b e^{-E_a/RT}$ come VERBATIM from cited mechanisms (GRI-Mech 3.0, Ó Conaire 2004, Glarborg 2018) in their native units – the engine converts and announces. Reverse rates from equilibrium: the kinetic end-state must land on the Gibbs equilibrium (the two-engine handshake the NOx tutorial pins at 98.9%). Expect stiffness ratios of $10^6\text{--}10^8$ (use Rosenbrock23), ignition delays in μs validated against shock-tube data, and thermal NO forming over SECONDS – the timescale gap behind quench-based NOx control. *Full derivation: Theory Guide, the combustion/kinetics chapter.*

ignition01_h2o2

tutorials/batch/combustion/ignition01_h2o2

bomb
 batchReactor

What it does: Adiabatic constant-V H₂/O₂ ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); Rosenbrock23 stiff solver; reports ignition delay

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

A closed rigid vessel holds a stoichiometric H₂/O₂ charge diluted in N₂ at 1100 K and ~5 bar. A tiny H-radical seed starts chain branching; after a short induction the radical pool explodes and the mixture ignites — a sharp temperature jump from 1100 K to ~2440 K.

and plot trajectory.csv (T and the radicals H/O/OH vs time). You will see:

The ignition delay τ_{ign} (the time of maximum dT/dt) is ≈ 0.9 ms here.

*The radical lifetimes are $\sim 10^{-12}$ – 10^{-13} s while the induction runs over $\sim 10^{-3}$ s — a timescale separation of $\sim 10^9$. That is stiffness. An explicit method (RK4) is stability-limited to a step of order the *fastest* scale, so it cannot take the $\sim 10^{-3}$ s steps this case uses. The solver here is Rosenbrock23, an L-stable linearly-implicit method that steps by *accuracy*, not stability; at verbosity 3 it prints its stiffness verdict and step counts (the no-silent- crutch credo).*

See it fail on purpose — the sibling case runs the same problem with explicit RK4 at the same step:

*It blows up to NaN within a few steps. (That case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail.) Switch the integrator back to Rosenbrock23 and the same step converges. *See, then decide*, applied to numerics.*

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bomb	P_final	500000
kpi	bomb	T_final	2641.48171412
kpi	bomb	n_H2O2_final	2.59406092875e-11
kpi	bomb	n_H2_final	2.44308154218e-06
kpi	bomb	n_HO2_final	1.97492765295e-10
kpi	bomb	n_H_final	1.4513832323e-06
kpi	bomb	n_N2_final	2.857145e-05
kpi	bomb	n_O2_final	1.02931747994e-06
kpi	bomb	n_OH_final	1.08725431027e-06
kpi	bomb	n_O_final	5.68688863565e-07
kpi	bomb	n_water_final	1.05706749995e-05
kpi	bomb	t_end	0.0002
kpi	bomb	totalMoles_final	4.57220738612e-05

ignition01_h2o2_rk4

tutorials/batch/combustion/ignition01_h2o2_rk4

bomb
 batchReactor

What it does: Adiabatic constant-V H₂/O₂ ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); EXPLICIT RK4 – fails on the stiff chemistry (the lesson)

The story: - Choupo - * | \{\} | C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} | H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} | O perations | Website: <https://choupo.org> | | \{\} \{\} | U nits | Licence: GPL-3.0-or-later | | \{\} | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

This is the failure half of the stiffness lesson. It runs the exact same adiabatic constant-volume H₂/O₂ ignition as the sibling case [ignition01_h2o2](../ignition01_h2o2/README.md), but with the explicit RK4 integrator instead of the stiff Rosenbrock23 — at the same fixed step ($\Delta T = 1e-5$ s).

trajectory.csv is a column of nan from the very first output step ($t = 1e-4$ s) onward — there is no clean induction-then-jump, just immediate divergence:

*That immediate blow-up is the result. The radical lifetimes ($\sim 10^{-10}$ – 10^{-9} s) are six orders of magnitude shorter than the induction timescale ($\sim 10^{-3}$ s) — the chemistry is stiff. An explicit method like RK4 is stability-limited to a step of order the *fastest* scale, so the $\sim 10^{-5}$ s step this case uses is far too large: the integration is murdered by stability long before accuracy matters.*

*Compare side-by-side with the stiff sibling: switch to (or open) ignition01_h2o2, whose L-stable adaptive sub-stepping integrates the *same* chemistry to a clean burned state (~ 2440 K). *See, then decide*, applied to numerics.*

This case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail, and it would otherwise trip the NaN guard.

Same as the sibling: the thermodynamics is real curated Choupo data, but the reaction rates are synthetic (hand-chosen Arrhenius constants reproducing the chain-branching structure, not a published mechanism). Do not quote any number from this case as quantitative.

ignition02_h2air_slack

tutorials/batch/combustion/ignition02_h2air_slack

bomb
 batchReactor

What it does: Stoich H₂/air ignition, 1100 K / 2 atm (Slack shock tube; O Conaire 2004 kinetics)

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Stoichiometric H₂/air, 1100 K, 2 atm, constant volume — the reflected-shock condition of Slack (1977) as compiled in ? Conaire et al. 2004 (Fig 8). Kinetics: the ? Conaire H₂/O₂ mechanism, Table I verbatim (19 reactions; two Troe falloffs; two summed pairs; A in cm³-mol-s, Ea in kcal — the engine converts and announces every one).

Result: τ_{ign} (max dT/dt) = 43 μ s vs the measured ~40–60 μ s window.

Two lessons: 1. No seed. This mechanism self-initiates (R5: H₂ + M = 2H, Ea = 105 kcal/mol — slow, but that slowness IS the induction chemistry). With the ignition01-style trace-H seed the delay collapses to 12 μ s: initial conditions are physics, not knobs. 2. The engine now prints tau_ign in the epilogue (time of max dT/dt on the write grid) whenever a vessel ignites — no CSV spelunking.

Run ignition01_h2o2 (GRI kinetics, 5 bar) for the sibling case and the stiffness lesson (_rk4).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bomb	P_final	202650
kpi	bomb	T_final	3041.74576271
kpi	bomb	n_H2O2_final	3.94467753319e-11
kpi	bomb	n_H2_final	9.82828113933e-07
kpi	bomb	n_HO2_final	4.27531541611e-10
kpi	bomb	n_H_final	3.53516153966e-07
kpi	bomb	n_N2_final	1.391515e-05
kpi	bomb	n_O2_final	3.90750822458e-07
kpi	bomb	n_OH_final	4.44937008544e-07
kpi	bomb	n_O_final	1.5493529764e-07
kpi	bomb	n_water_final	6.00759209227e-06
kpi	bomb	t_end	0.0003
kpi	bomb	totalMoles_final	2.22501764671e-05

nox01_zeldovich_postflame

tutorials/batch/combustion/nox01_zeldovich_postflame

postflame
 batchReactor

What it does: Thermal NO_x in post-flame H₂/air gas, 2125 K / 1 atm (Glarborg 2018 N chemistry)

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Hot burnt gas (lean H₂/air, 2125 K, 1 atm — Homer & Sutton Flame 1 as compiled in Glarborg 2018) held isothermal, starting from the equilibrated H/O radical pool with all nitrogen still as N₂. The Glarborg 2018 nitrogen chemistry (extended Zeldovich + NNH + N₂O routes, Tables 2/5 verbatim) then forms NO.

*1. The cross-engine pin. The kinetic plateau lands on the Gibbs equilibrium computed by a *different engine* (steady Gibbs minimisation vs stiff batch integration). Kinetics choose the path and the clock; the destination belongs to the thermodynamics. Reverse rates come from the same equilibrium constants — so this agreement is a genuine consistency check of the mechanism transcription + thermo, not a tautology. 2. NO_x forms in engineering time. Ignition takes microseconds (ignition02: 43 μs); thermal NO takes seconds at 2125 K. That timescale gap is the whole basis of NO_x control by quench: cool the gas before the Zeldovich clock runs.*

The burnt-gas initial pool was computed with this same thermo by a 14-species gibbsReactor at 2125 K/1 atm (25 % H₂ / 17.5 % O₂ / 57.5 % N₂ feed); the equilibrium NO share was returned to N₂/O₂ elementally.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	postflame	P_final	101325
kpi	postflame	T_final	2125
kpi	postflame	n_H2O2_final	1.05023679159e-12
kpi	postflame	n_H2_final	4.85355503797e-09
kpi	postflame	n_HO2_final	1.61492364784e-11
kpi	postflame	n_H_final	6.02372866884e-10
kpi	postflame	n_N2O_final	1.37258448136e-12
kpi	postflame	n_N2_final	3.72505182272e-06
kpi	postflame	n_NH_final	2.50521342942e-15
kpi	postflame	n_NNH_final	1.01561106781e-15
kpi	postflame	n_NO_final	2.94524618017e-08
kpi	postflame	n_N_final	2.26160617192e-14
kpi	postflame	n_O2_final	3.06851934926e-07
kpi	postflame	n_OH_final	2.0389175016e-08

*... and 4 more reference values in the case's **expected**.*

reactor

The theory you need. Three idealised steady reactors. Conversion reactor: stoichiometry only, $n_{\text{out}} = n_{\text{in}} + \nu \xi$, duty from the ONE enthalpy base $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$ (formation datum – no separate “heat of reaction” input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate $r(c_{\text{out}}, T)$, so $F(c_{\text{in}} - c_{\text{out}}) = rV$. PFR: the same rate integrated along the volume, $dF_i/dV = \nu_i r$. Arrhenius kinetics $k = AT^b e^{-E_a/RT}$, orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential T sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

batch01_first_order

tutorials/batch/reactor/batch01_first_order

reactor

batchReactor

What it does: Isothermal batch Fischer esterification ($\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$) at 350 K, pseudo-first-order in ethanol; analytical exp-decay comparison

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	3.75075346157e-16
kpi	campaign	element_H_closure_rel	3.00060276926e-16
kpi	campaign	element_O_closure_rel	2.50050230771e-16
kpi	campaign	element_worst_closure_rel	3.75075346157e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	-210.150638715
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	0
kpi	campaign	energy_dH_vessels_kJ	-210.150638715
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	energy_residual_kJ	0
kpi	campaign	mass_closure_rel	4.5240474843e-16
kpi	campaign	mass_kg_external_out	0
... and 20 more reference values in the case's <i>expected</i> .			

batch02_adiabatic

tutorials/batch/reactor/batch02_adiabatic

reactor

batchReactor

What it does: Adiabatic batch esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$), exothermic, starting cold (320 K); thermal runaway to ~428 K then plateau as ethanol is consumed

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.2212252
kpi	reactor	n_aceticAcid_final	9.87885051063e-31
kpi	reactor	n_ethanol_final	9.87885051063e-31
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch03_consecutive

tutorials/batch/reactor/batch03_consecutive

reactor

batchReactor

What it does: Isothermal batch reactor: consecutive $A \rightarrow B \rightarrow C$ with $k_1 > k_2$; intermediate B peaks then decays

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	1
kpi	campaign	energy_records_valid	0
kpi	campaign	mass_closure_rel	5.55111512313e-16
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.5
kpi	campaign	mass_kg_initial	0.5
kpi	campaign	mass_residual_kg	2.77555756156e-16
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0
kpi	campaign	transfers_logged	0
kpi	reactor	P_final	101300
kpi	reactor	T_final	350
kpi	reactor	n_compA_final	3.05902320887e-09
... and 4 more reference values in the case's <i>expected</i> .			

batch04_reversible_wgs

tutorials/batch/reactor/batch04_reversible_wgs

reactor

batchReactor

What it does: Batch reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) at 600 K; conversion relaxes to X_{eq}

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	1000000
kpi	reactor	T_final	600
kpi	reactor	n_CO2_final	0.000420715762123
kpi	reactor	n_CO_final	7.92842378768e-05
kpi	reactor	n_H2_final	0.000420715762123
kpi	reactor	n_water_final	7.92842378768e-05
kpi	reactor	t_end	1200
kpi	reactor	totalMoles_final	0.001

batch04_transient_dirs

tutorials/batch/reactor/batch04_transient_dirs

reactor batchReactor

What it does: Adiabatic batch esterification with OpenFOAM-style REAL-TIME instant dirs: watch conversion climb $0 \rightarrow 1$ and T runaway $320 \rightarrow \sim 428$ K by scrubbing 0/ 30/ 60/ ... s

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.2212252
kpi	reactor	n_aceticAcid_final	9.87885051063e-31
kpi	reactor	n_ethanol_final	9.87885051063e-31
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch05_crystalliser

tutorials/batch/reactor/batch05_crystalliser

cryst
 batchCrystalliser

What it does: Batch crystalliser: supersaturated sugar relaxes, PBE by method of moments

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} // O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	6.81588841762e-15
kpi	campaign	element_H_closure_rel	4.18663266355e-15
kpi	campaign	element_O_closure_rel	4.18663266355e-15
kpi	campaign	element_worst_closure_rel	6.81588841762e-15
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	mass_closure_rel	5.07045231785e-15
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	1748.87224776
kpi	campaign	mass_kg_initial	1748.87224776
kpi	campaign	mass_residual_kg	8.86757334229e-12
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0

... and 13 more reference values in the case's *expected*.

batch06_adaptive_runaway

tutorials/batch/reactor/batch06_adaptive_runaway

reactor

batchReactor

What it does: Adiabatic esterification runaway (AcOH+EtOH); ADAPTIVE Rosenbrock23 time-stepping — the step grows after the fast transient (glass-box step history)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.221319512
kpi	reactor	n_aceticAcid_final	8.88140514762e-19
kpi	reactor	n_ethanol_final	1.10133235387e-34
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch07_lhhw_methylAcetate

tutorials/batch/reactor/batch07_lhhw_methylAcetate

reactor

batchReactor

What it does: LHHW in a BATCH reactor: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	333.15
kpi	reactor	n_aceticAcid_final	0.00216454708498
kpi	reactor	n_methanol_final	0.00216454708498
kpi	reactor	n_methylAcetate_final	0.00708545291502
kpi	reactor	n_water_final	0.00708545291502
kpi	reactor	t_end	3600
kpi	reactor	totalMoles_final	0.0185

batch08_isothermal_duty

tutorials/batch/reactor/batch08_isothermal_duty

reactor

batchReactor

What it does: Isothermal jacket duty from a DECLARED dH_{rxn} : $E = dH \cdot dX_i$ via extents (lumped species, no formation data)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	mass_closure_rel	1.11022302463e-15
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.05
kpi	campaign	mass_kg_initial	0.05
kpi	campaign	mass_residual_kg	5.55111512313e-17
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0
kpi	campaign	transfers_logged	0
kpi	reactor	E_reaction_seg0_kJ	-49.8760623911
kpi	reactor	E_reaction_total_kJ	-49.8760623911
kpi	reactor	P_final	101300
...and 5 more reference values in the case's <i>expected</i> .			

recipes

The theory you need. A recipe is time-triggered discrete control over batch units: charge, heat/cool, hold, transfer, sample. Between events the DAE integrator runs; at each event the state jumps (a transfer moves holdup between vessels). Expect exact event times on the trajectory and holdup bookkeeping that conserves mass across transfers.

recipe01_react_then_distill

tutorials/batch/recipes/recipe01_react_then_distill



What it does: Two-vessel batch recipe: esterify ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$) in a reactor, transfer the charge to a still, then distil

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	1.12522603847e-15
kpi	campaign	element_H_closure_rel	1.05021096924e-15
kpi	campaign	element_O_closure_rel	1.12522603847e-15
kpi	campaign	element_worst_closure_rel	1.12522603847e-15
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	-149.260408348
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	1.98711999947e-11
kpi	campaign	energy_dH_vessels_kJ	-149.260408078
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	6
kpi	campaign	energy_residual_kJ	2.69901306638e-07
kpi	campaign	mass_closure_rel	9.0480949686e-16
kpi	campaign	mass_kg_external_out	0
...and 31 more reference values in the case's expected.			

recipe02_setpoint_ramp

tutorials/batch/recipes/recipe02_setpoint_ramp

reactor

batchReactor

What it does: Single-reactor recipe with T_setpoint steps: 320 K → 350 K @ 200 s → 380 K @ 500 s

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	3.75075346157e-16
kpi	campaign	element_H_closure_rel	3.00060276926e-16
kpi	campaign	element_O_closure_rel	2.50050230771e-16
kpi	campaign	element_worst_closure_rel	3.75075346157e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	-104.365240634
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	0
kpi	campaign	energy_dH_vessels_kJ	-104.365240634
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	5
kpi	campaign	energy_residual_kJ	0
kpi	campaign	mass_closure_rel	4.5240474843e-16
kpi	campaign	mass_kg_external_out	0
... and 21 more reference values in the case's expected.			

recipe03_conditional_partial

tutorials/batch/recipes/recipe03_conditional_partial



What it does: Conditional + partial recipe: react until $x_{\text{ethanol}} < 0.30$, then transfer 50% to a still

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	350
kpi	reactor	n_aceticAcid_final	6.33853236066e-05
kpi	reactor	n_ethanol_final	6.33853236066e-05
kpi	reactor	n_ethylAcetate_final	0.00456161467639
kpi	reactor	n_water_final	0.00456161467639
kpi	reactor	t_end	1200
kpi	reactor	totalMoles_final	0.00925
kpi	still	P_final	101300
kpi	still	T_final	390.833339675
kpi	still	n_aceticAcid_final	0
kpi	still	n_ethanol_final	0
kpi	still	n_ethylAcetate_final	0
kpi	still	n_water_final	0

...and 2 more reference values in the case's expected.

recipe04_charge_enthalpy

tutorials/batch/recipes/recipe04_charge_enthalpy

hot

batchAccumulator

cold

batchAccumulator

What it does: H-equality recipe charge: T_mix from the first law, not the molar average; balance CLOSES across a transfer

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What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	2.13028428746e-12
kpi	campaign	energy_dH_vessels_kJ	5.81991344006e-10
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	0
kpi	campaign	energy_residual_kJ	5.81991344006e-10
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
...and 18 more reference values in the case's expected.			

still

The theory you need. A batch still is Rayleigh distillation: $d(n x_i)/dt = -\dot{V} y_i$ with y_i in equilibrium with the pot – the pot composition DRIFTS as the light component leaves (contrast the steady column, where stages hold profiles constant). Expect the distillate to start rich and fade, and the recipe layer (charge, heat, switch receivers) to drive the run. *Full derivation: Theory Guide, the batch-distillation section.*

still01_benzene_toluene

tutorials/batch/still/still01_benzene_toluene

still

batchStill

What it does: Rayleigh batch distillation: benzene/toluene 50/50, 1 atm, ideal Raoult

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	2.66880534766e-16
kpi	campaign	element_H_closure_rel	1.23908819713e-16
kpi	campaign	element_worst_closure_rel	2.66880534766e-16
kpi	campaign	energy_H_external_kJ	38.7543769459
kpi	campaign	energy_Q_ledger_kJ	0.942780166065
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	4.58362894565e-16
kpi	campaign	energy_dH_vessels_kJ	-37.8115967798
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	1.7763568394e-14
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0.0760287317788
kpi	campaign	mass_kg_final	0.00909876822121
... and 15 more reference values in the case's expected.			

still02_ethanol_water_azeo
tutorials/batch/still/still02_ethanol_water_azeo

still
batchStill

What it does: Rayleigh batch distillation: ethanol/water 40/60 with NRTL, residue concentrates in water

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	still	P_final	101300
kpi	still	T_final	372.446750369
kpi	still	n_ethanol_final	3.38836968904e-11
kpi	still	n_water_final	9.99999650927e-05
kpi	still	t_end	600
kpi	still	totalMoles_final	9.99999989764e-05

still03_with_receiver

tutorials/batch/still/still03_with_receiver



What it does: Rayleigh still distilling INTO a receiver tank (dischargeTo accumulator)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	2.66880534766e-16
kpi	campaign	element_H_closure_rel	1.23908819713e-16
kpi	campaign	element_worst_closure_rel	2.66880534766e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.942780166065
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	3.00056993907e-11
kpi	campaign	energy_dH_vessels_kJ	0.942780163768
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	2.2974155911e-09
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275
...and 28 more reference values in the case's expected.			

still04_rectifier_benzene_toluene

tutorials/batch/still/still04_rectifier_benzene_toluene

still

batchStill

receiver

batchAccumulator

What it does: batch rectification: 3 stages + total condenser, R=3, with receiver tank (dischargeTo accumulator)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.534303338765
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	7.08707897844e-10
kpi	campaign	energy_dH_vessels_kJ	0.534303321211
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	1.75539529579e-08
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275
...and 26 more reference values in the case's expected.			

still05_rectifier_constant_xD

tutorials/batch/still/still05_rectifier_constant_xD

still

batchStill

receiver

batchAccumulator

What it does: batch rectification, constant-composition policy: $R(t)$ holds x_D until $refluxMax$, then $D=0$

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.690222586439
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	8.89386285064e-10
kpi	campaign	energy_dH_vessels_kJ	0.69022261392
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	2.74809703882e-08
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275
...and 28 more reference values in the case's expected.			

still06_ledger_mixed_validity

tutorials/batch/still/still06_ledger_mixed_validity



What it does: Ledger H-validity is MONOTONIC: mixed valid/invalid packages never resurrect a poisoned record

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	mass_closure_rel	2.1198000621e-16
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.26186975
kpi	campaign	mass_kg_initial	0.26186975
kpi	campaign	mass_residual_kg	5.55111512313e-17
kpi	campaign	transfers_H_invalid	4
kpi	campaign	transfers_H_valid	1
kpi	campaign	transfers_logged	5
kpi	dump	P_final	101300
kpi	dump	T_final	372.484229602
kpi	dump	n_benzene_final	0.000249727144577
kpi	dump	n_compA_final	9.64864338217e-05
kpi	dump	n_toluene_final	0.000353786421588
kpi	dump	t_end	600
...and 43 more reference values in the case's expected.			

Category IV: Dynamics and control (choupoCtrl)

Continuous processes in time with feedback: PID loops, setpoint schedules, disturbances and responses.

ctrl01_cstr_temp_control

tutorials/ctrl/ctrl01_cstr_temp_control

reactor

dynamicCSTR

What it does: Dynamic CSTR (compA → compB) with PID jacket-temperature control; T climbs from 320 K into the +/-2% setpoint band within ~20 s (no overshoot) and tracks toward 350 K across the 150 s window

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	TC1	MV_final	348.54491648
kpi	TC1	PV_final	348.468973192
kpi	TC1	SP_final	350
kpi	reactor	T_final	348.481893564
kpi	reactor	n_compA_final	0.0116161140368
kpi	reactor	n_compB_final	0.00038388596316
kpi	reactor	t_end	150

ctrl02_disturbance_rejection

tutorials/ctrl/ctrl02_disturbance_rejection

reactor

dynamicCSTR

What it does: PID disturbance rejection: T_in steps down then up, PID maintains T_setpoint

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535585
kpi	TC1	PV_final	350.00207645
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059115
kpi	reactor	n_compA_final	0.0110806465498
kpi	reactor	n_compB_final	0.000919353450202
kpi	reactor	t_end	2000

ctrl03_adaptive_disturbance

tutorials/ctrl/ctrl03_adaptive_disturbance

reactor

dynamicCSTR

What it does: PID disturbance rejection with ADAPTIVE plant integration — controller samples on the fixed 1 s grid, Rosenbrock23 sub-steps between samples (MV held)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535626
kpi	TC1	PV_final	350.002076468
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059134
kpi	reactor	n_compA_final	0.0110806465416
kpi	reactor	n_compB_final	0.000919353458409
kpi	reactor	t_end	2000

ctrl03_startup_transient

tutorials/ctrl/ctrl03_startup_transient

reactor

dynamicCSTR

What it does: Open-loop CSTR START-UP transient with OpenFOAM-style real-time instant dirs: charged cold (320 K), no control, watch T(t) and the holdup approach the natural steady state by scrubbing 0/ 50/ 100/ ... s

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | H eat-transfer | Version: Choupo-2607 | O perations | Website: <https://choupo.org> | U nits | Licence: GPL-3.0-or-later | P roperties | Tutorial case file – part of Choupo | O ptimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	T_final	320.759374011
kpi	reactor	n_compA_final	0.0117988900072
kpi	reactor	n_compB_final	0.000201109992805
kpi	reactor	t_end	1000

ctrl05_disturbance_transient

tutorials/ctrl/ctrl05_disturbance_transient

reactor

dynamicCSTR

What it does: PID disturbance rejection with OpenFOAM-style real-time instant dirs: T_in steps down (700 s) then up (1300 s); scrub 0/ 40/ 80/ ... s to watch the PID track the 350 K setpoint and the jacket MV move

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535585
kpi	TC1	PV_final	350.00207645
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059115
kpi	reactor	n_compA_final	0.0110806465498
kpi	reactor	n_compB_final	0.000919353450202
kpi	reactor	t_end	2000

ctrl06_sine_disturbance

tutorials/ctrl/ctrl06_sine_disturbance

reactor

dynamicCSTR

What it does: Forced SINUSOIDAL inlet-T disturbance: $T_{in} = 320 + 15 \sin(2 \pi t / 600)$. The PID chases a moving target → the reactor temperature settles into a forced sinusoidal response (attenuated + phase-lagged). Sweep the period to build a one-point Bode by trial-and-error (the frequency-response seed).

The story: - Choupo - * | | C hemicals | Choupo: open-source, glass-box simulation | | H eat-transfer | Version: Choupo-2607 | | O perations | Website: <https://choupo.org> | | U nits | Licence: GPL-3.0-or-later | | P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	319.842923238
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	319.842923238
kpi	TC1	MV_final	350.026347162
kpi	TC1	PV_final	350.170890591
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.173436842
kpi	reactor	n_compA_final	0.0110844237702
kpi	reactor	n_compB_final	0.000915576229809
kpi	reactor	t_end	3000

ctrl07_lhhw_inhibition
tutorials/ctrl/ctrl07_lhhw_inhibition

reactor
dynamicCSTR

What it does: Product inhibition meets a PID: an LHHW reaction whose exotherm fades as its own product covers the catalyst

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	TC1	MV_final	349.740705746
kpi	TC1	PV_final	348.328939265
kpi	TC1	SP_final	350
kpi	reactor	T_final	348.342145073
kpi	reactor	n_compA_final	0.0117152677233
kpi	reactor	n_compB_final	0.000284732276726
kpi	reactor	t_end	150

ctrl08_prbs_ident

tutorials/ctrl/ctrl08_prbs_ident

reactor

dynamicCSTR

What it does: PRBS identification experiment: $T_{in} = 320 \pm 10$ K by a maximal-length LFSR (n=4, period 15 bits x 120 s) – deterministic, seed-declared, period-VERIFIED; the broadband excitation the sine of ctrl06 cannot give.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedPRBS	MV_final	330
kpi	FeedPRBS	PV_final	0
kpi	FeedPRBS	SP_final	330
kpi	TC1	MV_final	348.579621229
kpi	TC1	PV_final	350.006520504
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.006465191
kpi	reactor	n_compA_final	0.0110804887195
kpi	reactor	n_compB_final	0.000919511280459
kpi	reactor	t_end	3600

ctrl09_stream_disturbance

tutorials/ctrl/ctrl09_stream_disturbance

reactor
dynamicCSTR

What it does: Inlet-field disturbance: componentMolarFlow.compA steps $3.5\text{e-}5 \rightarrow 5.25\text{e-}5$ kmol/s; F and z_in change by the DERIVED values ($F = 6.75\text{e-}5$, $z_A = 7/9$) – the disturbance never hides from the other components

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

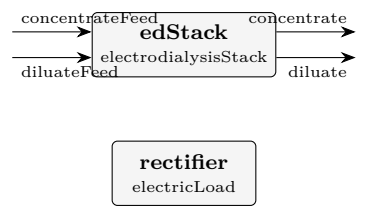
What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedComp	MV_final	5.25e-05
kpi	FeedComp	PV_final	0
kpi	FeedComp	SP_final	5.25e-05
kpi	TC1	MV_final	350.204904403
kpi	TC1	PV_final	350.000124372
kpi	TC1	SP_final	350
kpi	reactor	F_in_final	6.75e-05
kpi	reactor	T_final	350.000123673
kpi	reactor	n_compA_final	0.00879292389459
kpi	reactor	n_compB_final	0.00320707610541
kpi	reactor	t_end	1800
kpi	reactor	z_in_compA_final	0.777777777778
kpi	reactor	z_in_compB_final	0.222222222222

Category V: Electrochemical systems

ed01_nacl_desalination

tutorials/electrochem/ed01_nacl_desalination



What it does: Electrodialysis stack: NaCl diluate desalination, 100 cell pairs

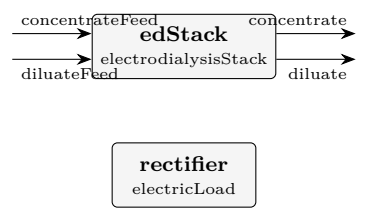
The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrate	F	0.00501492454831
stream	concentrateFeed	F	0.005
stream	diluate	F	0.00498507545169
stream	diluateFeed	F	0.005
kpi	edStack	E_mem_pair	0.0794573845707
kpi	edStack	I	8
kpi	edStack	N_cellpairs	100
kpi	edStack	R_pair	0.00507753048751
kpi	edStack	U	13.5077628471
kpi	edStack	W_electric	108.062102777
kpi	edStack	W_electric_kW	0.108062102777
kpi	edStack	current_efficiency	0.9
kpi	edStack	demin_ratio	0.831450327631
kpi	edStack	i_density	40
... and 9 more reference values in the case's <i>expected</i> .			

ed02_over_limiting_current

tutorials/electrochem/ed02_over_limiting_current



What it does: Electrodialysis stack driven ABOVE its limiting current (loud warning)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} / H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} / O perations | Website: <https://choupo.org> | | \{\} \{\} / U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrate	F	0.00500205212539
stream	concentrateFeed	F	0.005
stream	diluate	F	0.0199979478746
stream	diluateFeed	F	0.02
kpi	edStack	E_mem_pair	0.156590406589
kpi	edStack	I	5.5
kpi	edStack	N_cellpairs	20
kpi	edStack	R_pair	0.0130038886532
kpi	edStack	U	6.06223588363
kpi	edStack	W_electric	33.34229736
kpi	edStack	W_electric_kW	0.03334229736
kpi	edStack	current_efficiency	0.9
kpi	edStack	demin_ratio	0.142508707778
kpi	edStack	i_density	27.5
... and 9 more reference values in the case's <i>expected</i> .			

Category VI: Multi-sector plant showcases

The fractal flowsheet at full size – read them like worked examples.

ChemicalPlantTutorial

The theory you need. The full multi-sector plant showcase: feed treatment, reaction, separation, utilities – the fractal flowsheet at its largest, run-only (no golden), for reading and dissecting rather than reproducing numbers.

CONCENTRATION

tutorials/plant/ChemicalPlantTutorial/CONCENTRATION

The story: - Choupo - * | / C hemicals | Choupo: open-source, glass-box simulation | / Heat-transfer | Version: Choupo-2607 | / Operations | Website: <https://choupo.org> | / Units | Licence: GPL-3.0-or-later | / Properties | Tutorial case file – part of Choupo | | Optimization |

DRYING

tutorials/plant/ChemicalPlantTutorial/DRYING

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\}\{\} // H eat-transfer | Version: Choupo-2607 | | \{\}\{\}\{\} /// O perations | Website: <https://choupo.org> | | \{\}\{\} /// U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

FERMENTATION

tutorials/plant/ChemicalPlantTutorial/FERMENTATION

The story: - Choupo - * | C hemicals | Choupo: open-source, glass-box simulation | Heat-transfer | Version: Choupo-2607 | Operations | Website: <https://choupo.org> | Units | Licence: GPL-3.0-or-later | Properties | Tutorial case file – part of Choupo | Optimization |

JuiceSplitter

tutorials/plant/ChemicalPlantTutorial/JuiceSplitter

The story: - Choupo - * | / C hemicals | Choupo: open-source, glass-box simulation | / H eat-transfer | Version: Choupo-2607 | / O perations | Website: <https://choupo.org> | / U nits | Licence: GPL-3.0-or-later | / P roperties | Tutorial case file – part of Choupo | | O ptimization |

esterification2sector

tutorials/plant/esterification2sector

What it does: Esterification plant – two sectors (REACTION + SEPARATION) sharing components; per-sector constant/ data (S1 kinetics, S2 NRTL pair); ethanol-water pair inherited from the standard library

The story: - Choupo -*—————* esterification2sector – a TWO-SECTOR fractal plant, WIRED.

[REACTION] — esterification CSTR, its OWN kinetics in | sectors/REACTION/constant/reactions reactorOut

[SEPARATION] — isothermal flash; the ethylAcetate-water NRTL pair | \{\} is PARTICULAR to this sector liquid vapor (sectors/SEPARATION/constant/parameters/NRTL/...); ethanol-water is INHERITED from the standard library.

The point of the case is the FRACTAL ‘constant/’: components shared across the plant, particular data (kinetics here, a fitted pair there) living with the sector that owns it, and the rest walked up to the standard catalogue.

The root graph assembles the sectors and the inter-sector links, nothing more. Each stream’s state lives in its own file under 0/, owned by the sector that produces it.

A two-sector plant that exists to prove one thing: data lives with the sector that owns it, and a whole-plant run finds it there.

*Components are shared across the plant. The kinetics belong to REACTION and are never copied to the root. The ethylAcetate-water NRTL pair belongs to SEPARATION — a *particular* datum, fitted for this separation — while ethanol-water walks up to the standard catalogue. Three levels of ownership, one run.*

That perNode is the whole case. The sector’s pair beat the standard library, and the log says so rather than leaving you to wonder which one it used.

Feed 100 kmol/h equimolar acetic acid + ethanol at 360 K. The CSTR ($V_R = 0.5 \text{ m}^3$) reaches 34.2 % conversion — a PLACEHOLDER number: the kinetics in sectors/REACTION/constant/reactions are still the pseudo-first-order stand-in, not the 2nd-order fit this case’s own property evidence supports (see CLAUDE.md, "Pending"). The flash at 355 K, 1.013 bar splits the product into a vapour rich in ethyl acetate (bp 350 K) and ethanol, and a liquid rich in water and the unconverted acid. Mass closes exactly: $34.18 + 65.82 = 100.00 \text{ kmol/h}$.

This case shipped for weeks marked .known-broken, and it earned the mark twice over. Its flash carried P 1.01325; with no unit, which is 1.01 pascal — a hard vacuum — so the K -values exploded and everything went to the vapour. And it was never wired: the sectors existed as thermo regions with the connections commented out, awaiting a "curation phase" that never ended.

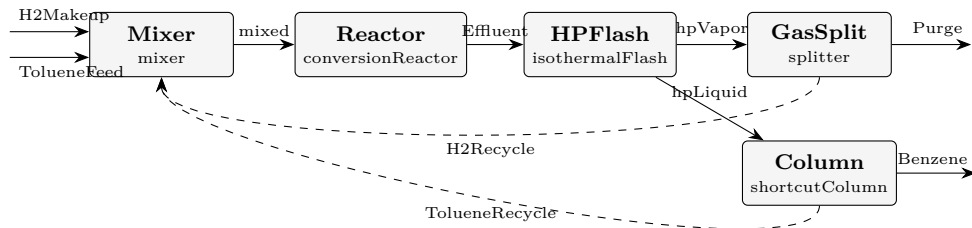
*Fixing it surfaced two real engine bugs, both of the same shape — data owned by a *sector* was invisible from a whole-plant run:*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.0277777777778
stream	liquid	F	0.00949461019202
stream	reactorOut	F	0.0277777777778
stream	vapor	F	0.0182831675858

hda

tutorials/plant/hda



What it does: HDA plant – toluene + H₂ → benzene + CH₄. A single FLAT flowsheet with two plant-level recycles (an H₂ gas loop and an unreacted-toluene liquid loop); cross-sector tears are unsupported, so it is not fractal. Furnace lumped into the isothermal reactor; light-gas K-values rest on Antoine extrapolation (see thermoPhysPropDict).

The story: - Choupo -*—————* flowsheetDict – HDA plant (single FLAT flowsheet).

toluene + H₂ → benzene + CH₄, with TWO coupled recycles: an H₂ gas loop (GasSplit 95 % → back to the Mixer) an unreacted-toluene loop (Column bottoms → back to the Mixer)

WHY FLAT, not the two-sector fractal we first attempted: a recycle closes only when its ‘tearStreams’ declaration AND both endpoints live in the SAME flowsheet node. The engine’s tear-closing override (Flowsheet.cpp) is leaf-only and same-level-only – it does NOT support CROSS-SECTOR (composite-to-composite) tears. Both HDA loops run from the separation train back to the reactor feed, so no single sector owns both ends; they are inherently plant-level and must be flat (the proven process03_recycle pattern: each tear name is BOTH a Mixer input AND a producer output, listed after the Mixer). The fractal sector showcase lives elsewhere (the sector-internal recycle in the two-sector demo).

Everything is at 35 bar so the recycles return at reactor pressure (the Mixer takes min-P). The furnace is lumped into the isothermal reactor (operation.T 900 K) – the engine’s heater is duty-specified, not T-target. The benzene product is ~89 mol% benzene but carries a non-trivial ~10% CH₄ (plus ~0.2% H₂ and ~0.3% toluene): the light non-keys dissolved in the flash liquid are lighter than benzene, so the FUG shortcut sends them all to the distillate. A real plant would put a stabiliser / light-ends column upstream of the product spec to strip that ~10% methane; the shortcut here is quantitatively approximate (it needs no rigorous liquid enthalpy, so unlike a Wang-Henke column it does not choke on a gas in the liquid).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Benzene	F	0.0308570333668
stream	Effluent	F	0.589422462034
stream	H2Makeup	F	0.0305555555556
stream	H2Recycle	F	0.522049697299
stream	Purge	F	0.0274762998579
stream	TolueneFeed	F	0.0277777777778
stream	TolueneRecycle	F	0.00903943150968
stream	hpLiquid	F	0.0398964648764
stream	hpVapor	F	0.549525997157
stream	mixed	F	0.589422462034
kpi	Column	B	0.00903943150968
kpi	Column	D	0.0308570333668
kpi	Column	N_min	8.3953506183
kpi	Column	N_theoretical	20.1717591626

... and 28 more reference values in the case’s *expected*.

lithiumBrinePlant

tutorials/plant/lithiumBrinePlant

What it does: lithiumBrinePlant – THE reference case: a FRACTAL lithium-carbonate plant from salar brine, WIRED into one solver. Five SECTORS, each a thermo WORLD of its own – BRINE (electrolyte/Pitzer), EXTRAC-TION (gamma-gamma NRTL LLE), CARBONATION (Li₂CO₃ retrograde solubility), HOTAIR (Gibbs combustion), FINISHING (drying + cyclone) – connected sector to sector, each sector declaring its own propertyPackage (its world) on the global component set, with optional per-unit thermo overrides where a single unit needs a different world. The proof that the architecture is fractal: sectors are composite boxes, streams flow between them, model boundaries audited.

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	carbLiquor	F	0.02777777777778
stream	cleanAir	F	0.0320991487231
stream	emulsion	F	0.0380203541089
stream	finer	F	0
stream	fuelAir	F	0.01085555555558
stream	halite	F	0.000179323046651
stream	hotAir	F	0.01085555555556
stream	humidExhaust	F	0.0320991487231
stream	liquor	F	0.0269092429978
stream	loadedOrganic	F	0.0124069292995
stream	organic	F	0.01111111111111
stream	product	F	0.00644893419035
stream	raffinate	F	0.0256134248094
stream	salarBrine	F	0.02777777777778

...and 1 more reference values in the case's *expected*.

polycaprolactonePlant

The theory you need. Ring-opening polymerisation plant: monomer to polycaprolactone with step-growth KPIs (Carothers, Flory) on the PFR, polymer properties from group contribution on the repeat unit.

properties

tutorials/plant/polycaprolactonePlant/properties

pclRepeat_Tg_yang2020
estimateComponent

What it does: Close the loop: ESTIMATE the PCL repeat-unit glass transition Tg by the Yang 2020 group-contribution scheme — the plant gives Mn/PDI, this gives the physical property from the SAME repeat unit -[O-(CH2)5-CO]-

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A choupoProps sub-case of ../ (the polycaprolactone plant). The plant gives you the chain statistics from the kinetics (Mn / Mw / PDI); here you estimate the polymer’s physical property — the glass transition Tg — from the same repeat unit -[O-(CH2)5-CO]- (M0 = 114.14, the exact number the reactor used for Mn = M0/(1-p)).

** Tg via Yang 2020 (CC-BY) — runnable. Tg(inf) = 85.8 K; the console prints the -59.7 % deviation from the measured ~213 K (an honest limit of an additive scheme on a flexible aliphatic polyester). M0 = 114 is goldened (an exact mass-balance fact); Tg is shown but not goldened as validated physics. * Density via Van Krevelen — *not* runnable on PCL with the shipped data (the Slice-1 group set has no ester -O-/-CO- group, so it would hard-error rather than invent a value). See the full explanation and the runnable hydrocarbon example linked from the parent ../README.md.*

The full teaching narrative lives in the parent README. See system/propsDict for the exact group decomposition and the estimator invocation.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pclRepeat_Tg_yang2020	M0_g_per_mol	114
diag	pclRepeat_Tg_yang2020	Tg_K	85.81
diag	pclRepeat_Tg_yang2020	YgSum_1e3gKmol	9.782

sugarPlantEconomicsSweep

tutorials/plant/sugarPlantEconomicsSweep

What it does: Sugar plant ECONOMICS SWEEP – IRR / payback / NPV vs RawJuice sucrose content (Phase-2 SweepDriver post-chain differentiator)

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | \{\} / P roperties | Tutorial case file – part of Choupo | | O ptimization | |

This is the flagship sugar plant (tutorials/plant/ChemicalPlantTutorial) driven by a SweepDriver instead of a single pass. It exists to show the one thing a single CAPEX number cannot: how the project economics move when a process variable changes.

The driver runs the FULL post-processing chain (sizing → costing → economics, in system/postDict) on every converged point, so economics.IRR, economics.paybackYears, economics.NPV, ? resolve as ordinary KPI look-ups (the Phase-2 SweepDriver post-chain wiring). The result lands in sweep_results.csv.

Sweeping the RawJuice sucrose content 0.09 → 0.16 mol fraction walks the plant from sub-economic, through break-even, to clearly profitable:

Richer juice → more crystalline sugar per unit feed → more revenue against an almost-unchanged plant cost → the IRR climbs and the payback appears and shortens. The NPV crosses zero between 0.104 and 0.118.

*Both use the same constant/economics prices (EU Sugar Market Observatory white sugar, ECB EUR/USD, *Chemical Engineering* CEPCI, US statutory tax) and the same system/postDict cost chain — only the run mode differs.*

tartaricAcid

tutorials/plant/tartaricAcid



What it does: Tartaric-acid plant (wine route): lime precipitation of calcium tartrate, then sulfuric acidulation to tartaric acid + gypsum. Molecular reactors; the solution chemistry sub-study is tutorials/props/electrolyte/-tartaricAcid_acidulation. HONEST LIMITS: H2SO4 declared nonvolatile by case overlay (P_{sat} ~1e-5 Pa at process T); precipitates ride as nonvolatile solutes with NO rigorous solid-liquid separation; reactor duties honestly UNAVAILABLE (species without ideal-gas C_p – a named gap, never a fake zero).

The story: - Choupo -*—————* Tartaric-acid plant – the WINE ROUTE chemistry, as a reactor train.

bitartrate (KHT) lime Ca(OH)₂ M1 mixer R1 precipitation (CaTartrate + KOH)

H2SO4 M2 mixer R2 acidulation product (tartaricAcid + gypsum)

R1: KHC₄H₄O₆ + Ca(OH)₂ → CaC₄H₄O₆ + KOH + H₂O R2: CaC₄H₄O₆ + H₂SO₄ → C₄H₆O₆(tartaric acid) + CaSO₄(gypsum)

SCOPE (honest). This demonstrates the REACTION chemistry and closes the MASS balance. The physical solid-liquid SEPARATIONS of the real process – filtering the calcium-tartrate cake from the mother liquor, and the gypsum from the acid – are NOT modelled: Choupo treats the precipitate as a dissolved nonvolatile, not a solid phase, so there is no F_{solid} mass for a filter to capture (a precipitator / solid-phase reactor is a roadmap feature). The ENERGY balance also refuses on the solid calcium tartrate (its formation datum is omitted – no curated dissolution enthalpy). The solution-chemistry of the acidulation liquor (Pitzer, gypsum saturation) lives in tutorials/props/electrolyte/tartaricAcid_acidulation.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	acidFeed	F	0.943266666667
stream	bitartrate	F	0.583333333333
stream	lime	F	0.166666666667
stream	precipOut	F	0.7766
stream	product	F	0.943266666667
stream	slurry	F	0.75
stream	sulfuric	F	0.166666666667
kpi	M1	F_out	0.75
kpi	M1	P_out	100000
kpi	M1	T_out	333
kpi	M1	n_in	2
kpi	M1	vf	0
kpi	M2	F_out	0.943266666667
kpi	M2	P_out	100000

... and 13 more reference values in the case's *expected*.

twoSectorDemo

The theory you need. The zippable two-sector demonstration plant – the case used to rehearse the online landing. Same physics as the esterification demo.

SECTOR_R

tutorials/plant/twoSectorDemo/SECTOR_R



What it does: SECTOR_R – the reactor sector. Runs its OWN compiled unit op (stoichReactor) on the plant-level thermo (cascaded from ../constant).

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SECTOR_S

tutorials/plant/twoSectorDemo/SECTOR_S



What it does: SECTOR_S – the separation sector. Runs its OWN compiled unit op (sharpSplitColumn) on the plant-level thermo (cascaded from ../constant).

The story: - Choupo - * | \{\} / C hemicals | Choupo: open-source, glass-box simulation | | \{\} \{\} // H eat-transfer | Version: Choupo-2607 | | \{\} \{\} \{\} /// O perations | Website: <https://choupo.org> | | \{\} \{\} // U nits | Licence: GPL-3.0-or-later | | \{\} / P roperties | Tutorial case file – part of Choupo | | | O ptimization | |